



Summary of all chap

Introduction Psychology and its Methods AI (Vrije Universiteit Amsterdam)



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Psychology Lecture 1 Notes

By Somansh – former Prince of Nigeria

Overview

- Genetic basis of life
- Evolution by natural selection
- Genes and behaviour linkage

Genetic Basis of life

- **Does violent person have violent genes**
 - Varied in past
 - **70s and 80s belief** – Behaviour (including violence) was **influenced by environment**. Because ideology influenced their scientific thinking. As if they **wanted to believe** that capitalism and strict parents created violent kids.
 - Dutch dude/prof called **Bowhouser** wanted **to find biological basis of crime**, like – does testosterone increase criminal behaviour. Fellow scholars and media **trolled** him and he quit science.
 - **In 90s, the heritability studies** became common. Studies found that **genes were the basis of violence** and other behaviour. Thus, refuted that environment was main cause of violence and other behaviour. Scholars like Rich, Udit ? wrote a book on it.
 - **Today** we know that **both genes and environment affect behaviour**.
- Genes and biology
 - Human cells have **23 pairs of chromosomes (46 in total)**. Chromosomes are mostly a coil of DNA strings/threads/wires.
 - **DNA is a large string of alphabets** [built of bases A,T,G, and C]. Combo of alphabets is called genes (words). Genes are **read by** a little chemical dudes (**tRNA ,mRNA**) in the cell and they **tell ribosomes** (located just outside nucleus of cell) **to make proteins** based on these words. These **proteins** help in **making** and working of **muscles, eyes, skin, immune cells, neurones, blood, brain** etc. Basically DNA builds and runs the body.
 - **'A' always pairs with 'T'. G always pairs with C**. These base pairs are held by Phosphate and sugar chemical structure.
 - Not all parts of DNA leads to creating proteins (**non-coding DNA**). We still don't know what it does, completely.
 - Basically, **genes give the blueprint of every part of our body, and how they will behave, mostly**. But food, exercise, social life, random events also affect our body and behaviour.
 - This is where **heritability** comes in.
 - It answers the question: **how much do genes affect a trait** (violence, eye colour, height, depression, IQ), **and how much does environment affect a trait**.
 - **Gene expression** - **not all genes are expressed at all times**. Every cell in body has same genes but the **genes that are ON in a cell determine it's function**.

Neural cells, liver cells, skin cells have same genes but very **different due to their gene expression**.

- **Methylation** – Methyl molecules come from environment/behaviour/random events. These Methyl molecules are attached to a gene, and turn it off. Thus restricting its expression. It stops whatever gene it touches.
- **Thus, cells are affected by environment**. Thus, environment affects genes, genetic expression and consequently affects behaviour.
- **Spielman ? 1967 study showed that cells from embryo become teeth, when planted on mouth area, and become skin when planted outside the embryo**. Clearly, the location (environment) of genes affect their expression.
- **So, not every protein encoded in the genes is created all the time**. Hence, a person with genes for good muscles may not have a fit body coz she didn't exercise. In other words, the genes for good muscles were not expressed.
- **Genotype – Set of genes of a specific organism**
 - **Phenotype – features/traits/behaviour of a specific organism. It is an outcome of genes and environment's interaction**.
 - **Monozygotic twins** (2 persons born out of one sperm fusing with one egg) have **100% genes in common**. But they **differ in many features – behaviour, hobbies, skin health, fitness etc**. Clearly, **environment, not genes, led to these differences**.
 - **Alleles** - Each baby gets **23 chromosomes from mother and 23 chromosomes from father**. Except sex chromosomes X and Y, the genes on every chromosome from mother's side is similar to genes on chromosome from father's side. For example, mother and father's chromosome number 11 will have similar pair of genes at same position. **These similar genes are called alleles. Think of alleles as genes that are located at the same position in chromosome, and do similar task. But vary slightly.**
 - **Note: NOT EXACTLY SURE ABOUT THIS ONE. VERIFY LATER.** There may be multiple alleles of a gene in a population, such as alleles of eye colour genes: blue, green, black, brown. But a person gets only two genes out of the total genes available. One from mother and another from father.
 - **If the two genes (alleles) at same position are identical, the person is homozygous for that gene pair. If they're somewhat different, the person is called heterozygous for that gene pair.**
 - In the alleles (genes), **one gene may be dominant and other may be recessive**. In that case, dominant gene will express, and recessive won't. To illustrate, assume a gene 'B' leads to blue eyes (when expressed), and gene 'G' leads to grey eyes when expressed. If G is dominant and B is recessive, the person will have grey eyes. Recessive genes are expressed when both alleles are recessive.
 - **Sometimes both genes are recessive or dominant. Then, a weird mixed expression of genes happens.**
 - Not every gene leads to one trait. **Many genes may influence one trait** (height/Schizophrenia) – called **Polygenic inheritance**.

- Or **one gene may influence multiple traits.**
- **How does an organism end up with a genome ?**
 - Due to **natural selection** – explained by Darwin.
 - Genes that are expressed and lead to **advantageous traits** (good hunting skills, sense of smell, vision) allowed **an individual to survive** long enough and **reproduce**. Thus **passing the genes**. With time, genes that help in life survive and thrive. Though, some not so useful genes also slip through, which may turn out to be useful after many generations, or turn out to be lethal.
 - ★ ○ **3 conditions for natural selection**
 - The **population** of organisms must have **variation in genes.**
 - **Individuals** with genes suited for the environment **survive and reproduce more** than others.
 - **Traits are thus passed** from parents to offspring.
 - **Example** – a **moth** of brown colour used to blend with the tree colour, so birds couldn't eat it. But in cities, the brown colour didn't blend well with dark polluted walls. So, moths of darker shades survived more in polluted cities. After pollution restrictions, brown moth was back.
 - **Evidences of evolution**
 - **Fossil evidence** – dinosaur, bird, human, fish fossils over millions of years.
 - **Similar genomes** of various organisms.
 - Chromosome 2 of humans is a fusion of two chromosomes of anthropoid apes that existed thousands of years ago.
 - **Pseudogenes** – **inactive genes in DNA**. Maybe functioned in the past but now just sits there. **Humans** have **2000 such genes**. We carry genes required to make **Vitamin C** from glucose. But one part is inactivated. Eating Vitamin C rich foods must've inactivated the gene. So, another proof of evolution.
 - **Distribution of species across Continents.**
 - Japan – Earlier connected to a continent. Diverse species.
 - Hawaii – Oceanic island, never connected to a Continent. Lacks diverse animals like freshwater fish, reptiles. Only long distance travelling species could colonise the island – like birds.
 - But how does it explain evolution ?
 - Evolution doesn't improve the species, nor does it create the best genome. It **only creates species adapted to the environment**. A fish has genes suited to water life but on land they are useless.
 - **Evolution doesn't create rigid behaviour patterns. It may change when needed.**
- **Nature nurture debate**
 - Scholars used to **argue if nature(natural build/biology/genes) or nurture(environment, outside events, parenting, random events) shaped the organism**
 - **Today we know it is both**, so the debate is irrelevant now. Nature and nurture interact and shape the organism

- Who we are and how we behave is determined by – how our genes behave in a certain environment.
- **MAO study**
 - **Persons with low MAO activity combined with parental mistreatment were more like to commit crime and show aggressive behaviour.**
 - Low MAO didn't automatically lead to higher aggression. Mistreatment alone did increase aggression, but when combined with low MAO activity, aggression shot up. Thus confirming: nature and nurture influence behaviour.

Psychology Topic – 2

By Somansh – successfully auditioned for Coldplay twice

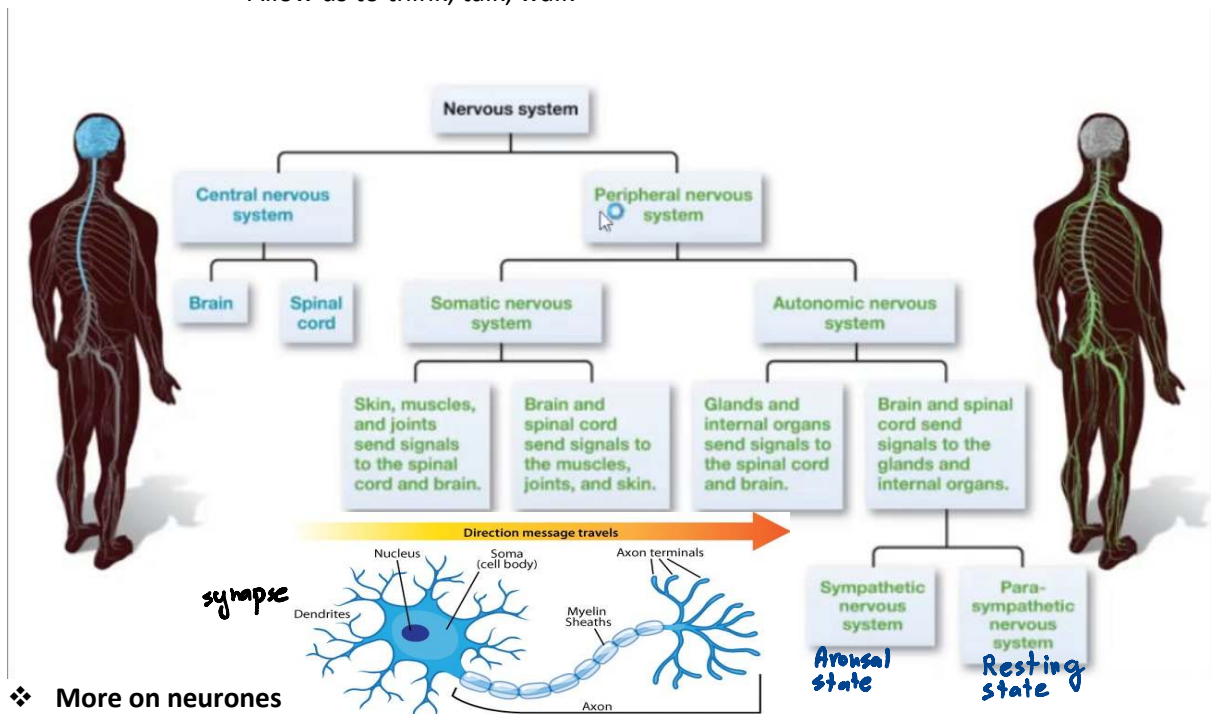
The Brain and the nervous system.

❖ Overview

- Nervous system structure
- Communication among neurones
- Communication of brain and body
- The Brain

❖ Genes and environment interact to build a trait of organism

- It also shapes our nervous system
- Nervous system has
 - **Central nervous system – Brain & Spinal cord (C.N.S)**
 - **Peripheral nervous system – Other nerves in body**
- Brain is critical for behaviour
 - Weighs 1.5 kg approx
- Nervous system – including brain – is made up of 2 basic cells
 - **Glia – supportive functions**
 - **Neurones** – Receive and send information via electric pulses.
 - Allow us to think, talk, walk

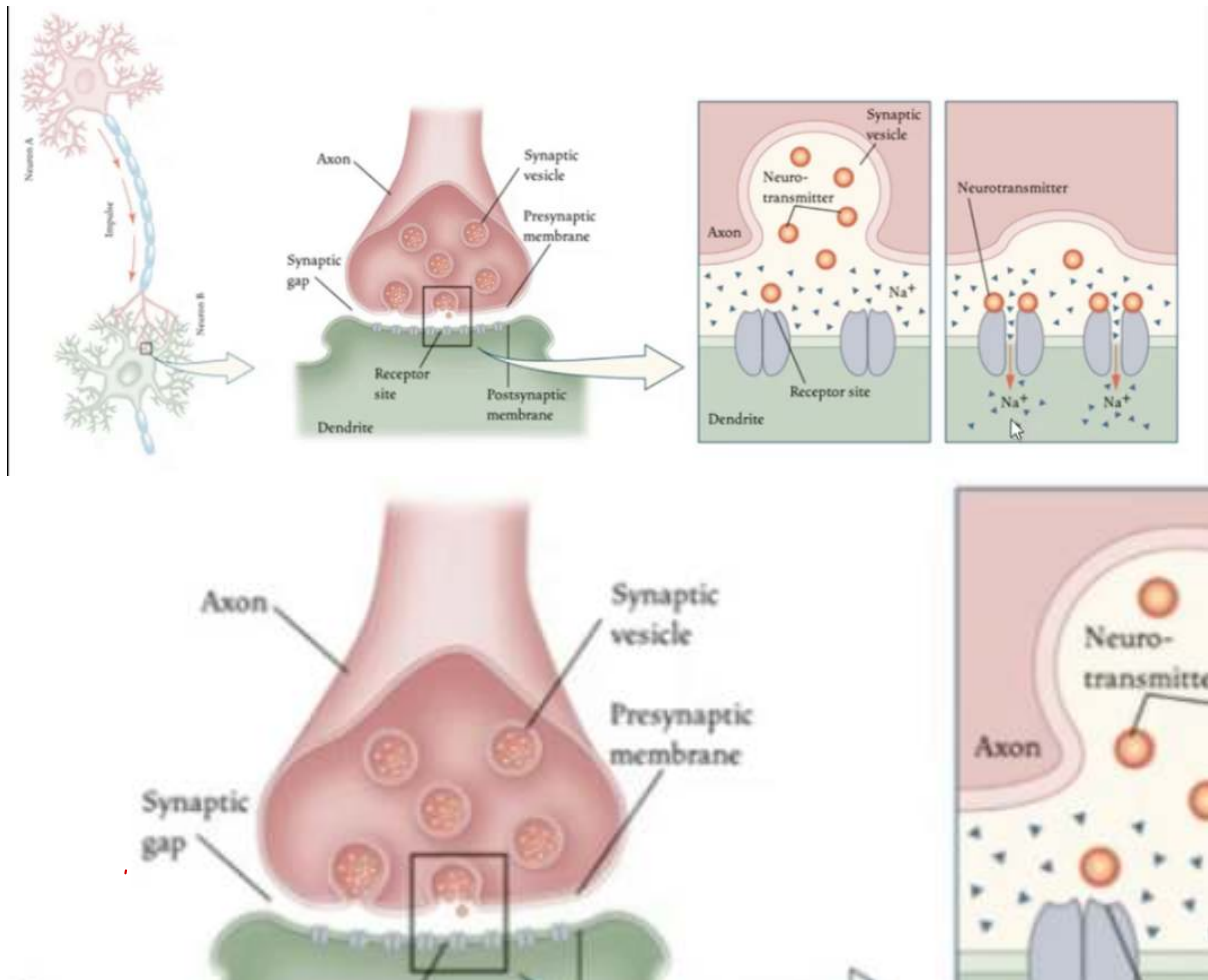


❖ More on neurones

- **Dendrites receive signal** from other neurones connected to it.
- **Axon transmits the information** received from dendrites to other neurones connected to axon terminals. The **output pipe** of neurone.
 - **Axon is covered by fatty substance called Myelin Sheath.**
 - Myelin sheath is **fatty**, so it **doesn't dissolve in watery surroundings** of the brain. It is **semi-permeable**, so that **some chemicals may enter or exit it. Ions (charged particles) can't freely move in and out of Myelin sheath. Ions can travel via ion channels.**
 - When the neurone is activated/fired/addressed there is **a change in potential difference between inside and outside of the cell. Potential difference**

means, amount of negatively and positively charged particles are different, inside and outside of neuron.

- **Voltage meters measure the potential difference** by inserting one needle inside and another needle outside of axon. The number comes out to be
 - **-70 millivolts when the neuron is inactive.** This state is called the **resting potential** of the neuron. **-70mV means the neuron is negatively charged on the inside** OR more positively charged particles (**Na⁺ Sodium ions**) outside the membrane than the positively charged particles on the inside (**K⁺ Potassium ions**).
 - This happens because **K⁺ ions get in and out of neurons more easily than Na⁺ neurons.** Sodium pumps and leakage channels are responsible for this.
 - When the membrane is stimulated, the potential difference changes between inside and outside a cell. **If the potential difference changes beyond a limit, we say that an action potential is generated. Action potential is a spike of positive charge inside the neuron. When the action potential exceeds -55mV, the neuron fires.**
 - **Suddenly, after the neuron fires, the inside potential goes back to -70mV (the resting potential)**
 - The **action potential happens at local level.** This means that there's **no big potential difference** in the full length of axon. **The action potential travels along the axon like wave travels over water.** This travel of action potential is called **propagation.**
 - **Myelin Sheath Membrane has ion channels where Na⁺ and K⁺ ions can travel in and out of neuron.** Process:
 - **When action potential arrives at that location, Na⁺ flows inwards into axon, from outside.**
 - **Shortly after the action potential leaves that particular point, K⁺ channels open and pump out K⁺ from the cell (axon) at that particular location on axon.**
 - **So, wherever action potential reaches the neuron, Na⁺ ions flood into the cell, and when it leaves, K⁺ ions are pumped out immediately.**
 - **Not every signal or stimulus to the neuron leads to an action potential. So, not every stimulus causes the neuron to fire.**
 - Add neuron firing diagram – Communication among neurons.



○ Steps

- Sodium channels open
- At 40mV Sodium channels close.
- Repolarisation
- Refractory period – No action potential possible for neuron in this period.

● Summary of neuron firing

- Neuron gets a stimulus from other neuron
- If stimulus is not strong enough to push potential above -55mV, nothing happens
- If level breached, Na⁺ ions on outside, flood in. In 1ms, K⁺ ions flood out the cell.
- Within 2ms, K⁺ enter back and Na⁺ flood out. Restoring the cell back to -70mV.

○ Major types of Neurons

▪ Sensory Neurones

- It's a Specialised cell that has sensory receptors capable of translating a physical stimulus to an electric signal. Like changing pressure change to electric signal.
- Sensory neurones transmit the electric signal from sensory receptor to Central nervous system (CNS) via Interneurones

- Motor or Efferent neurones send signal from central nervous system to muscles/other parts, thus executing the movement desired by brain.

- **Interneurons** make connections within the CNS, as well as connections to sensory and motor neurones.



○ Neuron firing features

- Either neuron fires or doesn't fire. Like 1 and 0 binary in Computers.
 - Intensity of stimulus leads to 2 possibilities
 - Number of neurons may be larger or smaller
 - **Rate of firing** may be higher with higher intensity
 - Neurons interact via chemicals travelling through synapses.

more frequency

more power

○ Synapse

- The place where signal passes from one nerve cell to another.
- Imagine signal travelling from neuron A to B
- **End of axon of neuron A is called pre-synaptic membrane.** Then there's synaptic gap. **Dendritic surface of neuron B is called post-synaptic membrane.**
- **Synapse working.** Note: Pre-synaptic neuron is sending the signal and post-synaptic neuron receives it.
 - The action potential arrives at presynaptic membrane, at the edge of terminal button of axon. The **action potential causes the synaptic vessels (located on presynaptic neuron) to release chemicals.** These chemicals are called **neurotransmitters**
 - Neurotransmitters flow to the receptor sites of the post-synaptic membrane of the neuron. Note : Post-synaptic membrane is at the edge of the dendrite of receiving neuron.
 - **As neurotransmitters attach to the receptor site, the Na⁺ ion channels may open up and a rush/flood of Na⁺ ions enter the signal receiving neuron. The sudden rise in Na⁺ ions in the neuron is the stimulus for the neuron.**
 - **If the stimulus has enough ions, it can push the action potential above -55mV in the receiving neuron. Thus will create an action potential in the neuron, and it will fire.**
 - The firing of this neuron will pass on signals to other neurons, which may or may not fire. This process keeps going on throughout our life.
- Since neuron is connected to many other neurons, **the sum of Na⁺ ions or stimulus from other neurons determines if the receiving neuron will fire.**

Axon terminal

○ Neurotransmitters. Made in USA ? No, made in the axon. Stored in vesicles at presynaptic membrane.

- Chemical substances that transmit signals from one neuron to the other.
- There are over 100 neurotransmitters – **dopamine, serotonin** etc
- Neurotransmitters **vary in shape (like keys), so they affect a receptor site, ONLY if they fit that shape.** And only if it fits, the receptor will open Na⁺ channels to let the ion in, and only then the neuron will receive stimulus.
- **So, if a neuron doesn't have receptors for serotonin, it won't receive stimulus even if the pre-synaptic neuron sends the serotonin signal.** This way, specific information is released to specific neurons.
- **After giving the signal via neurotransmitter, it has to be terminated.** Otherwise the neurons will keep firing and the person will have a **stroke** and die.
- Effect of neurotransmitter can be **terminated in 3 ways**
 - **Autoreceptors –**

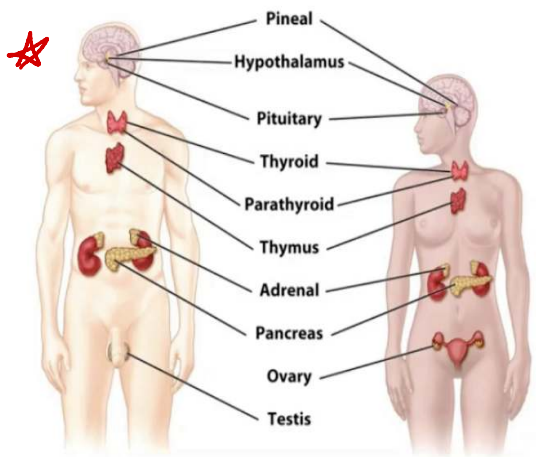
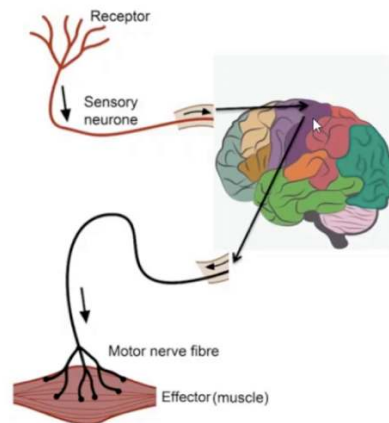
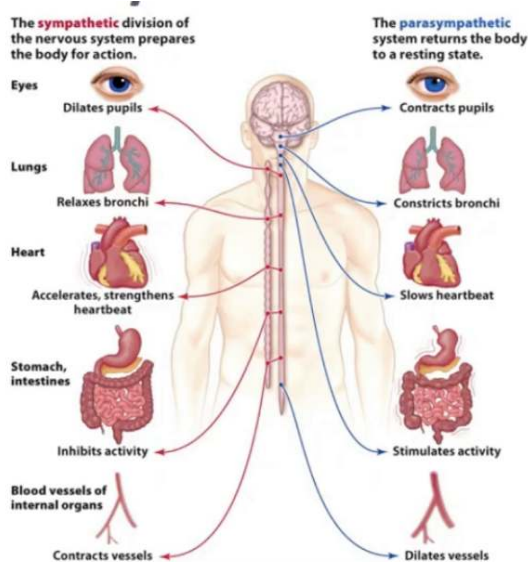
- **Receptors on presynaptic neuron** that monitor the amount of neurotransmitters released. When the released amount is large, these **autoreceptors send signal to the presynaptic neuron to stop sending any more neurotransmitters.**
- **Synaptic reuptake**
 - Since neurotransmitters can be used again, they are taken **back in by the presynaptic neuron.** Done by receptors at presynaptic membrane.
- **Clean-up Enzymes**
 - Clean-up enzymes **capture and deactivate** the neurotransmitters.
- But, neurotransmitters don't always lead to opening positive ion channels of a postsynaptic neuron. That is, **sometimes the same neurotransmitter may lead to inhibition or activation of signals to different neurons,** as they activate different receptors. Similarly **different neurotransmitters may excite or inhibit a neuron.**
 - **In such a case, neurotransmitters open channel for negatively charges ions like Chlorine ions Cl-**
 - This **makes the potential difference negative, inside the postsynaptic neuron, and reduces it's chance of firing.** For eg. A neuron D is connected to neurons A, B and C. Here, A and B send neurotransmitters that can increase the potential difference of D, above -55mV. But C releases neurotransmitters that bring the potential difference of D to -60mV. Then D won't fire
- Add neurotransmitter list image.

NEUROTRANSMITTER	PSYCHOLOGICAL FUNCTIONS
Acetylcholine	Motor control over muscles
Epinephrine	Learning, memory, sleeping, and dreaming
Norepinephrine	Energy
Serotonin	Arousal, vigilance, and attention
Dopamine	Emotional states and impulsiveness
GABA (gamma-aminobutyric acid)	Dreaming
Glutamate	Reward and motivation
Endorphins	Motor control over voluntary movement
	Inhibition of action potentials
	Anxiety reduction
	Enhancement of action potentials
	Learning and memory
	Pain reduction
	Reward

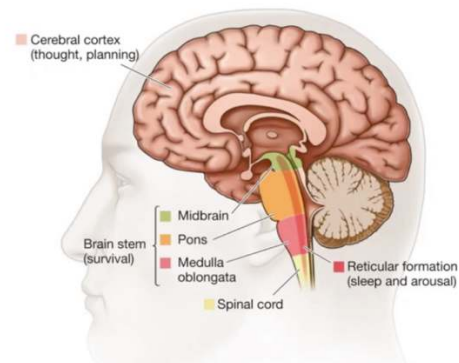
❖ **Drugs. For Brain**

- Drugs work **by inhibiting or enhancing the effect of specific neurotransmitters.**
- **Agonists - ENHANCE !**
 - They work by
 - **Increasing the precursor** i.e. forcing neurons to **release more neurotransmitters.** Done by **amphetamine. Meth ?**
 - **Counteract clean-up enzymes.**

- **Blocking reuptake.** Eg Cocaine and Prozac
- **Mimic neurotransmitter's action.** Eg Nicotine - tobacco
- **Antagonists**
 - Working
 - **Decrease precursor.** Eg AMPT
 - **Enhance clean-up enzymes'** activity.
 - **Enhance reuptake**
 - **Blocks receptors of post-synaptic neuron.** Eg Propranolol
- Parkinson's patients are short on dopamine. So, they're given L'dopa drug.

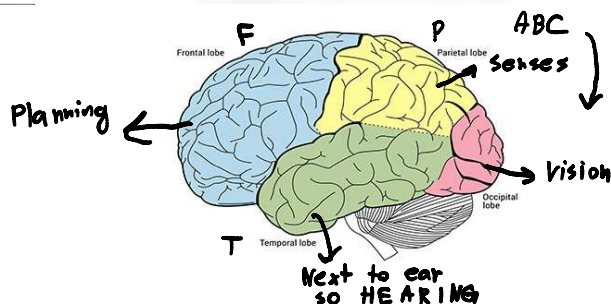


The brain



How Brain talks to body.

- **Central Nervous System**
 - Brain
 - Spinal Cord
- **Peripheral Nervous Sytem**
 - **Somatic Nervous System**
 - Skin, muscles, ears etc send signal to brain
 - Brain sends signals to skin, muscle, ears etc
 - **Autonomic Nervous system** – Autonomous and we don't need to think about it.



All in the pic

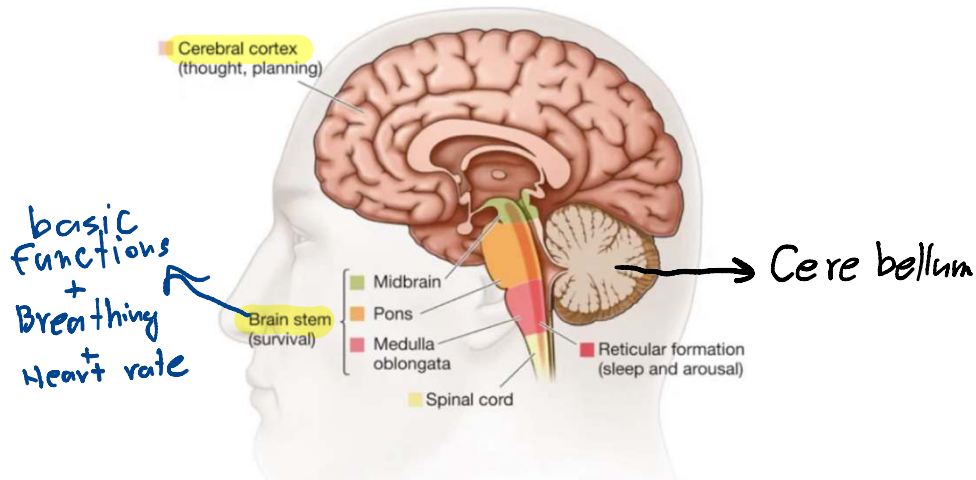
- **Sending signals from Brain to glands and internal organs.**
 - **Sympathetic Nervous System – prepares body for action**
 - Dilate pupil to see clearly
 - Relax bronchi to take in air
 - Increase heartbeat – to prepare for new memes
 - **Parasympathetic Nervous System - prepares body for resting**
 - Constricts pupils when things get boring.
 - Slows heart beat
 - Dilate blood vessels
 - **Receiving signals from glands and internal organs**
 - Brain directs endocrine system – a system of glands
 - Glands get the signal and release hormones like testosterone, oxytocin etc to parts of body. This creates certain events in the body. But this is long and slower process than neural communication
 - **But effects are longer lasting**
- ❖ Back to Brain
- **19th century theory – Franz Gall – If a person uses a trait, that part of brain will create a bump in the skull. False theory**
 - Bumps at back meant domestic and caring.
 - **Methods to study brain**
 - **Clinical neuropsychology**
 - **Compare injured part of brain to find which behaviour was changed.**
 - Patients who get damage to brain are studied
 - **P. Gage** had an accident that put a rod in his head, damaging the brain. This impaired his social skills and temper but he was still alive. This means, injured part of brain was important for social behaviour.
 - **Experiment** **invasive technique on animals**
 - **Studying animals with invasive techniques** on animals. Eg. **Damaging part of brain** of a rat and see the effect.
 - **Hubel and Weisel** (Nobel laureates) experiment 1982
 - Studied cat's visual cortex.
 - Registered activity of cortex by putting electrodes, and studied how neurons responded to the stimulus.
 - Specific neurons responded to specific orientation of images

only one ←
done on animals

- **TMS – Transcranial Magnetic Stimulation.**
 - **Temporary dysfunction created in brain by magnetic impulses.**
 - Done on humans
- **Electrophysiology – EEG and ERP**
 - **EEG caps measure electrical activity at various locations on skull**
 - **ERP is Event Related Potential recording**
 - **Many recordings of same event – like showing dog's image – are recorded and averaged over time.**
 - **Averaged ERP reduces the noise in measurements.**

- **MRI scans – fMRI** is functional MRI. fMRI measures **blood flow and oxygen**. Normal MRI reveals only structure, fMRI also reveals working.
- **PET scan – Positron emission tomography**. Shows activity in the brain after **injecting radioactive glucose** in subject. Indicates active areas.
- **Brain stem is bridge between spinal cord and the brain. Performs basic functions.**

The brain



- Brain stem is essential for survival – it controls breathing and heart rate.
- **Cerebellum** coordinates motor activity – **movement and balance**
- **Cortex is outermost parts – 80% of brain**
 - **Frontal lobe** – thinking, planning
 - **Temporal lobe** – hearing, memory.
 - **Parietal lobe** – touch, spatial relations and attention
 - **Occipital lobe** - vision
- **Sub- cortical regions** – as they are below cortex. Belongs to limbic system e.g. **motivation, memory, emotions, reward**.
- Brain is divided into **left and right hemisphere** – Connected by **Corpus callosum** – a bundle of fibres **carrying info** from one part to the other. Studied via split brain patient – patients whose Corpus callosum was cut, to avoid seizures.
 - **Left and right brains are mostly symmetrical but many differences**
 - **Eg. Language is in left hemisphere of right handed people.**
 - If **Corpus callosum is cut**, we can study which part of brain has which function, by designing experiments.
 - **Eg. Split brain patients** can't read words in left field of view because right brain reads the left view while language processing is mostly in left brain.
- **Further divisions of Cortex**
 - **Primary visual cortex** – visual info enters here
 - **Primary Somatosensory cortex** – info from **skin**
 - **Primary motor cortex** – moves parts of body
 - **Primary auditory cortex** – first audio info comes here.
 - Interesting division of body parts representation in brain.

Psychology Notes – Topic 3

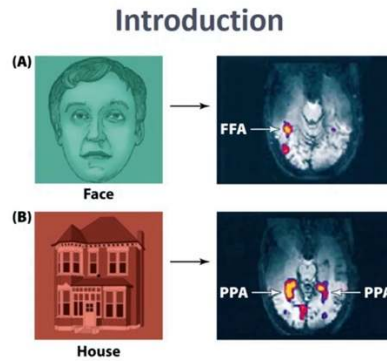
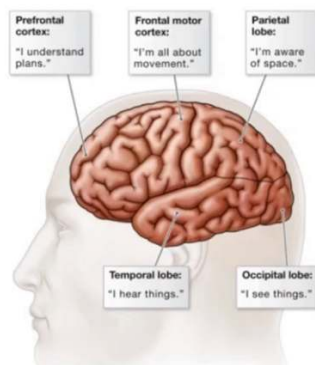
By Somansh – close friend of Al Pacino.

▪ Overview

- What is Consciousness
- Attention
- Sleep
- Drugs

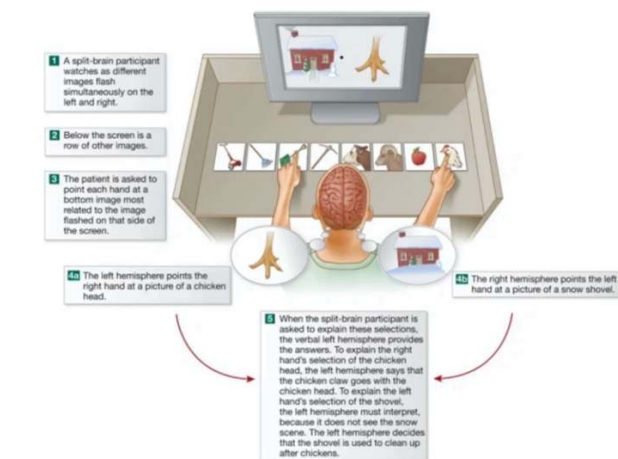
▪ Consciousness

- It is a **state or quality of awareness**. Or, extent of **being aware of an external object or something within oneself**. Or **One's subjective experience of the world, resulting from brain activity**.
- In reality **we aren't fully aware or conscious of everything around us**.
 - E.g. **Bonneh et al 2000** (Nature journal) Paper on **motion induced blindness**. Disappearing yellow dots in **green dot experiment**.
 - **Motion can suppress** the consciousness of **static stimuli**.
 - But can it **happen outside**? Yes. **At night a driver may lose sight of tail-lights of car** in front, due to motion induced blindness of other cars coming from opposite side.
- This shows us **that brain and mind are inseparable**.
- Note: In earlier times, people believed that mind and body are like soul and a vessel. The body (including the brain) has finite capabilities but the power of mind is limitless. Mind can do or imagine anything: go back in time, imagine the future and so on. This was called, **mind body dualism**. Rene Descartes believed brain and mind to be different.
 - But now we know that **mind or consciousness is just a phenomena of the brain**. Since brain is just like any bodily organ, with certain limitations. This means the mind is also limited by the capacity of the brain.
 - Today we know that consciousness varies with brain activity. Neural activity produces consciousness
 - **Everyone has their subjective experience of consciousness**. We cannot know if two people experience the world differently.
 - Doubt : Can it be that: Even if some people have exactly same neural network/ brain, in different situations, different neurons will fire and they may have completely different consciousness?
 - **Presently we can't translate biological activity (neural activity) into conscious experience. Thus, we can't study what a person experiences when certain neurons fire or when a certain brain activity is seen.**
 - We can ask people what they experienced, but again that is subjective. For instance, two people might see a color differently and experience it in a different way.



- **LOCATION** - Ok, so if brain creates consciousness, where exactly is it located in the brain?
 - As per **global workspace model**, it resides **everywhere in the brain**.
 - **Consciousness arises as a function of which brain circuits are active.**
 - **Parietal lobe** activity creates consciousness of **space**. **Occipital lobe** creates consciousness of **vision**.
 - This means, if a brain area is missing, a person won't be aware or conscious of that type of information.
 - In **visual neglect patients**, typically have lesion in right Parietal lobe. This area processes spatial information. Since right brain processes left visual field information, these patients **fail to process spatial info of their left visual field**. That's why they properly paint only right halves of what they see. These patients aren't even aware that they are neglecting left visual field.
- Other supporting **evidences of global workspace model**.
 - **Study by Tong 1998** – Participants shown superimposed **images of house and a face**. Shown to patient via filter such that one eye saw house and other saw face.
 - **Interestingly, participants didn't see both images, but ONLY either one of the images at a time.** They may see house for 10 seconds and 5 seconds a face.
 - Used MRI scanner to monitor brain activity.
 - **When they were aware of the face, brain activity was observed in Fusiform Face Area.** A dedicated area to perceive faces.
 - **When they were aware of the house, brain activity was observed in Parahippocampal Place Area.** A dedicated area to perceive places.
 - This shows, **same stimulus can lead to different consciousness depending on what the patient is aware of.**
- **But ONLY activity doesn't create consciousness. Other processes also happen** in the unconscious part of brain. Sometimes that **process may not even be known to our conscious part of brain**.
 - **Our conscious mind would struggle to explain what happened in the unconscious process.** This is called **after the fact explanation**. For e.g. we don't know how we retrieve info or memory from our brain. Yet our conscious mind would anyway explain the process.

Introduction



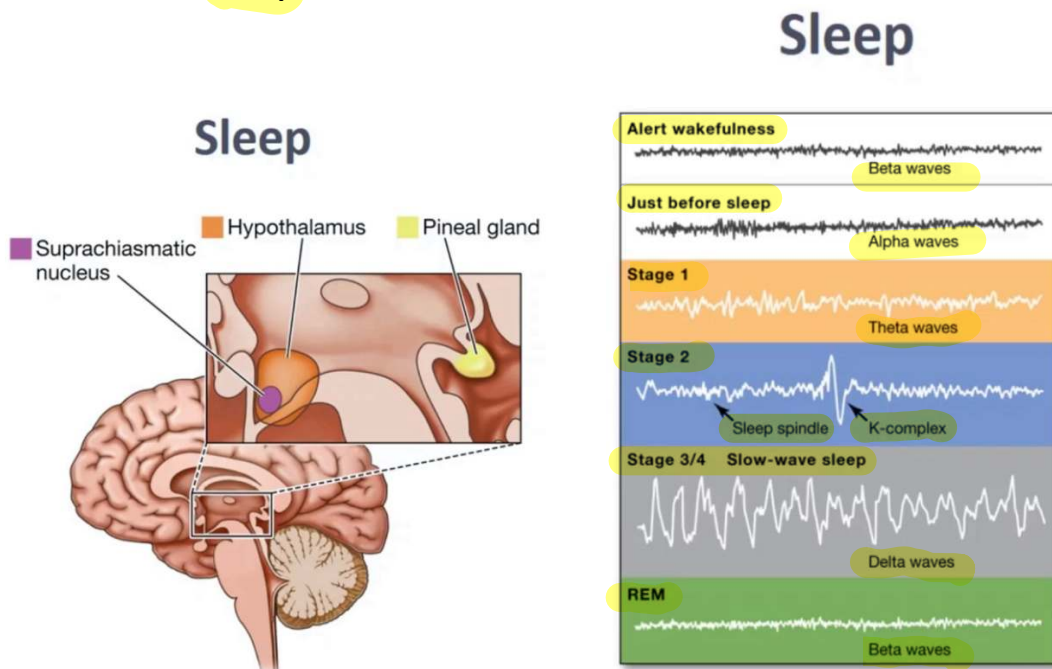
- **Experiment on split brain patient**
 - **Gazzaniga 1978 study.** They saw a house on left visual field and chicken foot on right visual field. While fixated on center of the field
 - Patient picked chicken with the right hand and shovel with left hand. Note: **right hemisphere controls left hand** and vice versa.
 - Then participant was asked about why they chose shovel and chicken. **As left hemisphere has speech capability, and knew about seeing the chicken foot, it clearly explained why chicken was picked. But being a split brain, left brain didn't know what right brain saw.** So, it was an unconscious experience for the left brain. Consequently, left brain explained that shovel was picked to remove chicken shit. An after the fact explanation. Note: **Left brain is called the interpreter of information and creates a story for us.**
- We can conclude that people **don't always know why they behave in such a way, and explain the unconscious action later on.**
 - **1977 paper by Wilson showed how we wrongly explain our mental processes.** Here, all night gowns were of same fabric. **Participants chose the rightmost gown.** But explained that they chose the ones with best fabric feel. Also, insisted that position didn't affect their choice.
 - This **shows we create an illusion of conscious behavior while in reality we acted on some unconscious thought.**
- **Study by Libet showed that voluntary free will behaviour may be an outcome of unconscious decisions.** And explained as a free will decision after the act itself.
 - Experiment asked people to choose freely when to move their hand. While noting the dot's position on the screen.
 - Used EEG to check readiness potential of brain. This precedes selection.

- Outcome: **The awareness of decision to move the hand occurred 204ms prior to the actual movement. BUT the readiness potential was already evident at 535ms before moving the hand.**
- This shows that **free will decision to move the hand wasn't the CAUSE of moving the hand but just an explanation of unconscious decision to move the hand.** Or does it?
- But sometimes unconscious behaviors make people struggle to explain that decision

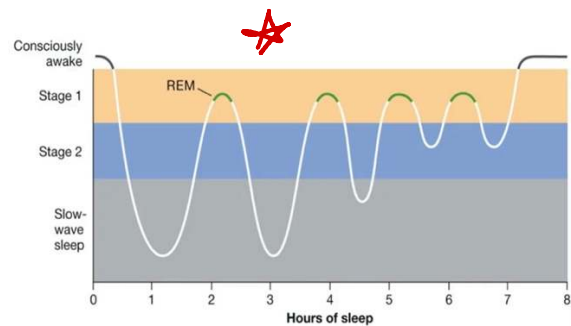
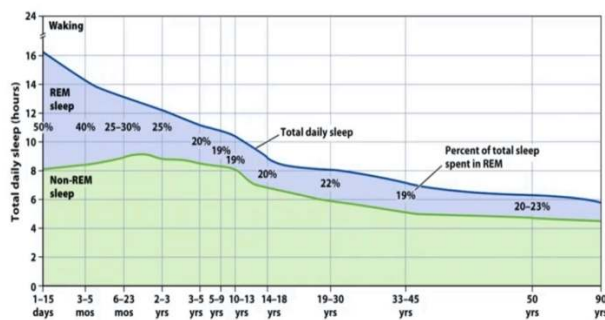
Attention

- We don't get aware of everything we see. **We are conscious of ONLY the things we give attention to.**
- Attention is the process that enables you to focus selectively on some things and avoid focusing on others. So, we have **limited awareness of our surroundings.**
 - We focus on one conversation in the party but ignore the others
 - **Cherry 1953 creates an experiment - shadowing task.**
 - Participants are given **two different streams of audio in both ears. Then, asked them to repeat what they heard.**
 - People are able to shadow messages that they didn't want to hear.
 - Further research showed that **info in unattended ear was processed but ignored.** Change in frequency of unattended messages was immediately detected. So, **physical changes in the signal was detected because they were monitored. But the signal contents were not monitored/decoded.**
 - Sometimes the **unattended audio info breaks through our attention. For instance, when someone calls our name, we immediately detect it.**
 - **Unattended information may sometimes affect our behavior. This is called subliminal perception.** E.g. A US political ad tried to alter people's perception by putting a repulsive message (Rats) flashing in between opponent's video clip. But this doesn't seem to influence long term behaviour.
- Phenomena of **change blindness** – how we selectively give attention to our environment. **We ignore even major changes, while focusing on something.**
 - **Daniel's experiment** showed we can fail to detect change of our conversation partner (if it's a stranger) and when gender or general clothing is similar. So, attend only a small proportion of our visual scope.
 - **We only store specific visual features of a stranger we are talking to: gender, what they want, peculiar features.** We may not always compare features even if we noticed them.
 - **In driving, accidents can happen when we fail to register changes in speed or direction of car ahead of us.**

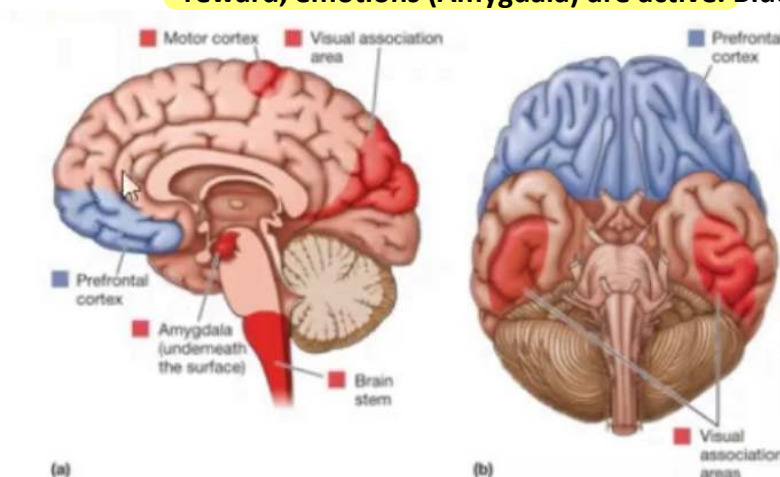
- In creating learning programs for people/children, we should not overwhelm the person with info. As, we have only limited attention span. In training Ai the same applies ?
- In website design, key changes and features should happen such that people can give attention to it.
- **Vision varies with context we are in. So, consciousness varies with context.**
- Affecting consciousness – Sleep
 - Many brain regions are active when we are asleep.
 - Brain never shuts down
 - Sleep and awakening are influenced by circadian rhythm.
 - Changes in light are registered by Suprachiasmatic nucleus of hypothalamus. This nucleus is just above the crossing of two optical nerve nerves. It signals to the pineal gland that it is time to go to sleep.



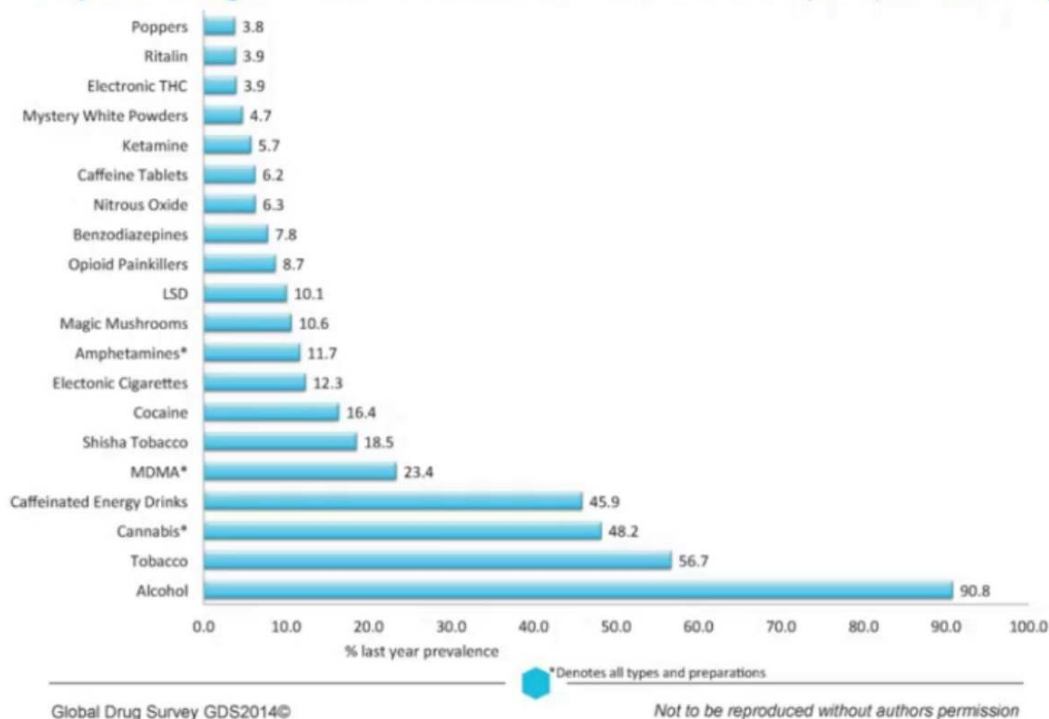
- Pineal gland secretes melatonin. This tells brain and body to go to sleep. Light suppresses production of melatonin and dark suppresses it.
- See sleep cycle EEG above. Alpha waves are same when a person is focused
- K Complex signals are spike in brain activity during sleep.
- **Rapid Eye Movement – REM** sleep has same beta wave pattern as alert wave forms. Dreaming happens but paralysis of motor functions during this stage.



- But WHY do we sleep
 - **Restorative theory**
 - Allows body to repair itself – Evidences: Exercise increases sleep activity; sleep improves immune system; sleep deprivation reduces vigilant tasks and causes problems in cognition, mood and even death. But sleep deprivation may help in reducing depression
 - **Circadian rhythm theory**
 - Animals need only few hours to eat, reproduce and live. In night there is danger. So, sleep helps us survive.
 - **Facilitation of learning theory**
 - Helps neurons connect, thus helps in learning.
 - Participants that slept well, performed better in exams.
 - Participants who dreamt of a concept were better at that concept than others who didn't dream. Also, students have more REM cycles during exams.
- **Dreams**
 - Dreams are a product of an altered state of consciousness in which the images and fantasies are confused with reality.
 - Only after we wake up, we realize we were dreaming.
 - REM dreams are different from Non-REM dreams. NON-REM is NOT COOL dreams. REM dreams are AWESOME, BIZZARE and INTENSE.
 - REM dreams are illogical yet seem logical. Because brain regions of planning and thinking (prefrontal cortex) is deactivated while areas of visuals, reward, emotions (Amygdala) are active. Blue is inactive, red is active.



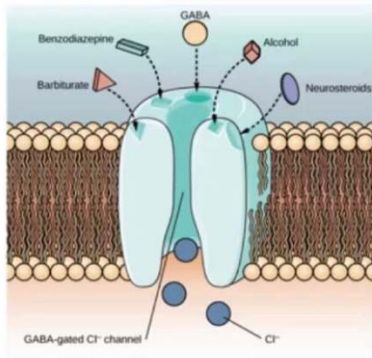
Top 20 Drugs – Last 12 Months – Whole Sample (N=78,819)



Drugs

- **Used to alter state of minds.** They may be used for recreation or treatment. Heroin and **codiene is same**. **Ritalin is ADHD drug, sold on street as smart drug.**
- Can treat a person when used correctly. When abused, can mess them up pretty good.
- **Addiction** – Compulsive drug use despite its negative effects. It creates **dependency** of the drug. A person can't literally live without it sometimes. **Withdrawal symptoms** – **anxiety, tension, craving, physical issues**. Also, drug users develop **tolerance** to a drug.
- Types of Drugs
 - **Stimulants** – I'm the one who ~~knocks~~ stimulates
 - **Increases sympathetic nervous system activity** – **Meth, Cocaine, Nicotine, Caffeine**. **Increases anxiety, insomnia, weight loss.**
 - **Cocaine inhibits reuptake of serotonin, dopamine, and neferine.**
 - **Meth is same as coke but stronger.** It inhibits uptake while also enhances release of neurotransmitters.
 - **This depletes dopamine levels and damages brain,** body and health. **Frontal lobe, temporal lobe, memory, emotion brain parts hurt.** Finally, death.

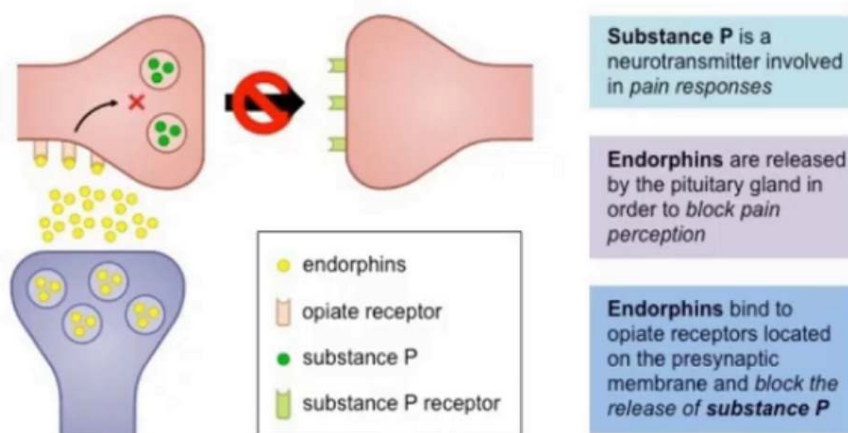
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○ Depressants

- **Alcohol, anti-anxiety drugs**
- **Reduces mental activity.**
- **Alcohol is a GABA agonist**
 - Enhances the suppressant effects of GABA receptors. Alcohol attaches itself to receptors and alters its functions. This allows GABA molecules fit easily in this receptor. This reduces neuron activity.
 - Alcohol withdrawal can cause: anxiety, seizure, stress.

○ Opiates. Narcotics *Para Symp. Ns*



- **Slow down Central nervous system, create pleasure, slow down pain. Effects the parasympathetic nervous system.**
 - **Heroin, Morphine and Codeine.**
 - They bind to natural opioid receptors in brain.
 - Opioid receptors are used by endorphins to **inhibit the sensation of pain**. Endorphins work by blocking the release of substance P. Opioids just enhance this pain blocking mechanism.
- ### ○ Hallucinogens *Perception*
- **Produce alterations in cognition, mood and perception.**
 - **LSD, Psilocybin mushrooms, Mescaline.**
 - Don't clearly affect the autonomic nervous system. But distort perception.
 - Prolong use can create recurrent hallucinogen perception – recurrent trips.

- Withdrawal symptoms are not as clear.
- Many drugs like Marijuana, MDMA don't fit into these categories.
 - Marijuana can show effects of stimulant, depressant and hallucinogen.
 - MDMA affects serotonin more, by blocking the reuptake. MDMA increases oxytocin or cuddle hormone.
 - Severe withdrawal symptoms of MDMA – fatigues, depression, damage to hippocampus and prefrontal cortex.
- In the long run drugs cause damage.

Psychology Notes – Topic 4

By Somansh – won the Nobel Meme Prize in 2013

Humphreys case study –

- We apply structure and classification on things. But John can't structure his perception. What the eyes see is ok, but the head can't focus on what he sees. Can't put together faces. Can recognize objects but has difficulty putting information together. Can see the details but fails to connect those details to their concepts.
- Can't give meaning to the patterns that he sees.

Lecture begins

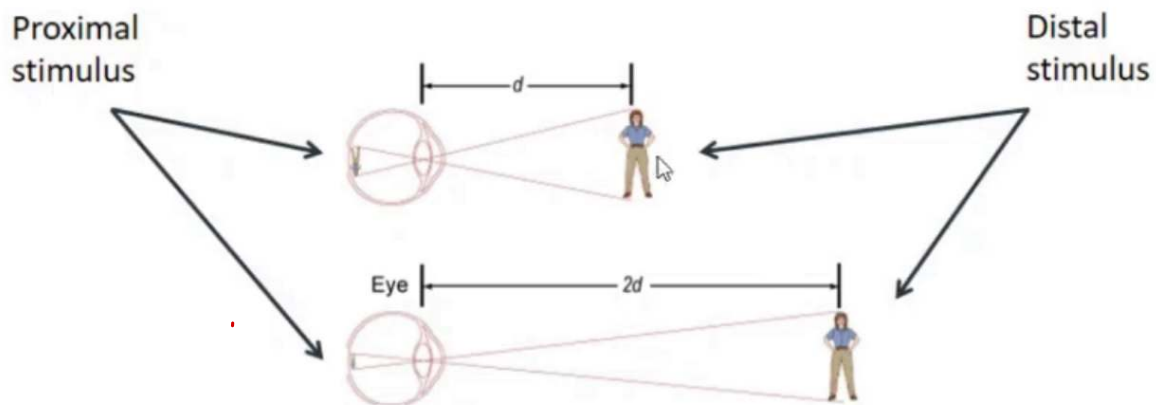
Sensation and Perception: Vision

Overview

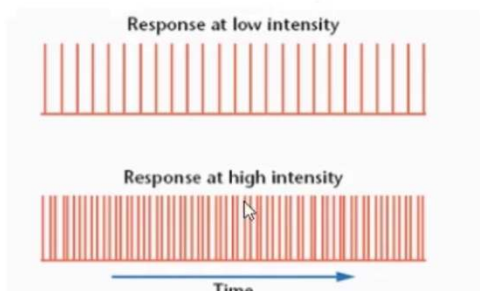
- ◆ The senses
- ◆ Vision – beyond the retina
- ◆ Vision – the retina
- ◆ Visual perception
- ◆ Perception organization
- ◆ Perception of depth
- ◆ Perception of size
- ◆ Perceptual constancies

The Senses

- **Are our perceptions a true reflection of reality or do they only represent reality.** Locke wrongly believed, our senses register reality truly, so they show our surroundings as they are. Eyes as camera.
- **Empiricists** believed a baby was a blank slate with nothing but ability to learn. Babies learn everything as they **experience** them.
- **Nativists like Immanuel Kant thought that important aspects of cognition were available at birth.** Certain hardwired presumptions about the world exist at birth. Perception can never be pure. **SO, perception is based on these innate (by birth) assumptions. We are born with skills to interpret the world.**

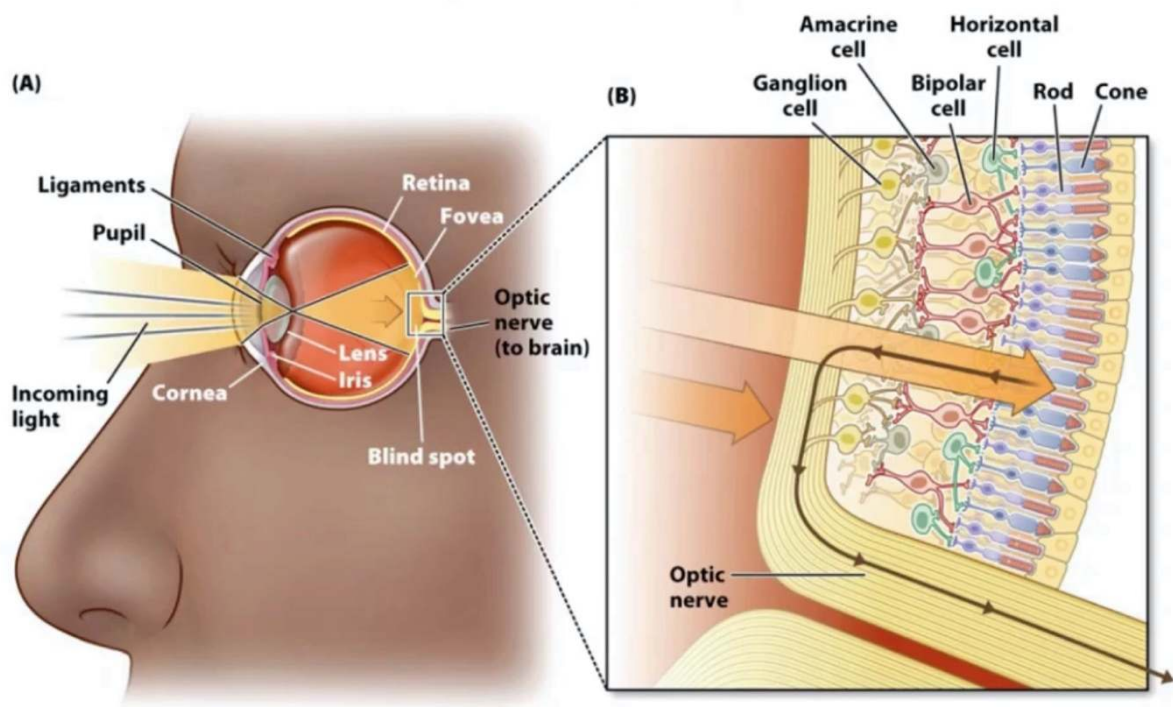
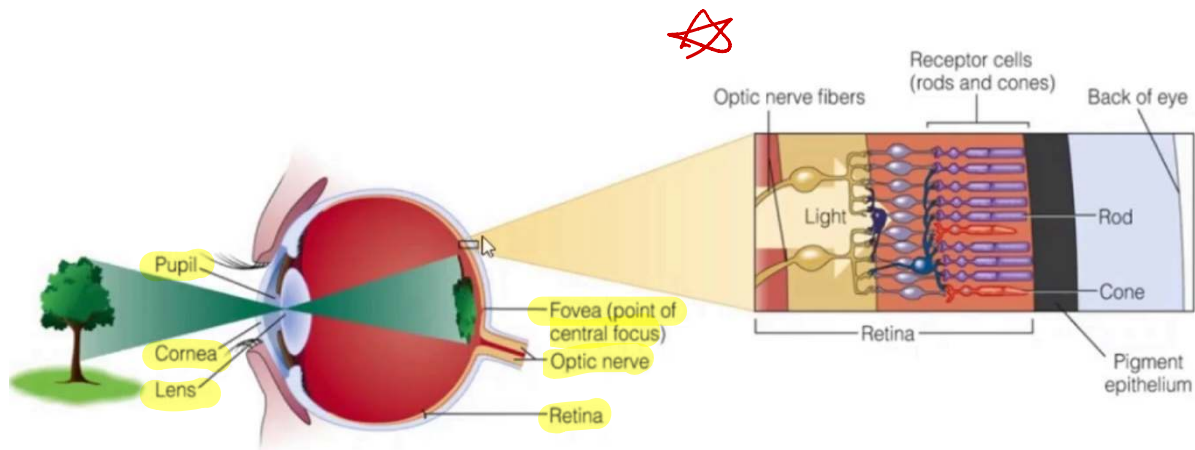


- **Proximal stimulus** (stimulus on our eyes or pattern of energy on our sense, in this case our eyes) and **Distal stimulus** (outside stimulus) – Empiricists believed that with experience we learn that smaller proximal stimulus means that distance is larger. But Nativists believed that this ability to perceive distant objects was pre-built at birth.
- **Sensation** – the detection of external stimuli and the transmission of this information to the brain.
- **Perception** – the processing, organization, and interpretation of sensory signals.
- Distal stimulus -> Proximal stimulus -> Transduction -> Sensation -> Perception
- **Types of senses:** Vision, Hearing, Taste, Smell, Skin senses (pressure, temperature, pain), **Vestibular sense** (sense of balance in the ear), **Kinesthesia** (conceives info about our body's orientation in space). Receptors are located in muscles, tendons and joints).
- **Quantitative variation/ Intensity of stimulus perceived by our senses.** Occurs by 2 means.
 - **Rate of firing in neurons** may increase with increased intensity
 - **Total number of neurons firing** – intense stimulus causes more neurons to fire.



- **Qualitative variations** – or **sensory quality** – like colors, texture of surface, sound differences. Two mechanisms: *by diff parts of the brain*
 - **Different sensory info signaled by different neurons.** Sound triggers auditory cortex and colors trigger visual cortex.
 - **Certain sensory qualities activate same set of neurons differently.** Color yellow results in an activity pattern in a set of neurons, while color red activates the same set of neurons, but activity pattern of those neurons is different.
- **Adaptation** – sensation is biased towards new events and changes. It is the tendency to **respond less to a stimulus** that has been present and unchanging for some time.
 - After some time, neurons stop firing to a unchanging/static stimulus. But not all stimuli disappear because we keep making eye movements, that keeps changing the stimuli.
 - Looking at clouds is just like looking at static stimuli.

Vision



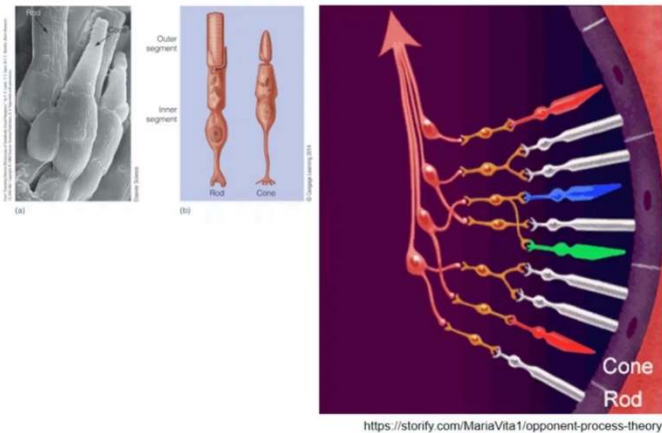
- The stimulus for vision is light. Light varies in brightness (intensity), color (quality/wavelengths).
 - What we perceive as colors is 400 to 700 nm wavelength of electromagnetic radiation. So, we only ever see a partial part of the world. We never get to see UV rays and radio waves.
- **Retina** – the light comes into the eye and creates proximal stimulus at the retina. Retina consists of light sensitive receptors called rods and cones.

Cones: small & conical
Rods: large & cylindrical



- Rods and cones are behind a thin layer of cells on the retina. Light reaches rods and cones.
- Rods and cones contain pigments that absorb the light, here the signal is converted into electrical signal – this process is called transduction.
- The electric signal is sent from rods and cones to Bipolar cells, to Ganglion cells to the optic nerve. This nerve reaches the brain.
- The Amacrine and horizontal cells modulate and regulate the electric signal from receptors to Ganglion cells.
- Rods are large and cylindrical. Cones are small and conical.

- **Rods have rhodopsin pigment (opsin + retinal)**
- **Cones have conopsin (photopsin + retinal)**
- When **light is absorbed** by these pigments, they **change shape**, this change of shape **creates electric signal**

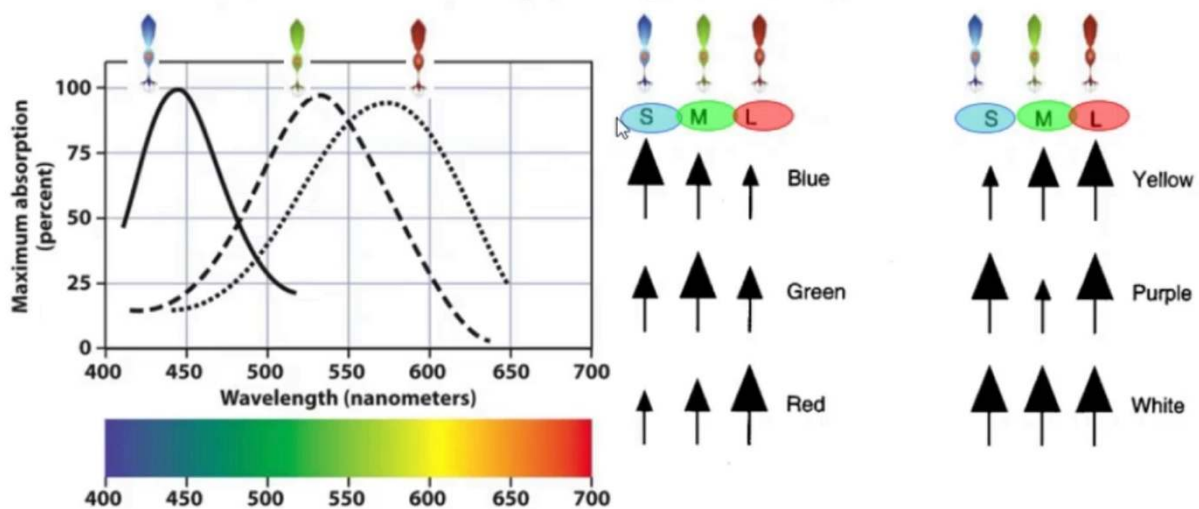


○ More on rods and cones

- **Rods are relied on during night** because they are **more sensitive to light**. Cease to fire during day as the light intensity is high.
 - They **have no color vision**
- **Cones are relied on during day**. More sensitive to color.
 - **Less light sensitive**
 - **Has color vision.**

• How we see color ?

○ Trichromatic Theory – by Young-Helmholz



- **Conopsin have 3 different pigments** or molecules with **different amino acids**.
- Three pigments are able to **transduce Red Blue Green color wavelengths**.
- **Young-Helmholz postulated the trichromatic theory** – since we can't have a receptor for every color, we have limited (three) receptors for RGB colors. *about 3 diff pigments* A mix of signals from these 3 pigments **creates different stimulus for hundreds of color we see**.
- **Color vision** arises due to pattern of activity across **all 3 cones**.

- after image
- opposing colors

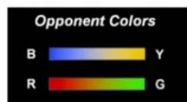


- So, **cones vary with respect to their sensitivity to light of different wavelengths. Blue cone can absorb more blue light than green and red cones.** However, we see blue because **blue cones fire more strongly than other two.**
- Note: **It is never the case that only one cone is activated by a certain wavelength. All three are active BUT at different intensity.**
- So, combined activity of these cones creates perception of color.
- **Color blind people lack pigments in either of these cones.**

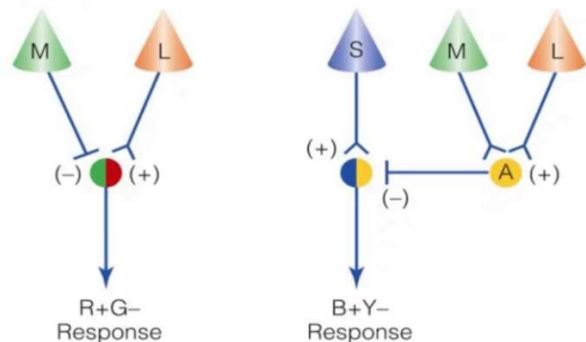
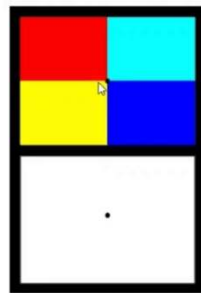
○ **Opponent-process theory on color vision by Hering 1892**

• Opponent-process theory (Hering)

- There is no such thing as "blue-yellow" or "red-green"

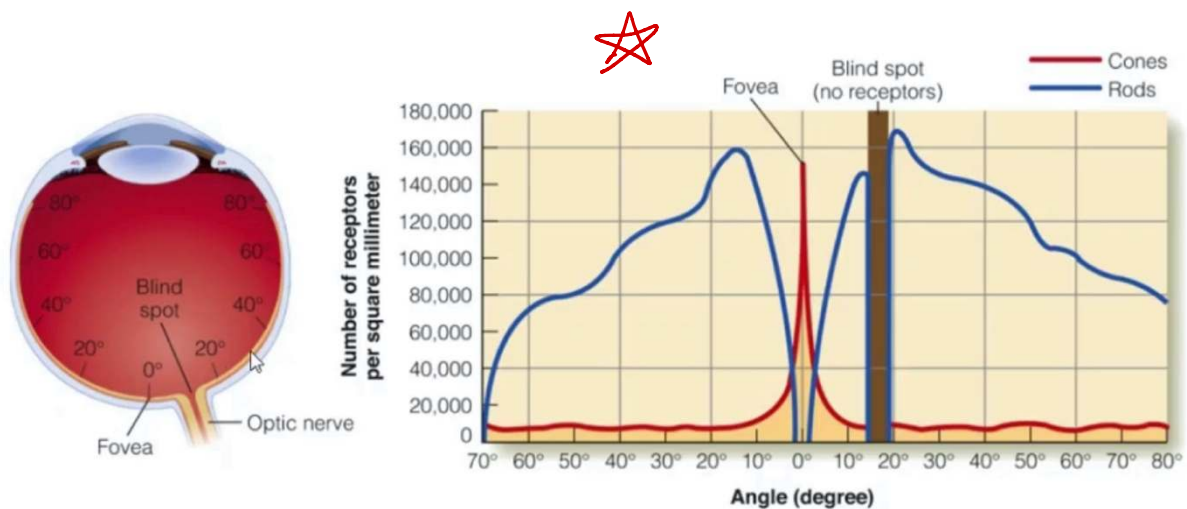


- Afterimages



- Some aspects of color vision can't be explained by trichromatic theory
- Certain colors seem to be complimentary. Some color combinations are just not possible like Bluish yellow or reddish green.
- He proposed that there are **3 opposing processes**
 - **Black(-) - white (+)**
 - **Red (+) - Green(-)**
 - **Blue (-)- Yellow(+)**
- When we **fixate on a color**, and after a while, look away. We will see **the complement of that color in after images**
- He argued, prolonged exposure to a process, the opposite of that process will begin. So, after viewing Red for a while, the Red+ process will be exhausted. In the after image, the Green (-) process will begin. So, we will see green - as its after image.

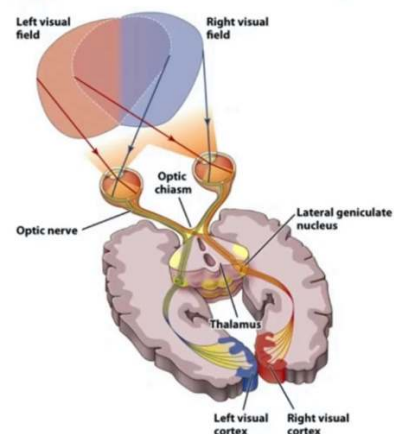
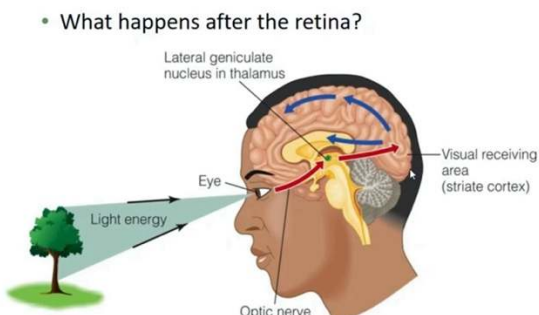
○ **Today we know both theories are correct.**



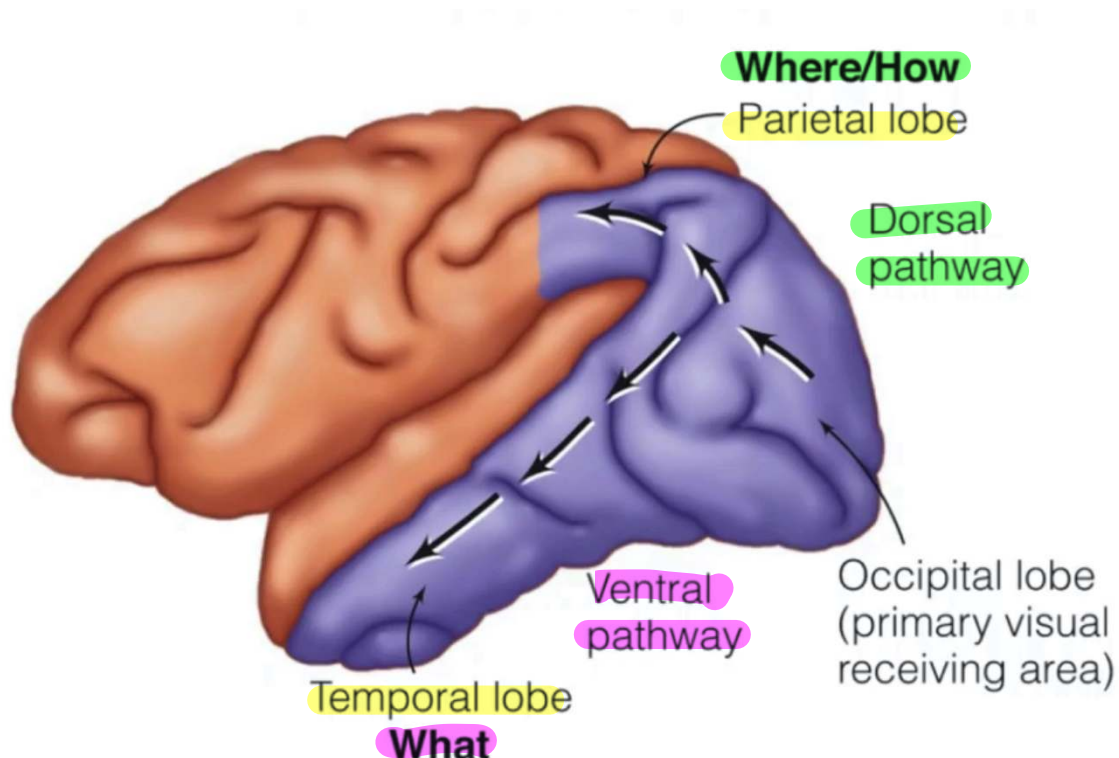
- Back to cones and retina
 - **Distribution of cones and retina in the figure above.**
 - **Fovea has cones ONLY. It is the point of fixation.**
 - There is **high resolution at this point**. Why? Because the cones (primarily responsible for high resolution perception) are concentrated in fovea region, while **less cones at periphery**.
 - So, **things must be bigger in our peripheral vision to be seen clearly.**
 - **Peripheral retina has more Rods there than Cones.**
 - **Blind spot has no cones and no rods.**
 - We don't feel the blind spot because
 - Info is filled by **other eye**
 - Info filled by **the brain**
 - We don't feel any absence of info because there is **no receptor there** to tell us something is missing

Beyond the Retina

- What happens after the retina?



- From **retina**, info goes to **Lateral geniculate nucleus in Thalamus**, then towards visual cortex. Then it **splits to main pathways: parietal and temporal**



- Info of **left visual field** ends up in **right visual cortex** and vice versa. Note the optic chiasm is located near thalamus.
- **What pathway - Ventral pathway to the temporal lobe. WHAT part.**
- **When pathway - Dorsal pathway to the parietal lobe. WHERE/HOW part.**
 - How we know? One study in **1982 by Ungerleider and Mishkin?**
 - He trained **monkeys to identify object shape. Failed this task when temporal lobe was removed.** But did well in Landmark identification test
 - Also trained them to identify landmarks/ **spatial relations. Failed it when parietal lobe was removed.** But did well in object recognition.
 - **Goodale 1995 study - DF patient couldn't copy the apple and book but could draw them both from memory.** Had no problem with WHERE and HOW part of tasks. But had trouble identifying the WHAT or object recognition.
- Within both these pathways there is strong modularity. Which means there are different functions within these different areas. E.g. PPA and FPA areas. Also there are different cross linkages with different areas. All the areas communicate with each other but we don't know how.

Visual Perception

- **How the information we receive is modulated, organized and interpreted. Beyond sensing the stimulus. We combine our previous knowledge and extract information from visual signals.**
- Therefore, **visual perception is not just bottom up process (from senses to our mind/consciousness) but also top down process (from previous knowledge to mind).**
- **Knowledge affects what we "see" through our senses.**

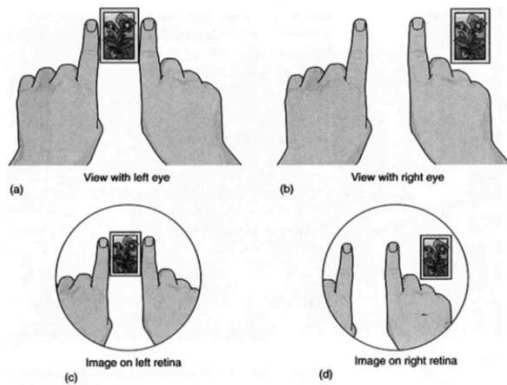
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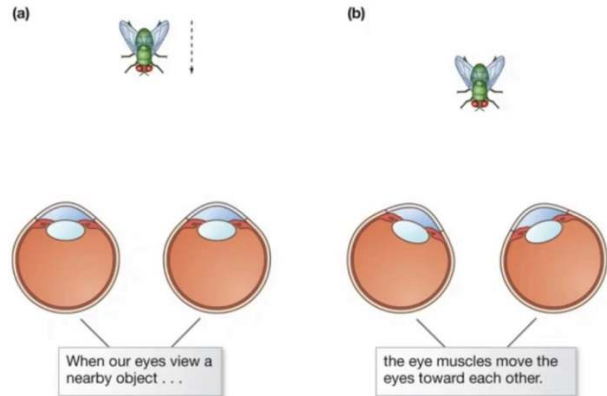
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- **Closure** – we perceive objects as closed and connected even if some part of its information is missing. Triangle image.
- **Perception of depth** – How we make 3D perception from a 2D image we receive. **Different cues used based on context/situation.**

• **Binocular disparity**

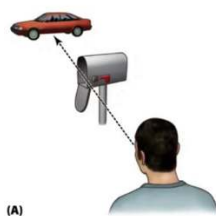


• **Convergence**

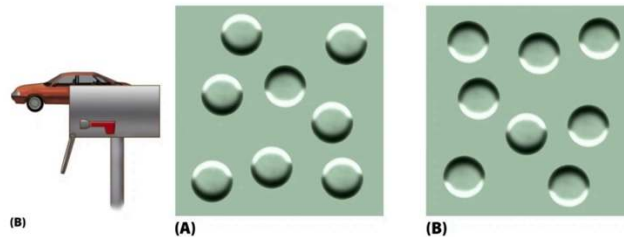


- **Binocular depth cues** – that arise due to us having 2 eyes
 - **Binocular disparity** – the slight difference in images that fall on to the retina of each eye. Brain cells are also dedicated to understand this binocular disparity. Used upto 10 m.
 - **Convergence** - information our brain receives from eye muscles. When we look at something closer, muscles move inward and brain perceives the depth of that object.

• **Occlusion**



• **Shadow**



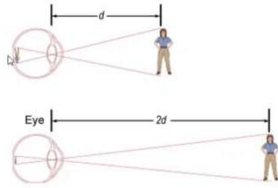
▪ **Monocular cues**

- **Occlusion** – we know that nearby objects obstruct/occlude far away objects – thus we deduce that occluded object is farther.
- **Linear perspective**. Seemingly parallel lines like railway tracks seem to converge.
- **Texture gradients**. Texture of different coloured objects like a field of flowers becomes denser with viewing distance. Brain can estimate that denser texture must be farther away.
- **Shadow**. Since in real life light comes from above, we perceive objects with shadow on top as depressed surface and objects with light on above as emergent surfaces.
- **Motion parallax** – objects moving at constant speed appear to move faster when they are closer, while farther away objects move slower. As we see during a train ride.

- **Perception of size**

- **Depth perception is used in estimating size, because size perception depends on perceived distance of an object.**
- **Same length of line seems bigger if it seems father away and seems smaller in it seems closer to us.** Because they create equal size of image on the retina, so depth cues are necessary to determine size.
 - Similarly lack of depth view can fool us about size. E.g. **Ames Room.**

• Size constancy



- **Perceptual Constancies –**

- **Size constancy** – we correctively perceive different properties of distal stimuli despite continuously changing proximal stimuli.
 - People walking towards us keep increasing in size of image on the retina but we don't perceive them as getting bigger because our we combine it with depth perception of that person to adjust our perception.
 - This proves that **we keep figuring out what we see.**
- **Change constancy** – we constantly perceive the same thing, despite changes in our viewing angles or object's orientation in space. **We keep interpreting the perception based on the context.**
- **Lightness constancy** – despite changes in illumination, we perceive the objects as same. We adjust our perception based on our knowledge that objects in light appear brighter than they are and darker than they are in the dark, and objects don't change their own brightness.
- **Color constancy** – we perceive colors as not changing despite the proximal stimulus might change. We simply **adjust our perception based on surroundings.**

Psychology Notes – Topic 5

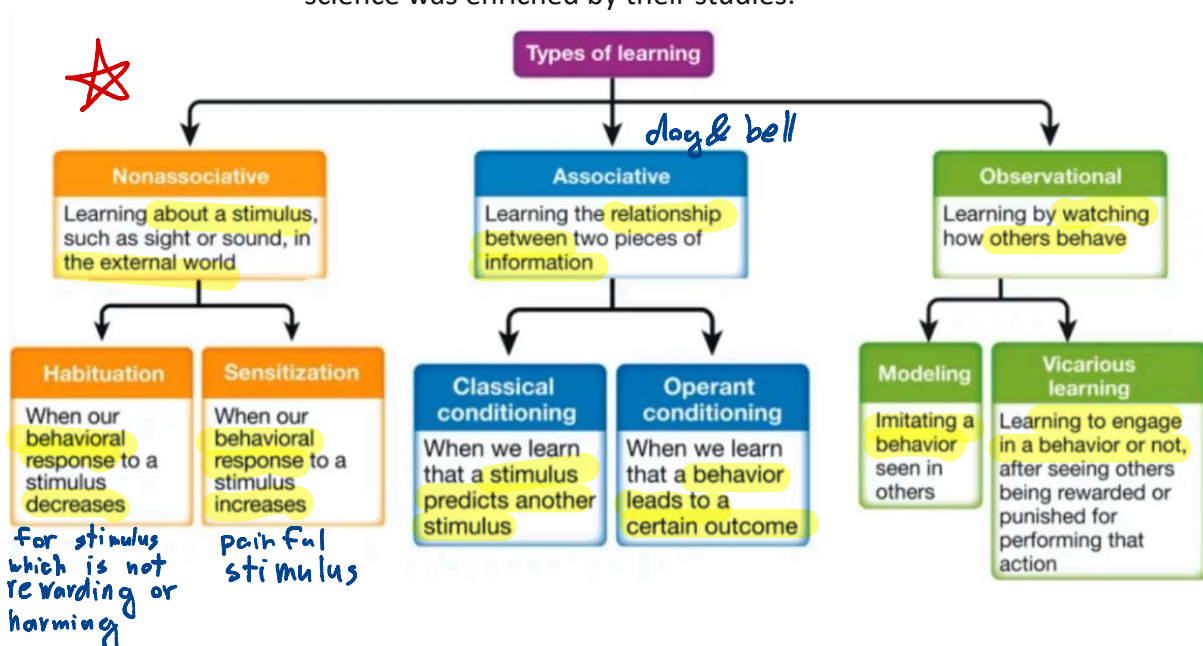
By Somansh – that's what she said

Overview – focus on acquired behavior this time.

- **Non-associative learning**
 - **Associative learning: classical conditioning**
 - Associative learning: **operant/instrumental conditioning**
- **Observational learning** – changing behavior after exposure to another person doing something.

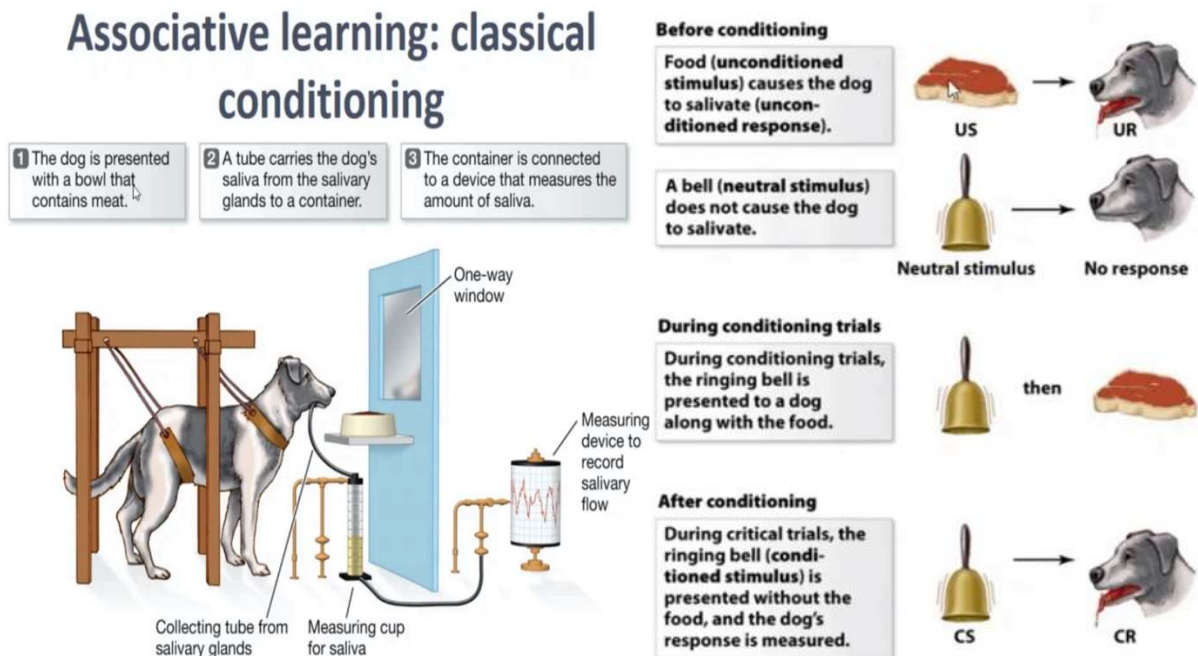
Learning

- Learning implies a **change in behavior resulting from experience**. Learning theories began in 20th century. Freud's different approach to human behavior ignited many theories.
 - **Earlier theories arose from the view that all behaviors are either reflexes or are the consequence of learning history or experience. This view is called behaviorism.**
 - It was dominant in 20th century and influenced every area in psychology. Focus was too much on environment, stimulus and reaction. Didn't focus on the mind that much. Mind was treated as black box, so its input, output was studied, not the mind itself.
 - **Used animals rather than humans in experiments. John Watson was the founder of this thought – the school of behaviorism. Watson once claimed**
 - "Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select—doctor, lawyer, artist, merchant-chief and, yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors. (1930)"
 - **But today we know they were wrong in many areas.** However, science was enriched by their studies.



Non-associative learning – responding after repeated exposure to a single stimulus.

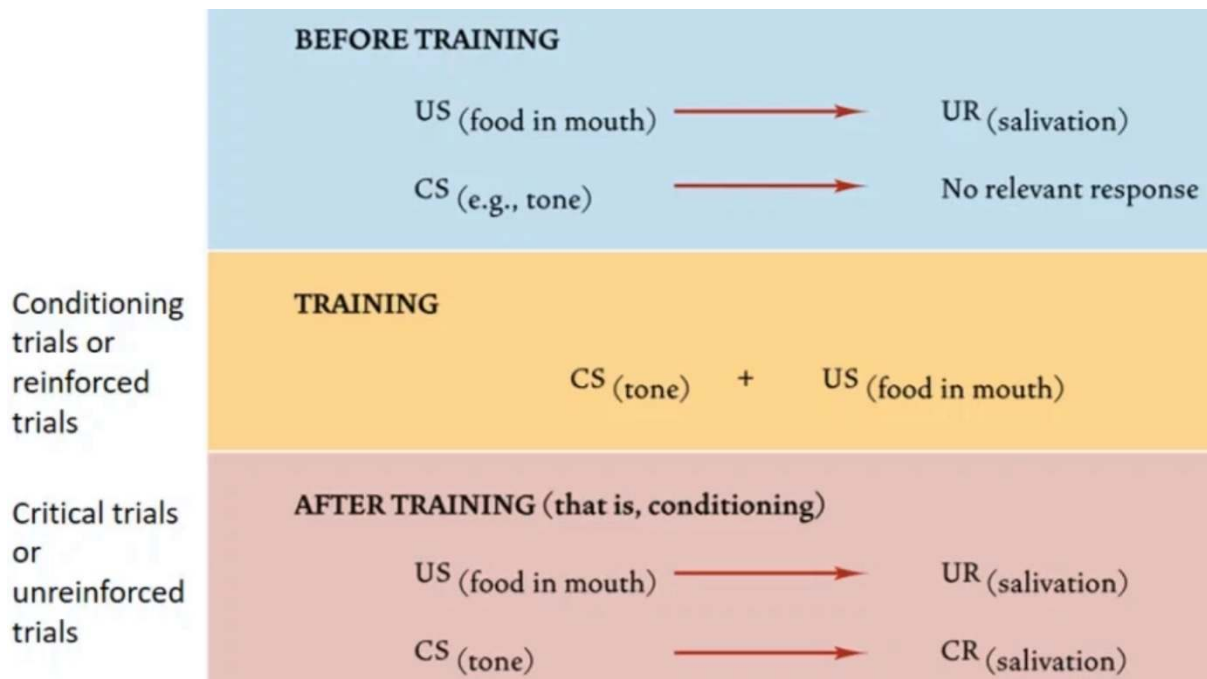
- **Habituation** – a **decrease in behavioral response to a stimulus**. A simple way of learning. Depends on memory, otherwise we won't know if something is repeating. It is **different from sensory adaptation** where a recurring stimulus is ignored by our perception. In habituation we are aware of the stimulus. **Usually occurs for stimulus that is neither rewarding nor harmful**. For example, getting used to **sound of planes**.
 - It allows us to focus on what's new and **ignore the familiar**.
 - **Dishabituation** – it's an increase in response to a change in something familiar, that was habituated previously. Might notice an airline sound if it sounds weird.
- **Sensitization** – an **increase in behavioral response after exposure to a stimulus**. Usually happens for **painful stimulus**. May become sensitive to pain in stomach after a stomach disease.



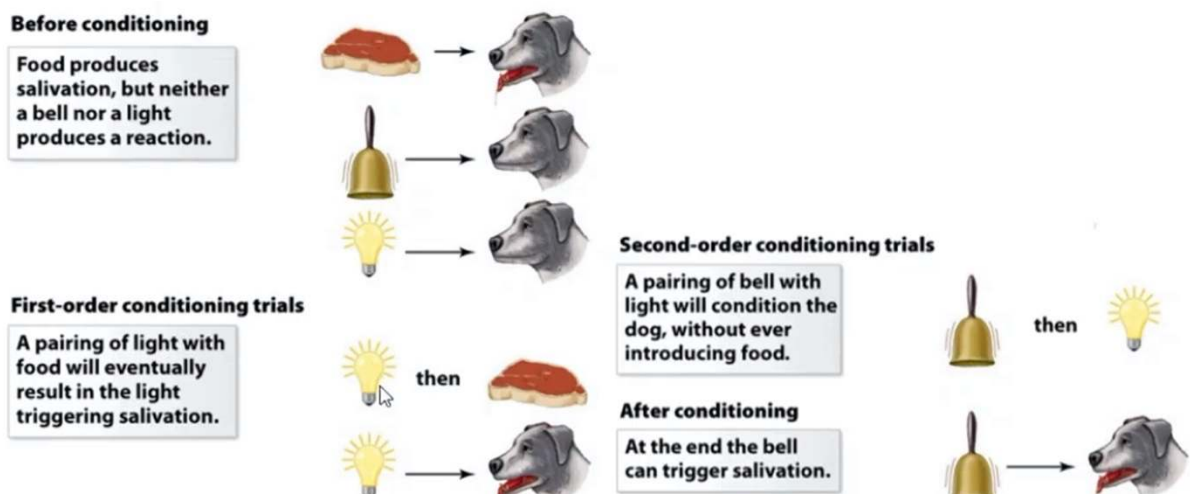
Associative learning: Classical conditioning

- A form of learning in which a **neutral stimulus comes to elicit a response when it is associated with a stimulus that already produces that response**.
- **Pavlov** was the grandfather of classical conditioning. Won Nobel prize in 1904 for his work on digestive processes. Dogs get reflexive saliva in the mouth when they have food in mouth. But later found that **sight of food bowl and sound of footsteps of caregivers led to saliva in their mouth**.
 - He studied these learning processes more deeply.
 - **See dog diagram above**
 - **Unconditional response [UR]** is the natural response, without having to learn it. **Unconditional stimulus [US]** triggers the **unconditional response [UR]**.
 - **Conditional stimulus [CS]** doesn't trigger the UR initially.
 - **After conditioning, the CS leads to a response similar to Unconditional response – called Conditional Response [CR]**. CR happens when the

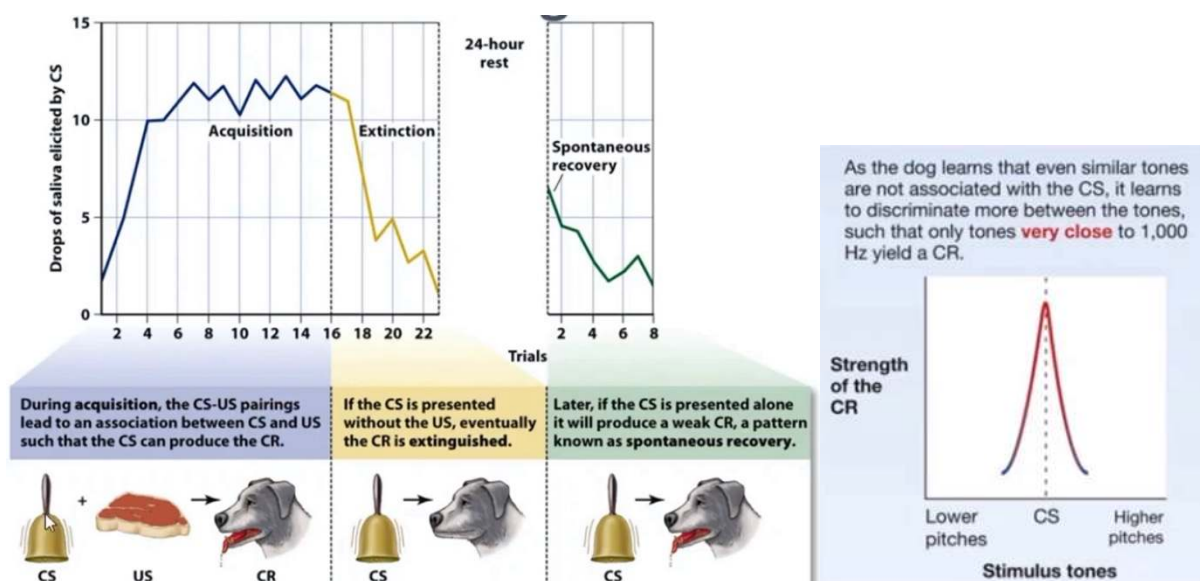
organism has learned of an association **between Conditional Stimulus and Unconditional Stimulus** like the Bell [CS] and the Food [US].



- **Classical conditioning not just in dogs**
 - **Observed in many animals** – cockroaches, dolphins, humans.
 - **Can happen in specific situations.** Humans can learn to associate certain smells, events and sounds with certain emotions.
 - Having repeated experiences of painful injections after hearing ‘This is not going to hurt’, a person will learn to associate this phrase with pain and fear.
 - Here the phrase, ‘This is not going to hurt’, becomes a CS that creates fear
- **Features of Classical conditioning**
 - **Second order conditioning** – when a CS is paired with a new stimulus S, that stimulus S produces CR.



- If being bitten by a dog is combined with the view of a dog, the fear [UR] will be associated with view of the dog [CS]. Later, just the view of a dog [CS] create a fear [CR].
- If a neighbor's dog is barking whenever it is seen, the person will associate barking with viewing the dog. Consequently, only by hearing a bark, the person will feel fear [CR].
- **Extinction** – where the association between the CS and CR can be eliminated by repeatedly presenting the CS in isolation.
- **Spontaneous recovery** – recovery of response after extinction and a resting interval. In dog's case, it will generate saliva after hearing the bell, but in very less quantity.



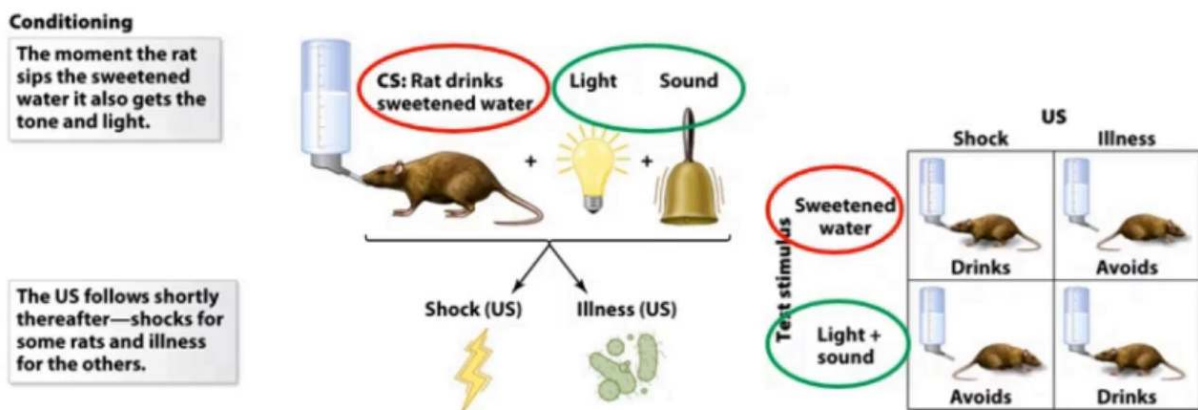
- **Stimulus generalization** – the CR can also be elicited by stimuli that are similar to the CS. For e.g. saliva production may be conditioned to 1000Hz sound but playing similar frequency sounds like 800 or 1100 Hz can also generate saliva but in lesser quantity.
- **Discrimination** – it is possible to teach animals to discriminate between different but similar stimuli.
 - If 1000 Hz frequency tone always result in food, and 800 Hz never gives food. There will be a strong response only in a narrow range around 1000 Hz or CS frequency. This reduces the stimulus generalization effect by narrowing down the range of CS.

Video clip on Watson – a child conditioned to a loud hammer sound [US] that creates anxiety [UR] whenever Watson touched a white rat [CS]. He started to fear [CR] after seeing rats [CS] and similar stimuli like rabbits and dogs even

when there was no hammer sound. Even repulsed by fur coat and furry objects. Today such studies are unethical.

Early behaviorists believed that humans and animals can learn any association between any US and CS.

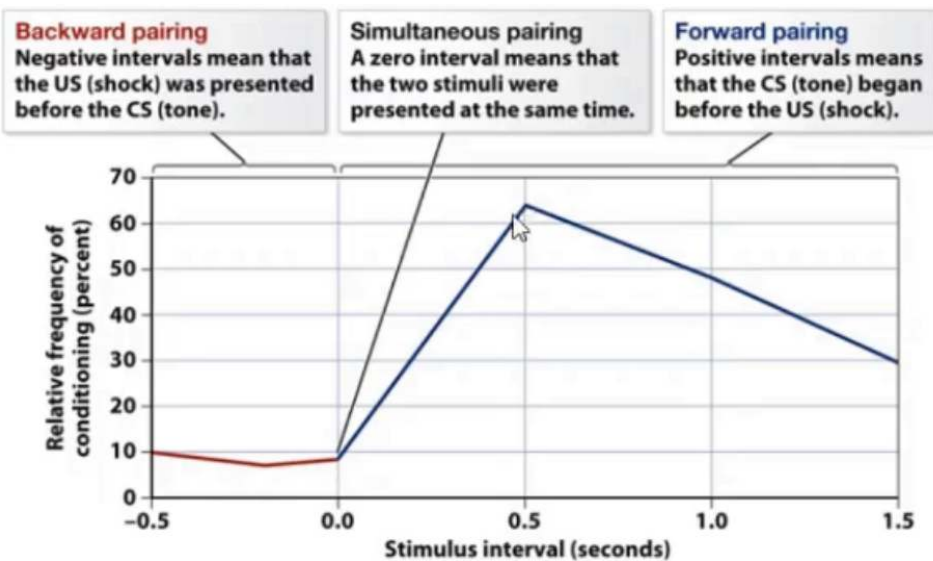
- But there are **biological constraints** to such learning. It was shown by a study in **1966 by Garcia**.
 - **Rats were given sweet water** to drink but whenever they drink, light and sound were turned on. There was either a shock to some rats or illness given [US] given to other rats shortly after.



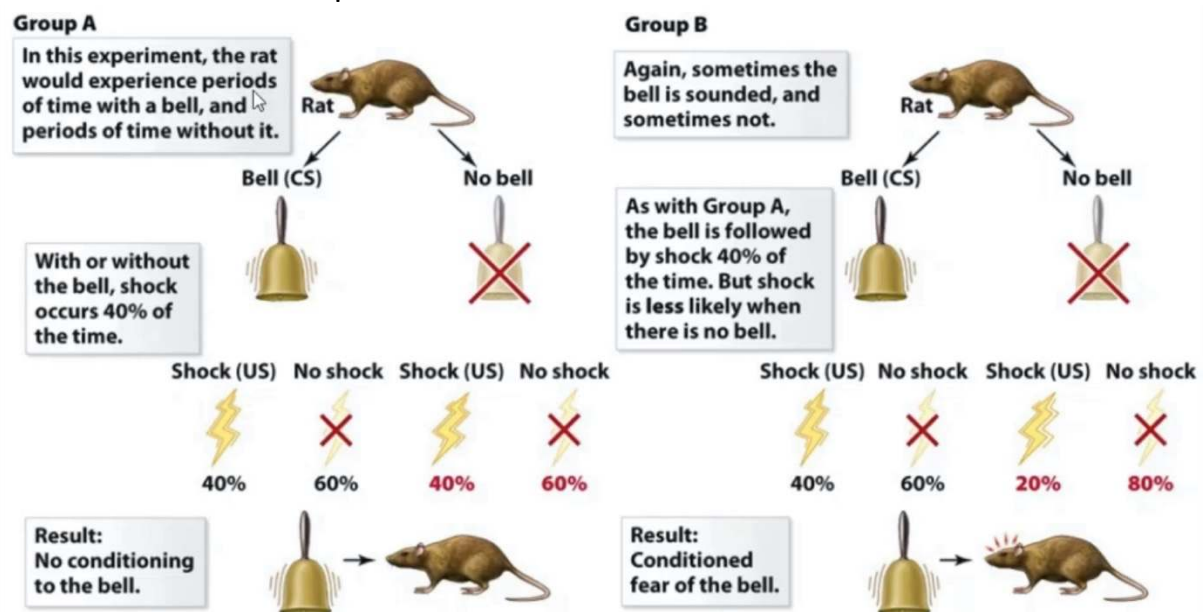
- **Associations made by rats**
 - **Rats that got shocked** – drank sweet water **but didn't drink** as soon as lights and sound were turned on. So, pain was associated with light and sound, not the smell of sweet water.
 - **Rats that got illness** – **avoided the sweet water based on its smell** but drank it when lights and sound were turned on. Doubts in this part. This means that rats **associated illness with sense of smell and taste rather than light**.
- This means that taste and smell goes with feeling and sickness while sight and sound go with pain.
- Later it was shown that **monkeys can be easily conditioned to fear snakes but very difficult to condition them to fear flowers or baby monkeys**.

Criticisms and features of Classical conditioning

- There needs to be a **contiguity** in presenting Conditional Stimulus. Which means, **CS must be presented before the US**. If CS is after or simultaneous with US, no learning can occur. So, **CS serves as some kind of warning for US**.

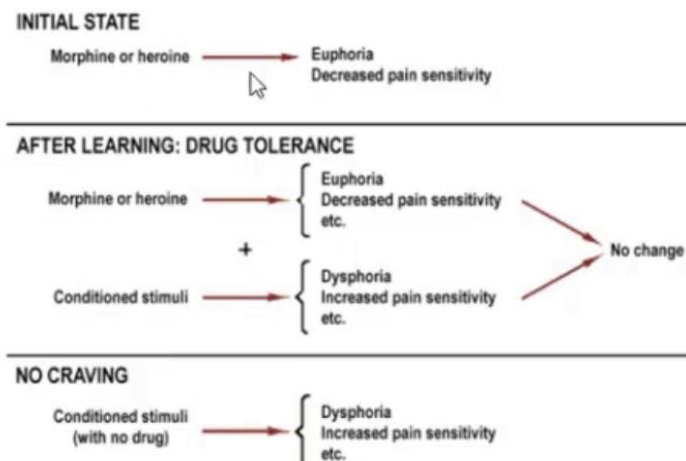


- **Contingency** – the CS should be informative with respect to the chance that US is offered. So, **CS must be a predictor of US happening**. As shown in the experiment below:



- **Contiguity and contingency features created support for cognitive view of learning**. That is, the animal is not learning to associate CS with UR but merely using CS to prepare for the arrival of US.

- **CR and UR are rarely the same.** Saliva when there's food is much more than CR.
 - Rats that get associate light with shock have different UR and CR. In case of shock [US], heart rate accelerates [UR] but when there is light [CS] heart rate decelerates [CR]. This means CR may have been a preparation for upcoming UR.



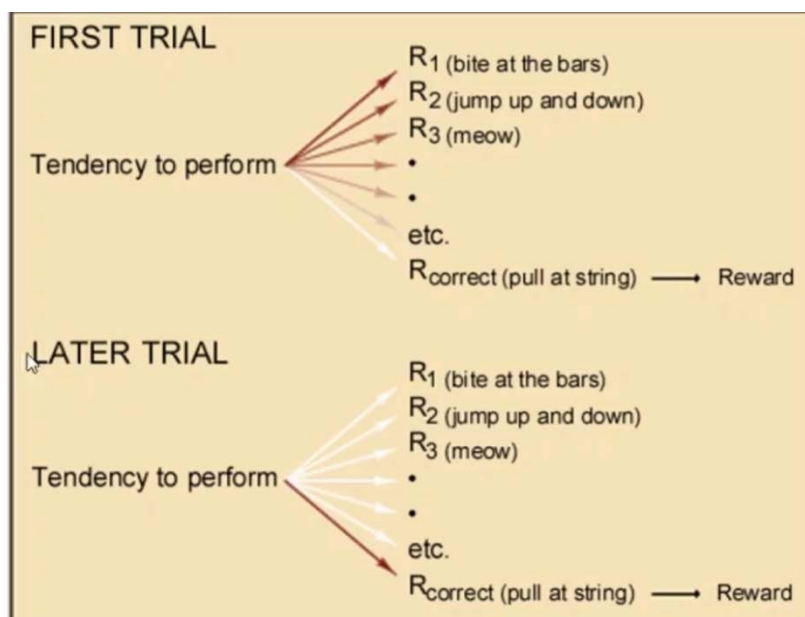
Drugs CR and UR

- Drug users take drugs in a regular place or use same objects to do drugs. So, they develop a CR which is opposite of drug effect [UR] to reduce the upcoming effects of drug. This creates tolerance in drug user, so the drug user takes more for getting the same effect.
- **When there's no drug, there is craving.** This craving may be more when person is in contact with objects associated with drugs or inside room where the person takes drugs. So, CR is present but UR may not be there.
- Moreover, when the person is not in the regular drug use environment, there may not be any CR before drug use. So, the **person is not prepared to dampen drug effects in this new environment.** This can lead to **drug overdose.**

Associative learning: Operant learning

- Operant or instrumental conditioning is a form of learning in which the **consequence of an action (response) determines the likelihood that it will be performed again.** Learned is the relationship between response and the consequence (reinforcer or punishment).
 - Here the **response is voluntarily generated by the organism**
 - **Law of Effect (Thorndike 1898)**
 - **Behavior is governed by its consequences.**

- **Performance** i.e. the tendency to give a response, is **strengthened if it is followed by a reward and weakened if it is not.**
- **Thorndike put a hungry cat in a box.** Cat had to push a lever to get out and eat food. After many trials, cat took less time on successive trials. But it was not a sudden decline. Only gradually did the cat learn. Which means, cat didn't gain any insight into the mechanism of trap door.
- As per law of effect, animals and humans have different tendencies to perform in certain ways. **By selectively rewarding a certain tendency, that tendency will become more common because it was rewarded.**

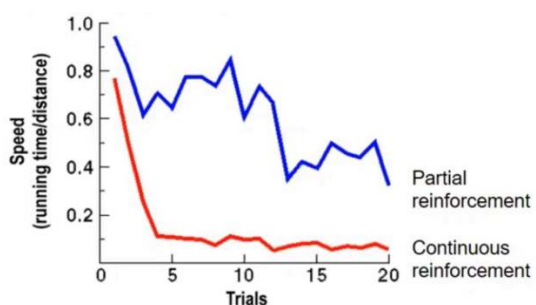
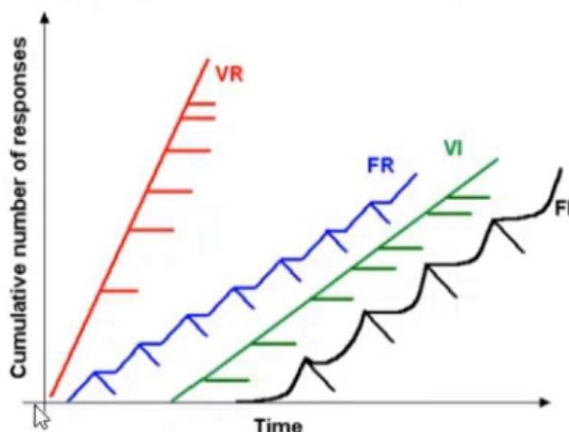


- **BF Skinner (1904 - 1990)** - law of effect was the inspiration for his later work. **Founder of instrumental learning theories.**
 - **Made a sharp distinction between classical and operant conditioning.** Because operant conditioning is due to voluntary decisions of animals
 - Used **Skinner boxes** or operant chambers. Animals were given certain stimuli and were given the opportunity to respond voluntarily.
- How did Skinner affect their behavior? By using the techniques:
 - **Shaping – Successive approximations by applying direct reinforcement.** For example reward a pigeon for a particular movement, and keep rewarding it for more subtle forms of that movement. This will shape the pigeon to do a complex trick. Used on circus animals and college students.

- **Reinforcers** – increase probability of increasing a particular behavior.
 - **Primary** (food and US) or **Secondary** (CS like 'Well done')
 - **Positive** (add something) and **negative** (remove something) reinforcers. Positive reinforcer need not always be good. Negative reinforcer may also be good, like removing a painful stimulus.

Process	Behaviour	Consequence	Result
Punishment			
Aversive punishment (also called positive punishment)	Response occurs (Two siblings fight over a toy)	→ An aversive stimulus is presented (Parents scold or smack them)	→ Response decreases (Fighting decreases)
Response cost (also called negative punishment)	Response occurs (Two siblings fight over a toy)	→ A stimulus is removed (No TV for 1 week)	→ Response decreases (Fighting decreases)

- **Punishment** *less effective than reinforcement*
 - **Positive** punishment – adding pain or shock to the organism
 - **Negative** punishment – removal of pleasant stimulus.
 - Punishment is less effective than reinforcement.
- **Continuous reinforcement** – after every desired act.
- **Partial reinforcement**
 - **Fixed ratio** – reward after every 4 responses.
 - **Variable ratio** – Reinforcer after an average of 4 responses but not always after same number of responses.
 - **Fixed interval** – food pellet after every 4 minutes
 - **Variable interval** – food after on average 4 minutes.
 - Most successful reinforcers are in Variable ratio of reinforcers



The Best ←

- **Partial reinforcement extinction effect** – **Partial reinforcement leads to less reinforcement** as shown above. So, partial reinforcement stays.
 - Useful for parenting – when a baby cries, parents may sometimes attend the baby and sometimes not. Unknowingly, they put the baby on partial reinforcement schedule. This makes crying stay longer. Continuous reinforcement [attending the baby every time] will make crying extinct sooner

More on operant conditioning

- But **operant conditioning is not independent of cognition of the animal**. An **organism CANNOT just learn any association**. What a pigeon can learn **depends on its biological design**.
- **Learning can also happen in absence of responses.**
 - **Tolman (1948)** claimed that learning is not just a change in behavior but also an acquisition of new knowledge. So, there's an **element of cognition involved in learning**.
 - He did a study that showed – rats that were driven through a maze solved it quicker than those who were not driven through it. Which means, rats made a **cognitive map** of the maze. This explains their better performance.
- **Operant conditioning only works when reinforcer has contiguity with behavior**. Otherwise wrong behavior can be shaped.
- **Contingency with the behavior is also required. Reinforcer must follow the desired behavior**, otherwise it won't work most of the time.
 - The **organism should have some sort of agency or control over the action**.
 - Study of **learned helplessness** shows – when organism feels lack of control, they start to give up and start to block the stimulus. Some dogs were taught to avoid shocks by jumping to other box while others were just given shocks and couldn't control it. In a later stage, in a different setup, dogs that had learned to avoid shocks were able to learn the desired action to stop shock. While dogs that didn't learn earlier just gave up.
 - **Humans may also learn helplessness** – in **depression** humans may give up on trying. People in situations where they couldn't do anything, may develop a feeling of helplessness.

Observational learning or Social learning

- The acquisition or modification of a behavior after exposure to another individual performing that behavior.
 - **Modelling:**
 - **Bandura** did observational studies. Showed that social behaviors like aggression can be learned by observation and imitation.
 - **Bobo doll experiments** – children saw two videos: adults playing with bobo doll; and adults attacking bobo doll.
 - Children who saw adults attack the doll, were more likely to attack the doll than children of the other group.
 - **Vicarious learning** – you learn the consequences of an action by watching others get reward or punishment for that action.
 - Does violence in **movies and games affect social behavior?**
 - Columbine shooting led to debate that video game 'Doom' caused aggression.
 - Studies in lab setting show that watching aggressive clips invoke aggressive behavior. But there is only a correlation. It is **not clear yet, if violence in media causes violence in real life.**
- **Neurological basis of observational learning:**
 - **Rizzolatti** showed that monkeys possess **mirror neurons** which are active not only when performing a specific action but also when another person is doing that action.
 - So, **mirror neurons may be the basis of observational learning.** However, firing of these neurons does not always lead to imitation of the action being watched.
 - **Mirror neurons may help an animal to understand behavior of others and feel empathy.**
 - Humans also have mirror neurons that get active when exposed to other's behavior.

Psychology Notes Topic- 6

By Somansh –knows Madonna and Madonna knows him

Memory

Video ted talk by Loftus. False memory. Hundreds of people convicted due to false memory of witness and victim. Our memories are constructed and reconstructed. We and other people can change our memory. In normal as well as stressful situations, memory can be altered. Misinformation can easily alter memory and today misinformation is everywhere. In the 90s the psychotherapy sessions of dream interpretations, imagination and hypnosis were creating false memories in the patients. Bizarre memories of satan, torture and abuse were drilled into the patient during these sessions.

Memory implantation influences future thought and behavior. So, a really dangerous tool which can be abused.

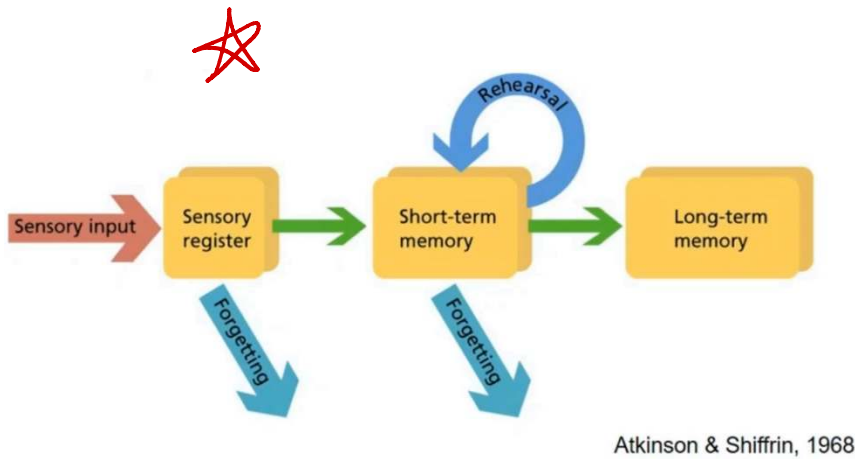
If a person describes a memory with confidence, detail and emotion doesn't mean that the memory was real. Memory like liberty is a fragile thing.

Overview

- The study of memory
- How memory operates
 - Encoding
 - Storage
 - Retrieval
- Variety of memory
- When memory fails

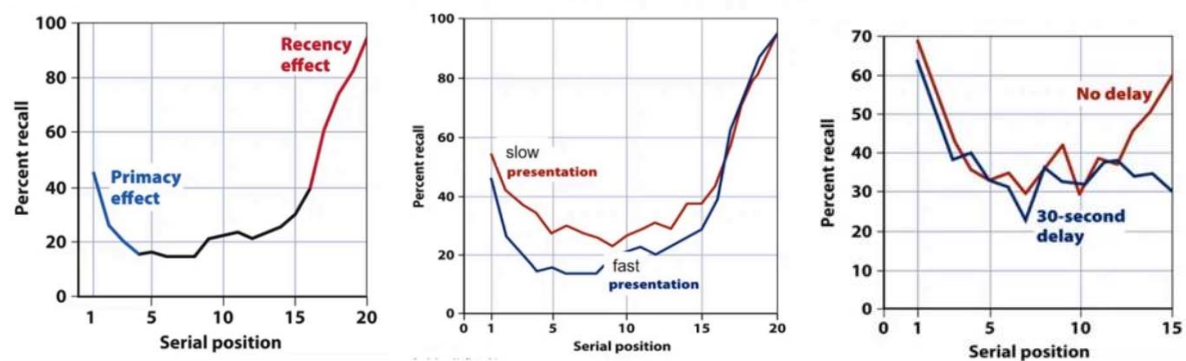
Memory

- **Perception** is the organization of current stimuli BUT **memory** is the organization of stimuli from the past (ideas and events).
- It is the nervous system's capacity to retain and retrieve skills and knowledge.
- We have multiple memory systems, and each memory system has its own 'rules'.
- Memory operates over time in 3 different phases
 - **Encoding:** Processing of information so it can be stored, and how will it be stored
 - **Storage:** retention of encoded representations over time. Consolidation is the neural process by which memory is stored in brain
 - **Retrieval:** recalling or recognizing stored memory when it is needed.
 - Recall
 - Recognition
- Not remembering may be due to error in any of these stages. Not paying attention to a person's name will cause the name to not be encoded, and not even be stored in memory. Having something in memory but still being unable to recall is retrieval error.



Encoding

- According to **Atkinson**, there are 3 types of memory that varies in capacity and duration.
 - Sensory register or memory** – has **iconic (visual) and echoic memory (audio)**
 - Contains all the raw input derived from senses.
 - Stays for about ~500 ms**
 - Can be sent to short term memory if it is selected
 - Short term** – remembers whatever we are working on.
 - Small capacity** and can be easily replaced
 - Rehearsal** helps to keep a short term memory alive
 - Long term memory**
 - If a memory stays in short term memory for a long time, it may get transferred to long term memory
 - It has **huge capacity and duration**
- Short term memory – STM** or *Working memory, info available for current use*
 - Has **limited capacity** as shown by memory span studies.
 - In **Memory span** task, people get to view a sequence of items, and have to recall the sequence accurately. We usually remember around **7 (+/- 2) number/items sequences**
 - Short term memory **disappears** as it **decays and replaced**.
 - But its **capacity can be expanded by re-encoding the sequence**
 - Eg UK EUCIAPHDFBIUSSR -> UK EU CIA PHD FBI USSR
 - Why? Because total number of chunks did not exceed 7.



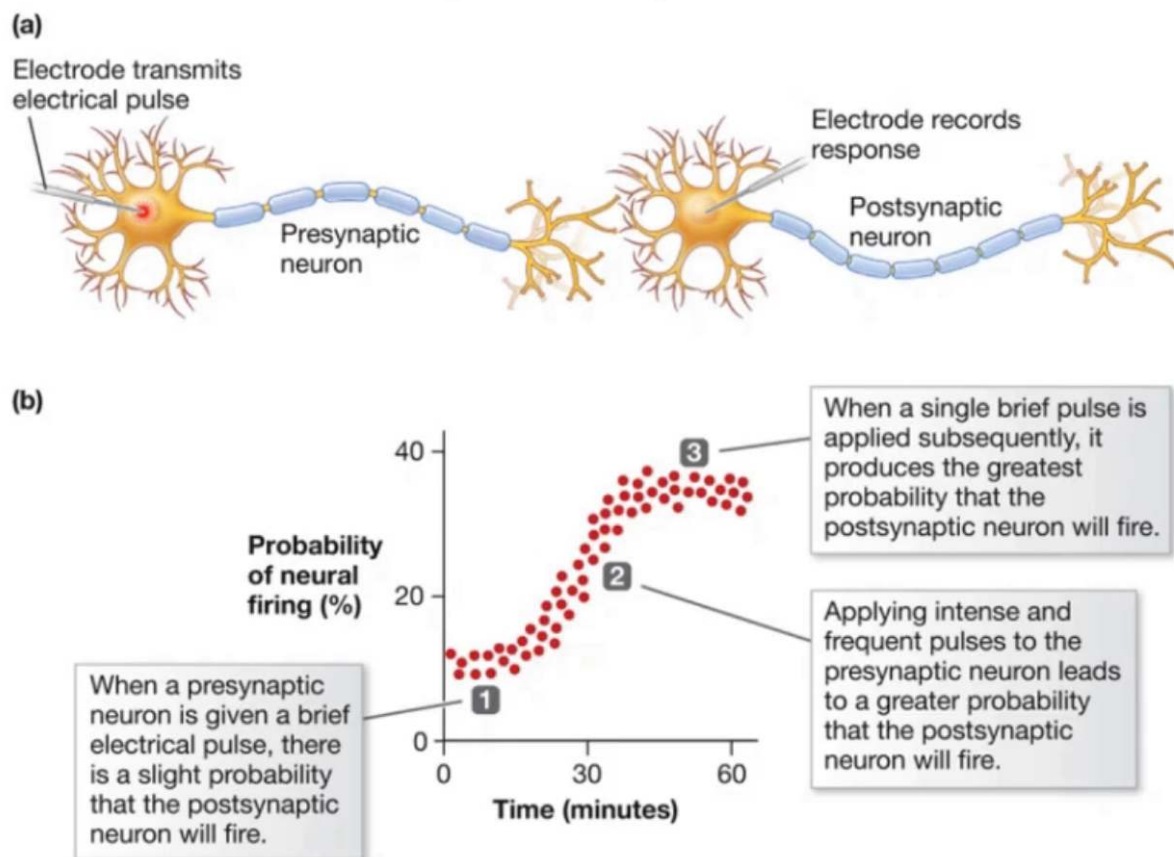
Figures show: Evidence for existence of STM and LTM

- **Long term memory – LTM**
 - **Unlimited capacity and relatively permanent.**
 - Repeating short term memory can shift the memory to long term memory.
 - In **free-recall task**, people get a series of unrelated elements. They recall most elements from **start of the list [LTM]** and **end of the list [STM]**.
 - **Initial items were rehearsed** and **put to LTM** because people cared to rehearse items in the start but give up around after 7 items – the middle items.
 - **Ending items were in STM so recall was easier.**
 - In faster presentation of items, items couldn't be rehearsed and thus items didn't go to LTM.
 - **Putting delay or giving participants a task after presenting last items on the list led to dip in recalling them.** This proves that **items in the last were indeed put in STM**
- The idea that rehearsal caused a memory to be shifted from STM to LTM was **challenged by later studies**. Studies by Craik and Watkins (1973) and Nickerson and Adams (1979)
 - **Craik and Watkins (1973)** – gave participants a list of words, and they had to keep repeating the most recent word that started with letter 'g' while other words were presented.
 - Surprisingly, participants did not remember 'g' letter words any better than other words that were presented. This means, **rehearsal by itself was not the key mechanism to shift a memory from STM to LTM.** → *more repetition*
 - **Nickerson and Adams (1979)**
 - Asked participants if **penny had Lincoln facing left or right**. People were only guessing the side. This means, people saw the penny every day for years, but still didn't remember. Again, it is proven that mere repetition is not the key mechanism to shift a memory from STM to LTM.
- Today we know, **memory is an active process requiring attention**. Memory is the **product of intellectual engagement and thinking**.
 - So, **STM is called Working Memory** – a form of active processing system that keeps information available for current use. Can only do a limited number of processing
 - **Working memory has the information being actively currently used. If the information in working memory [WM] is activated more and more, it goes to LTM.** So, it is not the amount of time an information spends in STM/ WM but the encoding process of information
- **Which encoding process promotes later recall**
 - **Depth of processing**
 - **Shallow or surface processing** – does word have an uppercase or lowercase. **Remembered less.**
 - **Deep processing or processing with meaning** – does a word fit in a sentence. **Remembered more**

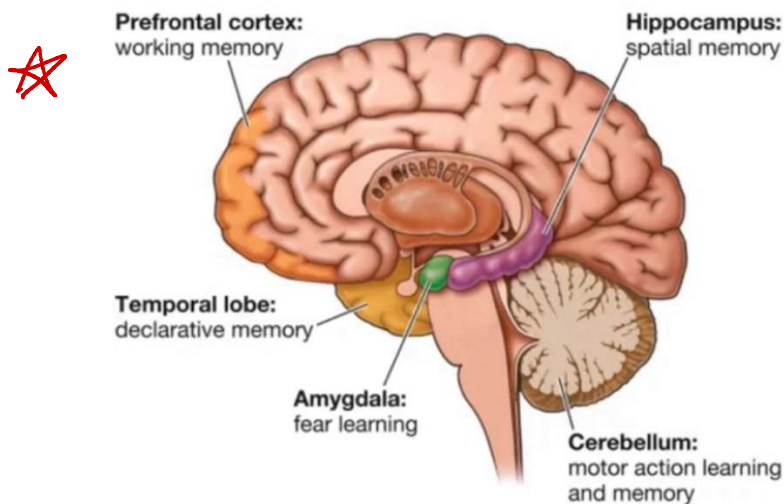
- **Understanding** – attach more meaning to it. This creates connections in brain and create more memory paths to that information. So, we remember much more complex information/story/concept if we understand it well.
 - **A schema helps us understand and subsequently remember information contents.**
- So, **good encoding** provides effective means to retrieve information later

Storage

- **Retention of encoded representations over time.**
- It is in **form of memory traces, done with memory consolidation**
 - **Memory consolidation** is the neural process by which encoded information becomes stored in the memory. Transient and fragile potential memory contents are transformed into a more permanent state.
 - **Process of memory consolidation takes some time.** When a person gets a blow to the head, they don't remember the recent events because they did not have the time to process it. May be that's why videos played on high speed are not remembered well enough?
 - **Memory consolidation restructures the neural setup and protein production is required for a memory trace to form. This is inhibited by the enzyme HDAC. When HDAC is active, proteins required for memory will not be formed and thus no memory created.** The HDAC is the natural brake for memory creation. HDAC can be created by drugs. May be useful in treating dementia.



- **Memory also requires neural adjustment called – Long term potentiation. LTP is, strengthening of a synaptic connection, making the postsynaptic neurons more easily activated by presynaptic neurons. Based on the work of Eric Kandel et al.**
 - Kandel and colleagues measured neural firing in a sea slug. If a presynaptic neuron is activated continuously, the post synaptic neuron's firing increases with time. As if the neural link was strengthened. Or, **cells that fire together, wire together.** Maybe it is due to increased number of neurotransmitters at their synaptic connection points.
- **Memory is also consolidated if it is reconsolidation i.e. memory is recalled, used and reconsolidated again. When we recall, erase, reconstruct, activate memory, the memory gets reconsolidated. Used to treat PTSD**

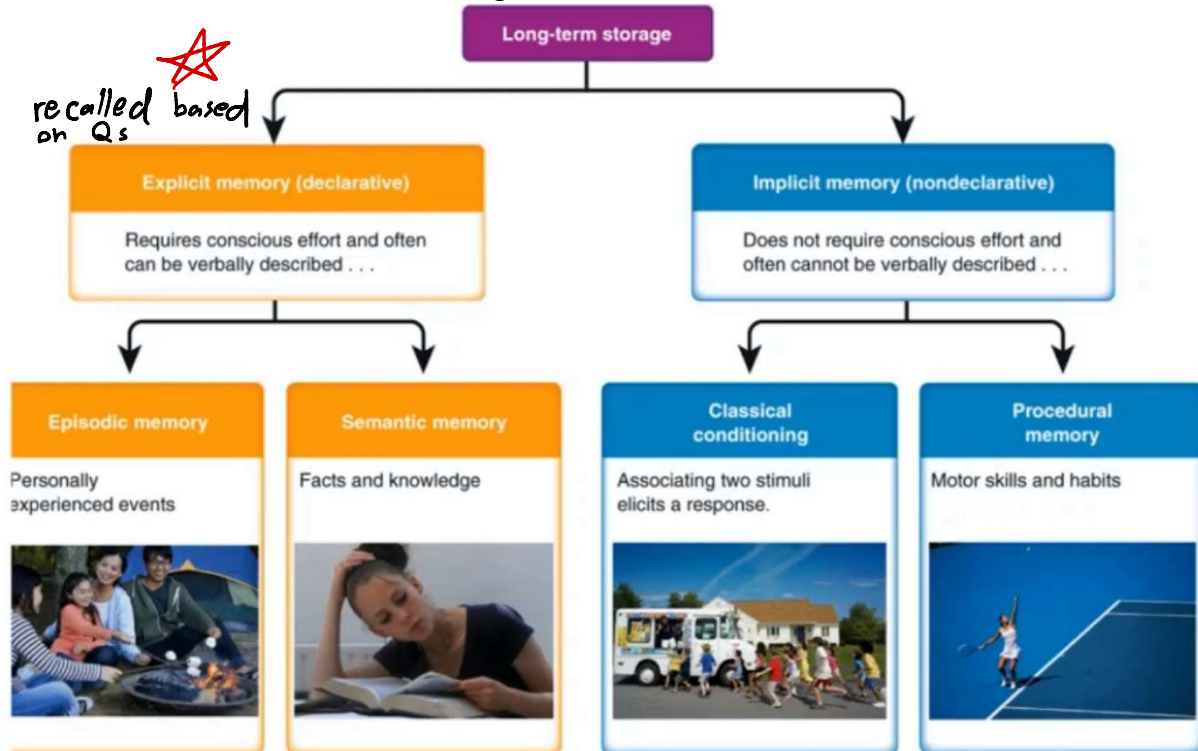


- **But where is memory stored**
 - **Everywhere in the brain.** Lesley trained rats to run a maze, and removed various parts of rat's brain. It was seen that the amount of brain removed affected their performance much more as compared to the area from which the brain was removed. So, memory is everywhere. **But, central part of brain is of key importance in making new memories.**

Retrieval

- **recalling or recognizing stored memory when it is needed.**
- How do we retrieve – by using good retrieval cues, cues that help us enter a memory trace. **Cues can work by:**
 - **Recreating the context** in which the original learning occurred aka **Context Reinstatement.**
 - E.g. Divers learned 2 lists of words on land and underwater. They recalled the underwater list better when they were underwater, and land list when on land.
 - Mainly about recreating the mental state when learning occurred. Even imagining the learning context is good enough to enter the memory trace easily.

- **Richer and deeper encoding results in better recall.**
 - This means that creating more paths connected to the memory trace will ease the retrieval.
 - Recreating or creating a related context helps create good encoding and cues.



Varieties of memory

- Long term memory also has various types
 - **Explicit – conscious memory that can be recalled based on questions.**
 - **Episodic** – specific individual events in one’s own life. Episodic and semantic memory have different brain structures.
 - **Semantic** – for memory in general. Moment or place we learned it is not important like, what’s capital of France? It is not dependent on when or how we learned it. Different contents are stored in different areas of the brain. Food knowledge is stored in different areas than facial knowledge.
 - **Implicit – unconscious memory that cannot be voluntarily recalled.** Requires indirect testing. No conscious attention needed.
 - **Classical**
 - **Procedural**

Memory failure

- **Forms of failure:**
 - **Inability to remember:**
 - **Amnesia** is the deficit in long term memory, resulting from disease, brain injury, or psychological trauma – in which the individual loses the ability to retrieve vast quantities of information.
 - **Retrograde amnesia** – lose past memory of events, people and so on. Mostly due to injury on head.
 - **Anterograde amnesia** – lose the ability to make new memories. Due to alcohol abuse, bad surgery. In a failed surgery, patient lost the ability to form explicit memory but retained the ability to form implicit memory.
 - **Absent mindedness** – inattentive or shallow encoding of events. In everyday life we are attentive/aware of only small portions of our thoughts and actions. Every information is not encoded.
 - Some parents forget to drop the children at day-care and they die in the car
 - **Memory reconstructions or distortions of memory** – may be due to bias, suggestions or misattribution. Very common in patients as well as normal people.
 - **Memory decay** – forgetting over time. Memory is better for recent events as compared to older ones.
 - If there is longer period of retention, the memory will decay as seen in **Forgetting curve by Ebbinghaus 1885**.
 - This is because of
 - **metabolic processes that cause decay in memory traces**. In Goldfishes, metabolism increases in warm water, thus they have larger memory decay in hot water.
 - Due to **interference from other information** –
 - That's why there is less forgetting after sleep as compared to wakefulness state because no interference happened in sleep.



- **Types of interference**

- **Proactive and retroactive interference** –

- **Proactive interference** is when **prior information interferes with new information.**
- **Retroactive interference** is when **new information interferes with an already stored information**
- **Memory bias** – the changing of memories over time so that they become consistent with current beliefs or attitudes
 - People revise memories when their beliefs were changed. Even societies alter or downplay their history after their beliefs have changed.
 - Even flashbulb memories like 9/11 memories can change over time as people talk about it. This constant reconstruction alters the memory
- **Source misattribution** – memory distortion that occurs when people misremember the time, place, person, or circumstances involved with a memory.
 - In a study, people were given 2 factual statements and a week later, they were given one fact and one absurd statement. People were likely to consider the absurd information as legit because it felt familiar or from same source.
 - So, we misattribute the source of our memory.
 - **Familiarity increases credibility.**
- **Suggestibility** – the development of biased memories from misleading information.
 - Research showed influence of suggestion on memory.
 - **Elizabeth Loften** proved that framing questions in a specific way, people's memory can be changed.
 - People saw a pretty lame car crash, with no glass shattered in ground. One group was asked how fast the cars were going. Another group was asked, how fast were the cars when they smashed.
 - Second group was more likely to remember that there was glass shattered on ground.
 - May be due to schema activated by the word 'SMASH'
 - This **framing can alter memory of witness and victim**, creating distorted memories and wrongful accusations.
 - **Suggestibility can even implant fake memories.** Like planting a balloon ride memory that never happened. Or planting repressed memories - planted by psychotherapist - that were never there.
 - **Can also be altered by intrusion of semantic associations.** Prior knowledge can cause memory to distort. Memories that don't fit our schema may be forgotten, while false things that may fit our schema can intrude in our memory. A student visiting a professor's room may remember seeing books in the room while there were none present.

Psychology Notes - Topic 7

By Somansh – hasn't seen Friends and Harry Potter yet.

Thinking and Language - Overview

- Mental Representations
- Decision making
- Problem solving
- Language

Mental Representations

- **Thinking involves** application of operations on our **mental representations**.
- **Mental Representations are components of our knowledge**. Two broad classes
 - **Analogical Representations**
 - **Symbolic representations**
- **Analogical Representations**
 - That capture many of the features of things they represent, but in the end they are just models/interpretations of the things they represent
 - Image of a cat may represent the shape, color or other features of a cat.
 - But they are Dissimilar from pictures
 - Because **pictures show what a thing actually is**, whereas **analogical representations** just have a partial or **model representation of things**.
 - **Mental representations are already organized and already interpreted**. We know the shape of a rabbit and seeing an outline that resembles it will trigger our mental representation of a rabbit. Only after we're told that the shape resembles a duck as well, will we change our perception.
 - **Mental representations is biased by a person's beliefs, schemas, knowledge**
- **Symbolic Representations**
 - **Do not resemble what they represent**.
 - The word 'cat' has no resemblance with the actual cat.
 - Symbol evokes a particular meaning, so **symbolic representations** are just a **fast way to retrieve information from our mind**. Like an index page.
 - Postulated to be organized in networks of associations or spreading activation model
 - Collins and Loftus model – in which each unit of information/ item/ concept is a single node in the network for e.g. vehicle, car, fire etc. The closer the node, the more likely they are to be associated. If one node becomes active, the close by nodes will most likely be activated. The amount of activation spreading around the nodes depends on strength of their association
 - Experiment – participants were given words and non-words and they had to quickly press a button to identify if it was a word or non-word. Interestingly, when the meaningful words like 'doctor' were presented after a related word like 'nurse' the reaction times were much faster as compared to non-related words like 'butter'. This happens because the word 'nurse' activated the network where the word 'doctor' was also located. Doubt – This also means, in our brain, the related concepts/words/symbols are located in close by neural networks?

Decision making

- Selecting the best alternative among several alternatives. Some are important and some are trivial. Some are taken once in life and others are taken repeatedly.
- How we decide?
 - Logic – May think rationally and logically before deciding. These are called normative decision theories.
 - Expected utility theory is one such theory that argues, decisions should be made to maximize the outcome.
 - It should depend on values of alternative options
 - Probabilities associated with alternative options
 - For example – buying a 10 Euro lottery ticket or not. Probability of winning lottery is 1 in a million with prize as 1 million. Using magic and maths, we calculate that, the value of buying such a lottery is minus 9 Euros and value of not buying a lottery is zero Euros. Thus we should decide not to buy the lottery.
 - But we do buy lottery.
 - Heuristics – not thinking rationally, but based on heuristics. Descriptive decision theories describe how people actually make decisions
 - Daniel Kahneman showed that we don't always take logical decisions, but we base our decisions on heuristics or rules of thumb.
 - Heuristics – this approach involves, following shortcuts or rule of thumb or informal guidelines to take decisions. It reduces amount and time to think and make decisions.
 - It happens unconsciously
 - It helps us decide quickly
 - Reduces use of cognitive resources, so that they can be used elsewhere.
 - Can result in faulty decisions and biases.
 - Anchoring – making decisions and judgements based on the first piece of information encountered or the first information that comes most quickly to mind.
 - Real Estate Agents use it. They anchor a property in buyer's mind by showing overpriced, not so good place. Then show a lower priced better location, so buyer thinks it's a great deal. But in reality it may not be a great deal.
 - Framing – to emphasize the potential losses and potential gains from at least one alternative. It creates a change in point of reference. People avoid losses much more than trying to gain something. Framing a decision as averting some loss will be given more importance by people than a decision where a person might gain something.
 - Availability heuristic – making decision based on the answer that most easily comes to mind. We think something is true if we have examples for it. Honda may make the best cars but knowing 2 friends having trouble with their Hondas will make us believe that Hondas are bad cars.
 - Representative heuristic – making a decision based on how typical an individual, object, or event belongs to a certain category. We believe that

smart and active person will be more likely to be a scientist or a professor. Whereas statistically it may be more likely that she is a banker.

- **Affective forecasting heuristic** – when people **overestimate how events in future will make them feel**. We predict how we'll feel when moving to a new city or how will we feel if our favorite team lose. But we estimate it wrongly, most of the time.
- **Paradox of choice** – when there are **too many choices, choices become less attractive**, we become indecisive and experience conflict, even become sad with our choice.
- **In decision making a good strategy is to satisfy our needs and wants** rather than just optimizing our decisions. **How the decision was taken affects us more than what the decision was taken**. Too much thinking can leave us dissatisfied as we focus on negative aspects.

Problem solving

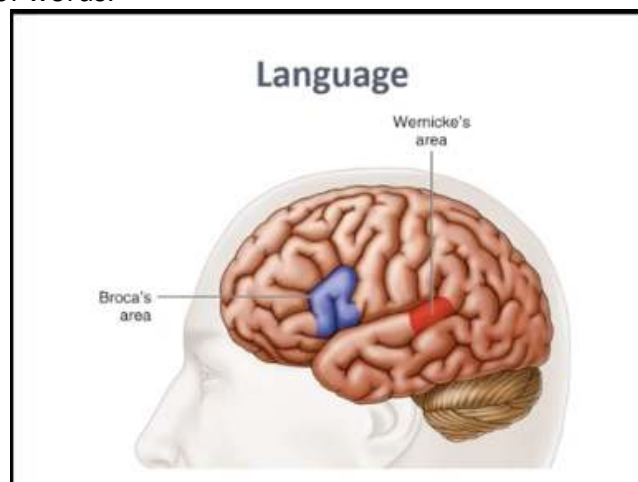
- Various types of problems exist. It may be **well-defined** or **ill-defined**. So, we need a thinking process to solve them.
- **Problem solving is a thinking process** in which we move from an initial state to a goal state.
- Obstacles to problem solving.
 - The problem may be very complex. **Complexity**
 - **Mental set** - person may be thinking in one way while **problem requires a different way of thinking**. For example, a problem may be solved by guessing rather than solving every possibility.
 - Some problems may be difficult due to biased perception. Like limiting ourself to thinking in 2 dimensions and not 3 dimensions
- How to solve
 - **Understand problem**
 - Create **sub-goals**. Not just focus on end goal.
 - **Restructuring the problem**. Try new perspective if one perspective doesn't work.
 - Use an **algorithm**
 - Use an **analogy**. Thinking of probability problem as a card problem.
 - Wait for a **sudden insight**.

Language

- **A system of communication using sounds and symbols** according to grammatical rules. There are thousands of languages. They all have certain properties
 - **Syntax** – a system of rules on how to **combine words into phrases and phrases into sentences**.
 - **Semantics** – **meaning**. Meanings that underlie words and phrases.
- Sentences are made up of phrases, phrases are made of words. Words are made of morphemes – the smallest meaningful unit. So, work is a morpheme, but 'wo' and 'rk' are not. Similarly, carjack is a combination of two morphemes – car and hijack.
- Phonemes are smallest unit of sound in a language. For example 'click' sound is a phoneme in '!Kung' language but not in English. Changing phoneme slightly can change the meaning of a word or phrase. For example, in the sentence, . "Call the

judge to get some fudge”, judge and fudge differ very slightly in phonetics but have very different meaning.

- **Development of language** – As brain develops, ability for language develops.
 - Language development occurs in an orderly way i.e. first children learn sounds or babble in infancy, then learn words at age 1, then rules.
 - At 3 to 5 years of age, children overgeneralize language rule like saying mouses rather than mice.
 - Listening preference in newborns is influenced by mother during pregnancy.
 - Newborns can distinguish between all phonemes but lose this ability at 6 months of age. Japanese children lose the ability to distinguish between L and R because the difference is not relevant in Japanese, so they get habituated to treating L and R as the same. Liver and River are same for that child.
- We produce language sounds by vocal tract, but we also use sign language, braille, and text.
 - Vocal tract and tongue modulate sound frequency, amplitude, volume etc. to create phonemes. However same phoneme like ‘e’ may have different sound frequency, amplitude and shape.
 - Speech can be seen as energy produced in a moment of time and can be visualized as spectrogram. Same words may have wildly different spectrograms.
 - Contrary to popular belief, there are no spaces of silence between every word we speak. Rather, the space is put by our brains when we interpret speech.
 - Our brain uses previous knowledge and context to interpret words. Recall top-down and bottom-up perception.
- **Aphasia** – a language disorder that results in deficits in language comprehension and production. Mostly due to brain damage or accident.
 - **Expressive aphasia (Broca’s aphasia)** – first reported by Broca. It is the interruption of the ability to speak.
 - **Receptive aphasia (Wernicke’s aphasia)** – difficulty to understand the meaning of words.



Psychology Notes – Topic 8

By Somansh – lifetime member of Cloud Appreciation Society

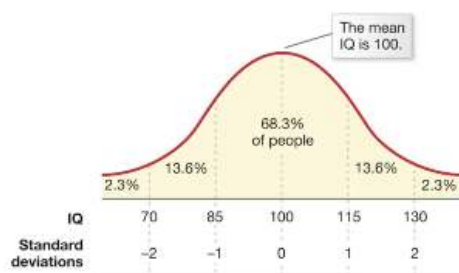
Intelligence

- Introduction
- Intelligence testing
- Psychometric approach
- Intelligence beyond IQ test
- Relation between IQ and cognitive performance
- Roots of intelligence

Introduction

- **General psychology**
 - The focus is on general psychological rules and theories applicable to all or general population
- **Differential psychology.** More individual focused psychological studies, for instance intelligence. IQ is studied on basis of individual differences.
- **Intelligence –**
 - the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience. [Gottfredson, 1997]
 - the ability to use knowledge to reason, make decisions, make sense of events, solve problems, understand complex ideas, learn quickly, and adapt to environmental challenges. [Gazzaniga, 2018]
 - No broad consensus on definition yet, as always.

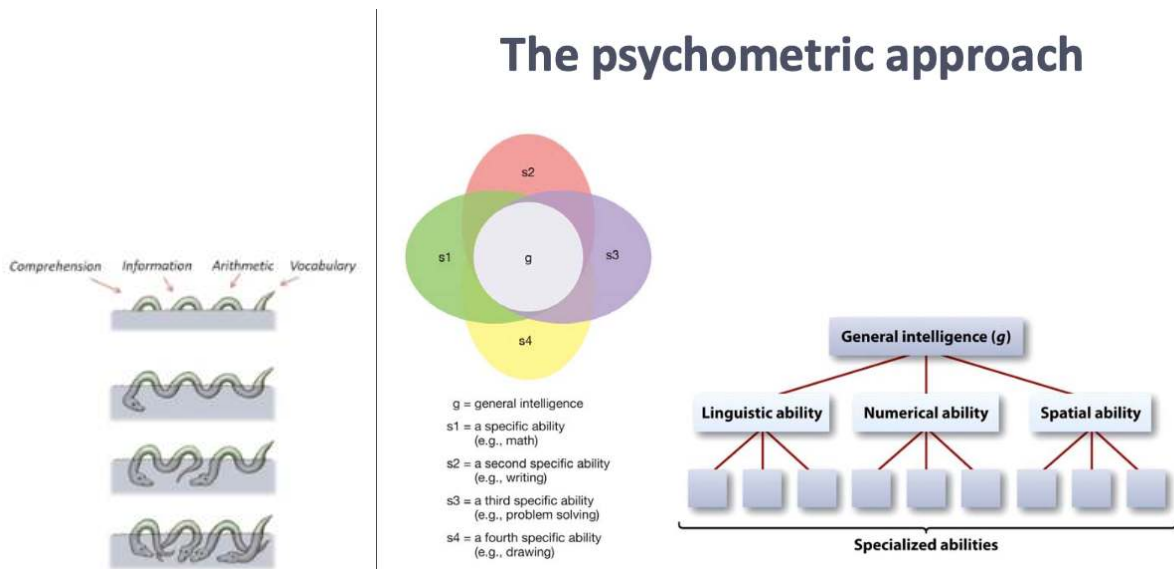
Intelligence testing



Intelligence testing

- **Measurement**
 - First intelligence test was developed by Binet in 1911
 - $IQ = (\text{Mental Age} / \text{Chronological Age}) * 100$
 - Used to determine which kids needed focus in teaching.
 - If an 8-year old can solve problems just like a 10-year old, her IQ would be $10 * 100 / 8$ or 125
 - But, after age of 16, mental age no longer increases, which implies, people's IQ drops as they age, as per this formula.
 - So, for adults, universities use deviation IQ.

- Scored relative to others. IQ is measured on basis of average adult and not on basis of age group.
- **Types of test**
 - **For children – Binet-Stanford test**, Wechsler Intelligence Scale for Children **WISC**
 - For adults – Verbal, geometric, arithmetic and other tests.
 - **Wechsler Adult Intelligence Scale**
 - **SAT, GRE**
 - **Raven’s Progressive Matrices Test** - non Verbal test and less biased by language and culture.
- **Reliability and validity of IQ tests**
 - **Very reliable and consistent.**
 - **High predictive validity** – good correlation with job performance. But only seen when job is complex.
 - **IQ also correlates with income, health, lifespan, car accident.**
- **Intelligence is not one thing but a collection of intelligences which may be social, emotional, spatial, and other types of intelligence.**



- **Psychometric approach** – tries to understand various forms of intelligence. It is an attempt to understand the nature of intelligence by studying the pattern of results obtained on intelligence tests.
 - Use outcomes of IQ tests like WAIS and analyze the outcomes.
 - One person should score similarly on all types of tests (Arithmetic, Vocab, Information, comprehension) if there’s only one form of intelligence. Whereas, score differently in different tests if there are different forms of intelligence.
 - Analysis of IQ tests show there is a correlation but it is not so strong, as the table shows.

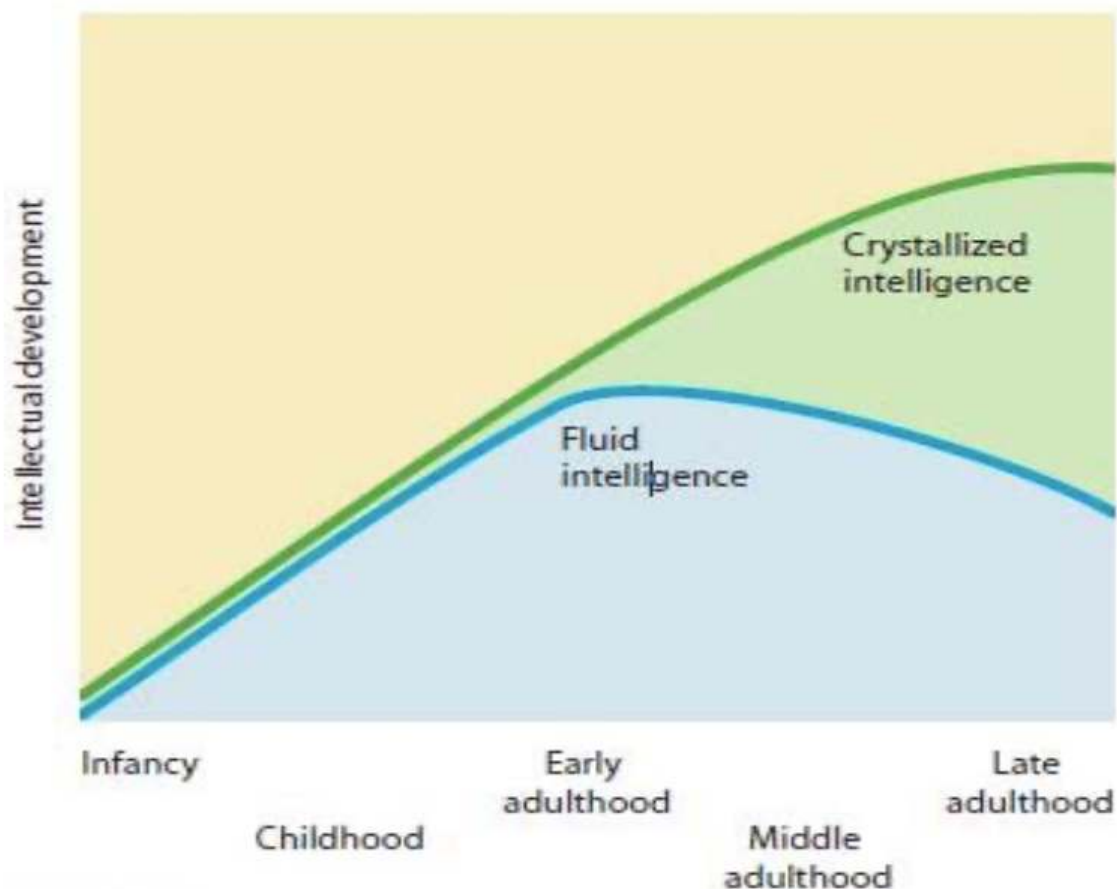
TABLE 14.3 CORRELATION MATRIX OF FOUR SUBTESTS ON THE WECHSLER ADULT INTELLIGENCE SCALE

	I	C	A	V
I (Information)	—	.70	.66	.81
C (Comprehension)		—	.49	.73
A (Arithmetic)			—	.59
V (Vocabulary)				—

NOTE: The matrix shows the correlation of each subtest with each of the other three. Note that the left-to-right diagonal (which is here indicated by the dashes) cannot have any entries because it is made up of the cells that describe the correlation of each subtest with itself. The cells below the dashes are left blank because they would be redundant.

SOURCE: From Wechsler, 1958.

- This means there is a **general intelligence as well as specific intelligences**.
- Factor analysis used by **Spearman in 1904**, and concluded that **intelligence is a mix of general intelligence and specific abilities**.
- **Cattell 1971 differentiated crystallized and fluid intelligence**. Past knowledge and experience create crystallized intelligence and fluid intelligence is used for solving new problems. Both are correlated, as one person having good crystallized intelligence has good fluid intelligence. But there are slight differences. With age fluid intelligence decreases but crystallized intelligence ~~decreases~~ **increases**.



- **Standardized tests fail to capture other forms of intelligence**. Traditional intelligence is captured like, general, linguistic, spatial, fluid, crystallized

intelligence. **But ignores non-academic intelligence like street smart, social intelligence.**

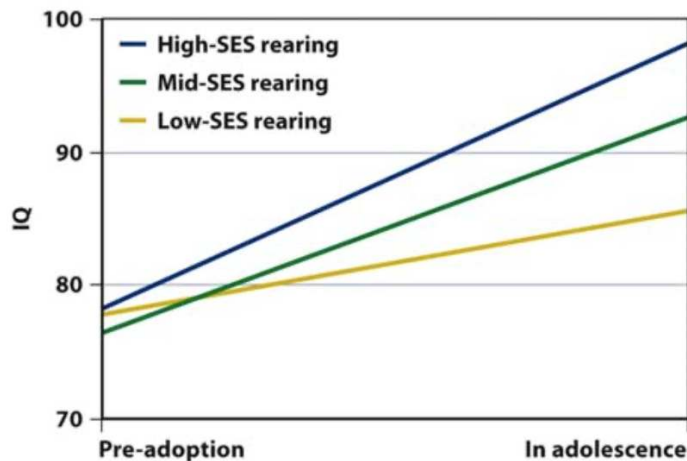
- Sternberg 1999 proposed we should include analytic, creative (thinking new and creative ways), and practical intelligence (sports, games, daily life). But there is not much evidence on these types of intelligence.
- **Salovey and Mayer 1990: created emotional intelligence test.** But EQ also correlates with IQ test, so may not be justified to keep it as separate entity
- Gardner 1983 argued there are different types of intelligence – music, linguistic, spatial, logical, bodily, inter and intra-personal intelligence. Some people are savants, and have specific intelligence for a specific task like memory, music etc.
- But since they also show correlation with IQ, so we may not need them as separate categories

Relation between IQ and cognitive performance

- **General intelligence is related to neural efficiency or speed of neural signal transfer**
 - Weak correlation of IQ with reaction time
 - **Strong correlation of IQ with choice-based reaction test**
 - **Correlation of IQ with discrimination task performance.**
 - **Strong correlation of IQ with working memory capacity** – people with higher IQ can repeat more of the the recently memorized words or numbers.
 - In **dual task situations**, performance strongly correlates with IQ
 - **Executive control** – people with high IQ can switch between tasks more easily than others.

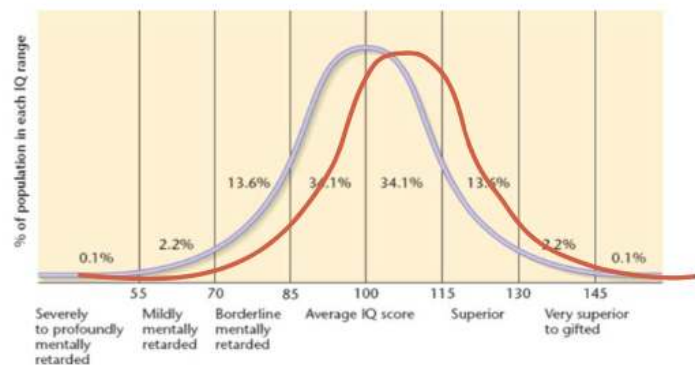
Roots of intelligence

- **Affected by both genes and environment.** An outcome of interaction of genes and environment.
- **Genetic influences**
 - IQ correlates between parents and children. So, there is a strong genetic component. But, parents and children also share their environment, so it may also have an effect. We study this via twin studies
 - **Identical twins have higher IQ correlation than fraternal twins, thus proving genetic contribution.**
 - Adopted children have lower correlation of IQ with adoptive mother as but high correlation with biological mother.
- But **environment also affects IQ**
 - **Impoverished versus enriched environment.** Expression of genes requires a minimal environment condition like good food, good social relations.
 - In rich environments, genes affect IQ more because basic needs are met. But in poor environment, bad environment affects IQ much more. An overall positive change in such an environment, the IQ improves.

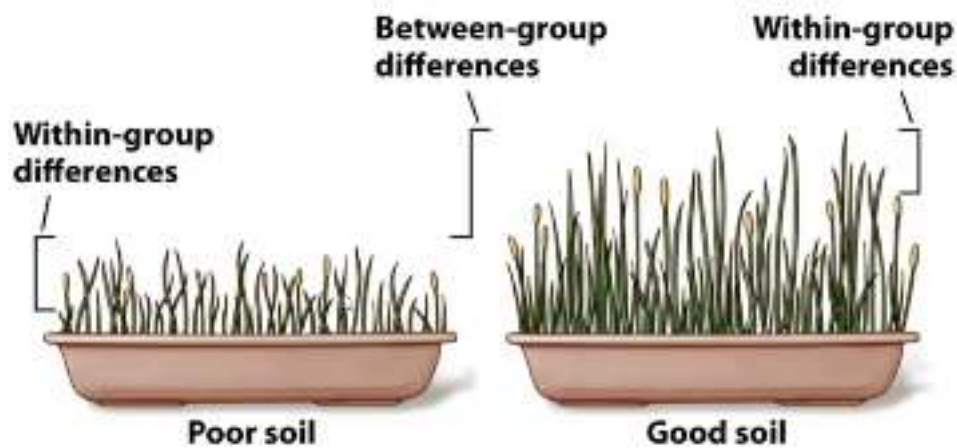


- **Flynn effect** – generational increase in IQ. Since our genes didn't change that much, it must be the result of better nutrition, good education and stimulation.

Flynn effect



- **Heritability** – It is the indication of the extent to which, variance in a trait in a certain population in a certain environment is determined by genetic variance.
 - **H = Genetic variance / Phenotypic variance**
 - For IQ, Heritability varies from 0.4 to 0.7 which is high
 - But this was studied for western middle class. For other cultures and environment, it may vary.
 - **Low heritability need not mean that genes don't affect a trait and high heritability doesn't mean that environment plays no role.**
 - In early China, women were made to wear tiny shoes, thus keeping their feet small. So, feet size would show very low heritability. Here genes were restricted. So, using this particular society and environment, it would be wrong to conclude that genes don't play a role in feet size.
 - Grass in poor soil varies largely from grass in good soil, due to environment. But grass in the good soil has within-group differences due to their genes because every blade has good environment to grow. Moreover, grass in poor soil has lower within-group differences.



- Similarly, in poor environments, the difference in IQ of children is not too much because brain couldn't develop properly, so the genetic differences didn't get to shine/express.
- **Group differences – usually variations within group is larger than variations among groups.** Inter-Gender, inter-ethnic differences are not that large as intra gender and intra ethnic differences. Moreover, correlation is not always causation. Also, studies may not take wide enough sample to make a scientific claim.
- Reports – men have higher IQ than women; whites have higher than blacks – but the differences are not necessarily due to genes but may be mostly due to socio-economic upbringing, how the subjects were testes, was the study biased, was the sample wide enough, were the test properly designed.

Psychology Notes Topic 9

By Somansh – firm believer in God, when in trouble

Scientific Reasoning – Methods of research

- Introduction
- Fixation of belief
- What is science
- The logic behind science
- Scientific reasoning
- Falsifiability criterion
- Two theories

Introduction

- **The Secret – scam**
 - Presents law of attraction as a law of nature
 - As per this law, like attracts like
 - If you don't believe in the law, the universe won't work for you. Although they just said that it had been working on people unaware of such a law.
 - Money can be made with this.
 - The ideas seem plausible but actually isn't. It is contrary to scientific belief.

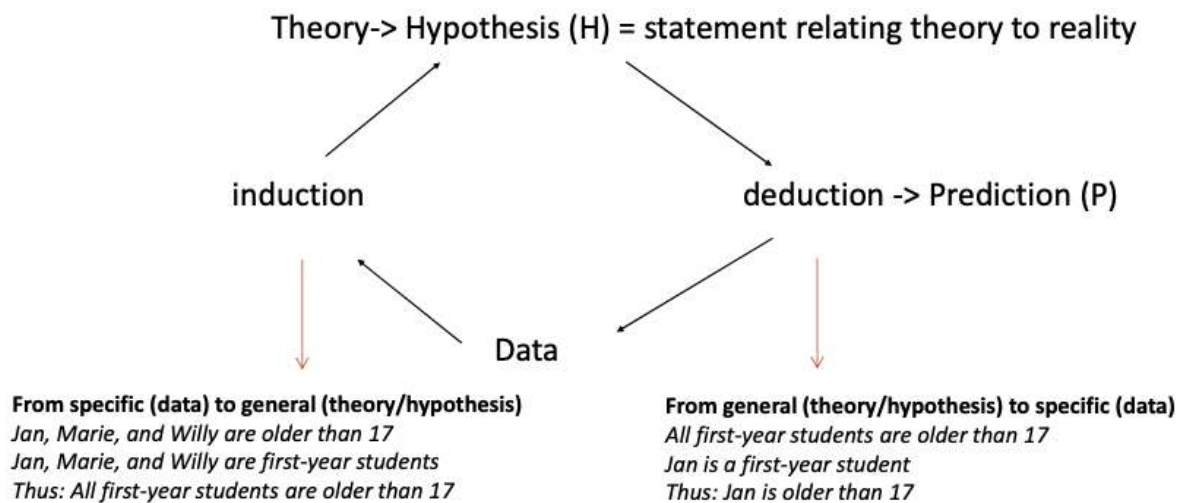
Fixation of Belief

- **Ronald N. Giere's Understanding Scientific Reasoning** – outlines features of scientific thought.
 - **What is belief** – an attitude of some person towards a particular claim. E.g. I believe Ferrari will win the F1 this year. I believe climate change is caused by humans
 - **Claim** – a statement which may be true or false
 - There is no direct connection between belief and truth – believing a claim provides no information about the truth or falsity of the claim itself.
 - So, how can we find out if a claim is true or false.
 - **Scientifically justified claims** such as – time dilation exists, or climate change is real – are much more valuable than unjustified claims.
- Scientifically justified claims are based on use of proper scientific method while unscientific claims are not
- What are methods to fix a belief
 - **Personal experience** – "Climate change is not real because it was cold yesterday". "Japan doesn't exist because I haven't seen it with my eyes."
 - **Intuition** – using instinct to believe a claim. Trump intuitively believes climate change is not man made. I intuitively believe 1 kg of steel is heavier than 1kg of feathers.
 - **Authority** – when an authoritative figure or institution makes a claim, we tend to believe it.
- But these 3 methods of fixating beliefs are not scientific. Only when we have concrete proof or evidence for a claim, do we accept it as scientifically justified and

only then it gets accepted and believed by scientific community. So, scientific method fixes belief on the basis of empirical research.

- **Peterson et al. (2019) – counted number of scientific papers and media articles on climate change.** Scientists have much more papers proving climate change exists, as compared to scientists who reject climate change. But in media, climate change contrarians publish much more than people who believe climate change is real
- **Scientific method fixes belief on the basis of empirical research.**
 - It combines **proper reasoning (using logic and arguments)**, and **systematic empiricism (use of verifiable evidence based on experience with the world) to reach conclusions.**
 - Anyone can verify the **evidence and conduct experiments.**
 - **Unbiased, systematic observation of world**
 - **Self-correcting** – as new evidence comes to light, the claim and theories change.
 - It **gives a procedure** to establish superiority of one belief over the other.
 - **Claims are published in peer reviewed journals.**
 - New beliefs are compared with old beliefs and they **are discarded if they don't conform to the evidence.**

• Empirical cycle (theory-data cycle)



What is science

- **Empirical cycle** – aka theory-data cycle
 - **Theory is a set of statements that describes general principles, about how variables relate to each other.**
 - **Hypothesis is just one statement that describes general principles about how variables relate to each other.**
 - **On basis of hypothesis, we deduce a prediction i.e. if the hypothesis is true, something specific must happen. E.g. if ice is less dense than water, ice must float on water.**

- **Deduction** – from **general hypothesis to specific data**. See above diagram.
 - Specifying a conclusion about something specific on the basis of a general rule. Conclusion follows inevitably from the rule.
 - Suppose we know that all swans in a park are white. So, we can conclude that next swan we see will definitely be white.
- **Induction** – from **specific data to hypothesis**. See above diagram.
 - If all swans we see in a park are white, we may conclude all swans are white. So, induction may not always be true, even though it is based on large data and specific observations. So, induction is often plausible but not inescapable.
- **Scientific method**
 - A method to provide **good justification for the belief that a claim or statement (hypothesis) is true or false**.
- **Goal of science**
 - To **find good justification for the belief that a claim or statement (hypothesis) is true or false, by means of a conditional argument**.

Conditional claim – it has the form as follows. This logic is same across scientific disciplines.

- **If A (antecedent) then B (consequent)**
- E.g. If you study, you'll pass the exam
- **Denying the consequent** – If not B then not A is also true.
- **Affirming the antecedent** – If A then B.
- For example, suppose a prof said, "if you all pass the exam(A), I'll buy you pizza(B)." Assuming the prof is honest and truthful, we can deduce:
 - If A then B – If students passed the exam, we can deduce, they will get pizza. True. **Affirming the antecedent**. So, whenever A is True, B must be True. **VALID DEDUCTION**.
 - If not B then not A – if students did not get the pizza, we can deduce, it means not all of them passed the exam. **True. Denying the consequent. Whenever B is False, A must be False. VALID DEDUCTION**.
 - If not A, then not B is also **not necessarily true**. It can be possible that, not all students passed the exam, yet they had pizza. So, just looking at the fact that all students passed or not, we cannot deduce that they must not have had a pizza. **Can be True or False. Denying the antecedent. Whenever A is False, we cannot deduce if B is True or False. INVALID DEDUCTION**
 - If B then A is not necessarily true. So, we cannot deduce that if students got the pizza, they all passed the exam. Professor only said he'll buy the pizza if they pass, but didn't say, they'll get the pizza IF AND ONLY if they pass. So, we can deduce if B is True, A may be True or False. Denying the consequent. **Affirming the consequent. Whenever B is True, we cannot deduce if A was True or False. INVALID DEDUCTION**.

Truth Table of – If A then B

A	B	NOT B	NOT A	A implies B	NOT B implies NOT A
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

H= Hypothesis

IC= Initial Conditions (e.g., description of state at beginning of experiment)

AA= Auxiliary Assumptions (e.g., registration is reliable)

P= Prediction (follows deductively from H and IC and AA)

- 2 conditions for a good test of the hypothesis
 - If **[H and IC and AA] then P.**
 - **Refuting a theoretical hypothesis (deductively).** If we can prove that a **Prediction is False** then it means **not [H and IC and AA]**. Which means either or all elements – H, IC and AA – are False. Since we **ensured** that in an experiment **IC** and **AA** are **definitely True**, it must mean, **H is False**.
 - If **[not-H and IC and AA] then probably not-P.** This is just as indirectly showing, **IF P then H.**
 - **Justifying a theoretical hypothesis (inductively).** If we can show a **prediction to be True**, we will prove that **H must be True**.
 - Still, if we showed that **Prediction is True**, there is no guarantee that **Hypothesis is definitely True**.
 - **Scientists have hypothesis and deduce predictions on basis of Hypothesis. Data either supports or rejects a hypothesis.**
 - **When Data supports the Hypothesis**, the reasoning is provided using **second condition** i.e. since Data showed Prediction to be True, **Hypothesis must be True**. But we are not absolutely sure about the hypothesis.
 - **If Data rejects the hypothesis, the first condition is met.** Since P is False, H must definitely be False. In this case, we are sure that Hypothesis was wrong, and we move on to the next hypothesis.

Falsifiability criterion

- People believe statements which
 - a. Can be **demonstrated to be false**.
 - b. Can be **demonstrated to be most likely true**.
 - c. Cannot be demonstrated to be false or true.
- Science can handle (a) and (b) type of statements, but not (c). Therefore **hypothesis must have some potential falsifiability**.
- More **precise the prediction**, the **more impressive the theory** is.

- Karl Popper 1961 rightly said, scientific hypotheses need to generate predictions that might not be true. This is the falsifiability criterion.
- Moreover, when we know we are wrong, we can learn and correct our mistakes.
- Evolution theory of Darwin is accepted because it generates predictions which can be falsified. But intelligent design theory creates no predictions, thus, science cannot test this theory, and it is a matter of faith, to accept this claim or reject it.

*“Those among us who are unwilling to expose their ideas to the hazard of refutation do not take part in the game of science.”
(Popper, 1934)*

*“The essence of science is making mistakes in public.”
(Dennet, 1995)*

Psychology Notes Topic 10

By Somansh – author of “How to write a self-help book”

Validity

- Basic knowledge
- Construct validity
- External validity
- Internal validity
- Statistical validity

Condition

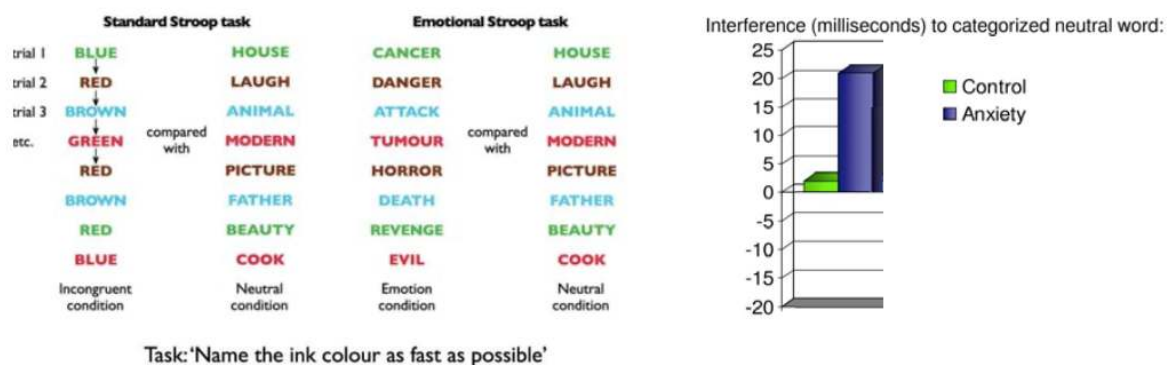
Incongruent	Neutral	Congruent
blue	zyp	green
brown	htklt	red
red	ygtde	blue
blue	gtopr	red
red	frangf	brown
green	hyfop	blue
Time _i	Time _n	Time _c

$$\text{Time}_i > \text{Time}_n > \text{Time}_c$$

Basic Knowledge

- **Stroop effect** – Time required in incongruent condition is greater than neutral. Time required in neutral condition is greater than congruent condition.
- **Independent variables** – They are manipulated variables. Here it was Congruence, which was varied as: Congruent, Neutral, and Incongruent states.
- **Dependent variables** – corresponds to the measured variable. It is a variable whose value depends on another variable (independent variable). Here it was the reaction time. Typically we alter the independent variables and measure the change in dependent variables.
- **Control variables or constants** – the variables that are kept constant on purpose. Here, it was the colours that were constant. Need it to prevent confounding.
- **Confounding variable** – it changes simultaneously with independent variable. Here, time of testing and order was: Incongruent, Neutral, and Congruent states. So, we are not sure if the response time reduced due to independent variable (the Incongruent, Neutral, and Congruent states) or the congruent variable (the order of presenting the variables). So, we should shift the order to remove confounding variable.

- **Random error** – Noise during testing. Presence of random variable affects the dependent variable measurement across all levels of the independent variables. In an ideal experiment, one or more variables are manipulated (independent variables) while other important variables are kept unchanged (control variables).
- So, to measure an effect of a variable on dependent variable, other variables should not be manipulated, otherwise we get confounding variables. Also minimise random errors.
- **Anxiety as a variable of interest**



Example

- **Is Anxiety a variable of interest? Do anxious people have an interest bias for anxiety inducing words.**
- Anxious people focussed more strongly on threatening words than normal people.
- But Anxiety is an abstract concept, what are we even measuring here. This is where operational and conceptual variables comes in.
- **Construct or Conceptual variable** – here anxiety is a conceptual variable. It is an abstract (theoretical) concept. **Conceptual variable is operationalised in a study via operational variable**
- **Operational Variable** – The specific way in which a construct is manipulated or measured in a study. "How did the researcher manipulate or measure the construct. "How did the researcher measure and manipulate the construct?"
 - Here, anxiety was operationalised using a standardised test. So, test scores were the operational variable to measure anxiety.

Goal of science

- To find good justification for the belief that is a specific claim or statement (hypothesis) is true or false.

Three claims

↗ low internal validity

- **Frequency claims** – focus on one variable. Often derived and justified on the basis of descriptive research. E.g. 4 in 30 teens text while driving.
- **Association claim** – focus on the relationship (association, correlation, covariation) of two variables. Justified on basis of correlational research. E.g. Single people eat fewer vegetables.

- **Causal claim – focus on causation.** Change in one variable is responsible change in another variable. Needs experimental research. E.g. Music lessons increase IQ.

Construct Validity

- **An indicator of how well a conceptual variable is measured or manipulated in a study.** How well a construct is operationalised.
 - E.g. What is the influence of familiarity on attractiveness. Do people like what they know. How it was operationalised – do people like original or mirror image of Monalisa. Familiarity was operationalised by using familiar (Original) and unfamiliar (mirror image). Attractiveness was operationalised by scoring how much people liked the picture.
- Threats
 - **Inadequate operational definition.** E.g. does anxiety test measure anxiety or low self-confidence? Is IQ really about solving math problems?
 - **Mono-operation bias – a single test may not be good enough.** Mental workload is used to see if a job is mentally stressful. We can use heart-rate, ask people, hormone level measurements. If we use just one, it may not be good enough.
- Example
 - A researcher is interested in the influence of culture on the ability to solve complex problems. He performs a quasi- experiment in which he measures how long people from different countries need to find their way to the baggage belts in Schiphol airport. The study shows that Dutch travellers are generally faster to find their way in Schiphol airport than travellers from other countries. The conclusion of his research is that the Dutch culture contributes to the ability to solve complex problems. What can you tell about the construct validity in this study?
 - **Culture** is operationalised by country of origin, but is confounded with finding way around Schiphol airport which is a Dutch airport. Finding baggage well is not a good indicator to solving complex problems.

External Validity

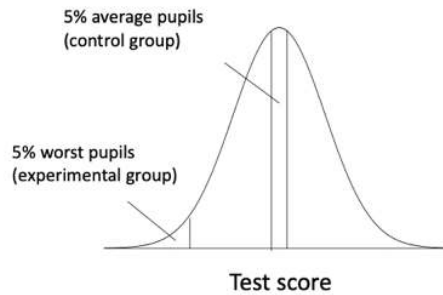


- **External Validity** is an indication of how well the results of a study generalize to, or represent, individuals, settings, places, and times (contexts) besides those in the study itself
- Artificial sweeteners enhance cancer risk. But this was done in rats. It is externally valid if same is observed in humans.
- Threats
 - **Selection bias** – when we want to make a claim on a population. So, every type of member must be in the study. But sometimes sample is biased. Study on marijuana may attract people who smoke marijuana. Thus, whole population will not be represented. So, the study represents only a group of that population, and it cannot be generalized for that population.
 - **Setting** – study setting is different from other settings in real life. Participants may react differently. E.g. responses may be socially acceptable ones, not real opinions.
 - **Study place** different from other places. Study in village and city may vary the results.
 - **Time of study** is different from other times. Study after a recent scandal may produce different results.
- Example
 - A traffic psychologist is asked to test which of several alternative signs for "carpool lane" is best understood. He performs an experiment in which he presents the various alternative signs to a group of students. The students have the task to indicate which sign is the best sign for carpool lane. What can you tell about the external validity of this study?
 - Not much external validity

Internal Validity

- The extent to which it is possible to rule out alternative explanations for a causal relationship between two variables. Usually for studies outside lab conditions, and inadequate research design.
- Threats, and there is already improper research design
 - **Maturation** – change in behaviour that emerges spontaneously.
 - **History** – change in external factor that may affect outcome of study
 - **Testing** – change in behaviour as a result of taking the same test twice.
 - **Instrumentation** – errors in measuring instrument
 - Examples. Impact of study program on math ability. Issues
 - Pupils may have matured independently, regardless of program.
 - Also, there's no control group that was not exposed to the study program. Plus, this group should be similar to experimental group.
 - Students may have improved just by giving the same test twice.
 - **Regression to the mean** – people or things regress to the mean eventually. Maybe the study measured the people when they were away from the mean. Maybe the person giving a test was ill on test day. Maybe the group of people were facing situations that lowered their score, they will revert to mean in the next test.

Internal validity



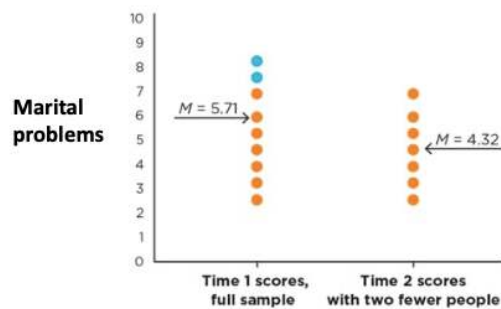
Score on test before the introduction of the new learning method	
experimental group	40
control group	80
Score on test after the introduction of the new learning method	
experimental group	50
control group	80

- **Attrition** – reduction in participant numbers as people drop out of the study.

E.g. Study on effect of couple counselling on divorce. But, couples seeking divorce may drop out of a study, thus, distorting the study or change the average score

- **Attrition / Experimental Mortality**

- *e.g., study on the effect of couple counseling on the divorce rate*



- **Selection effects** – a difference in participants at different levels in different

conditions. E.g. **Simpson's paradox**. Many more men in pill group than women. Men also healed more easily than women, regardless of the pill. So, there was a **confounded by gender differences in the group, and selection bias**.

Simpson paradox

	no headache	headache	% cured
N=80			
pill (N=40)	20	20	50
no pill (N=40)	16	24	40
men			
pill (N=30)	18	12	60
no pill (N=10)	7	3	70
women			
pill (N=10)	2	8	20
no pill (N=30)	9	21	30

-

- **Design confounds** – the presence of another variable that unintentionally varies systematically with the independent variable. E.g. Stroop effect. E.g. study of bowl size on calories consumed. But if quality of pasta in different bowls, the design of study was poor.
- **Observer bias** – caused by researcher's expectations. E.g. a doctor might be biased to expect a new drug to work on patients. The results may be intentionally or unintentionally altered by the researcher.
- **Demand characteristics** – participants may behave in certain way, to make the study a success.
- **Observer bias and demand characteristics** can be removed by using double blind tests where researchers and participants have no idea if they are in a control group, or what is the research all about.
- **Placebo effect** – that arises when people believe they are receiving a valid treatment, and improve their health, even though they didn't receive active ingredients in the drug. Note: Placebo is real, not an imaginary effect. In new medicine trials, placebo effect is removed by using three groups: experiment group (get real medicine), placebo group (get fake meds), control group (get nothing).
- Example
 - A manufacturer of skin care products brings a new day cream on the market. He claims that this day cream leads to significant skin improvement in women with skin problems. He justifies his claims on the basis of a study using a sample of 50 women with a problematic skin and 50 women with a normal skin. The purity of the skin was precisely measured in both groups at the beginning of the study (using a purity index).

Following both groups applied the new day cream on a daily basis for two weeks. The results were as follows:

	Average purity index	
	Before treatment	After treatment
Group with problem skin	40	60
Group with normal skin	80	80

The results show a clear improvement in the condition of the skin after treatment compared to before in the group with skin problems. There were no effects in the group with normal skin.

The conclusion of the study was that daily use of the day cream has a positive impact on the quality of a problematic skin. What can you say about the internal validity of this study?

- The groups were not chosen randomly. May've been affected by regression to the mean. Needs control group for each study group.
- **Statistical validity** – extent to which a study's statistical conclusions are accurate and reasonable.
 - They require certain assumptions to be true, and not violated.

- Fishing and the error rate problem. Seeing things that aren't there. If researchers fish for a hypothesis in their data.
- Low statistical power – probability of correctly rejecting the null hypothesis when it is true. In low statistical power experiments, this probability of rejecting is low. Happens when looking for tiny effects or large groups, and unreliability of measures.

Psychology Notes Topic 11

By Somansh

Overview

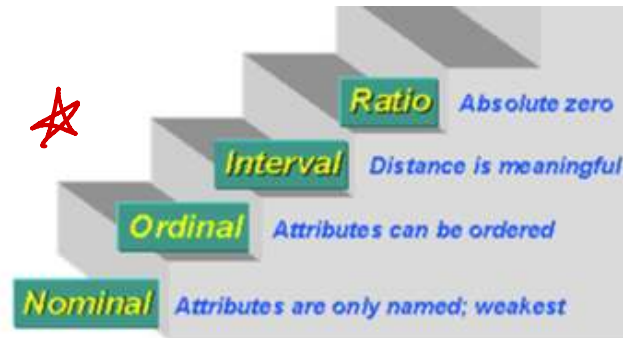
- Measurement
- Reliability
- Descriptive research

Measurement

- **Measurement is a systematic way to assign numbers or names to objects and their features.**
- When we perform a measurement, we can make distinction between different measurement scales. It can be:
 - **Nominal scale** – a scale in which we assign numbers or names to different categories
 - E.g. If we want a variable to have value as either 'man' or 'woman' as the value, then the variable may be called 'gender'. The variable 'gender' has 2 possible values 'man' or 'woman' but we can also convert it to numerical values 0 and 1, or give it symbol such as, M and W.
 - So, nominal scale only indicates that categories are different, but does not information on how much the difference is.
 - The book mentions it as categorical variables.
 - **Ordinal scale** – ranked order.
 - Assign numbers which are meaningful in terms of quality.
 - E.g. number of stars for a restaurant indicate their quality.
 - However, the difference between 0 and 1 star is not equal to 3 and 2 stars. There is a ranking here, but that does not mean that the quality is even.
 - Imagine Usain Bolt, David Beckham and a Tortoise running in a race. Usain may be 2 seconds faster than Beckham, but Beckham may be 50 minutes faster than the tortoise. Still, their ranking would be 1,2, and 3.
 - **Interval scale** – equal intervals. *NO absolute zero*
 - Assign numbers that are meaningful in terms of differences.
 - Like temperature differences can be used as values in interval scale.
 - But we cannot compare intervals without an absolute zero. We cannot say 20 degree Celsius is twice as hot as 10 degree Celsius because zero degree Celsius is not absolute zero, zero kelvin is the absolute zero. So, 20 degree Celsius is just 293 K and 10 degree Celsius is 283 K.
 - No absolute zero here.
 - **Ratio scale** – there is an absolute zero in the ratio scale, unlike interval scale above.
 - E.g. measuring reaction times or measuring weight.

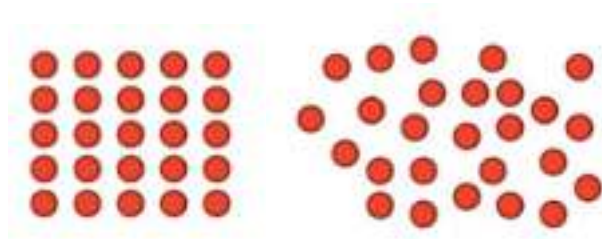
has absolute zero

- **Note:** Book mentions Ratio, interval and ordinal scale as just quantitative scale. The scales defined in notes are better.



■ Importance of measurement scales

- **Convey different types of information**
- **Type of scale determines the possible statistical operations** that can be done on a scale.
 - E.g. we cannot use parametric statistical tests on certain scales.
- **Determine the possible conclusions.**
 - E.g. concluding one person with IQ 150 is twice as smarter as other person with IQ 75, because **IQ is an interval scale, with no defined absolute zero.**



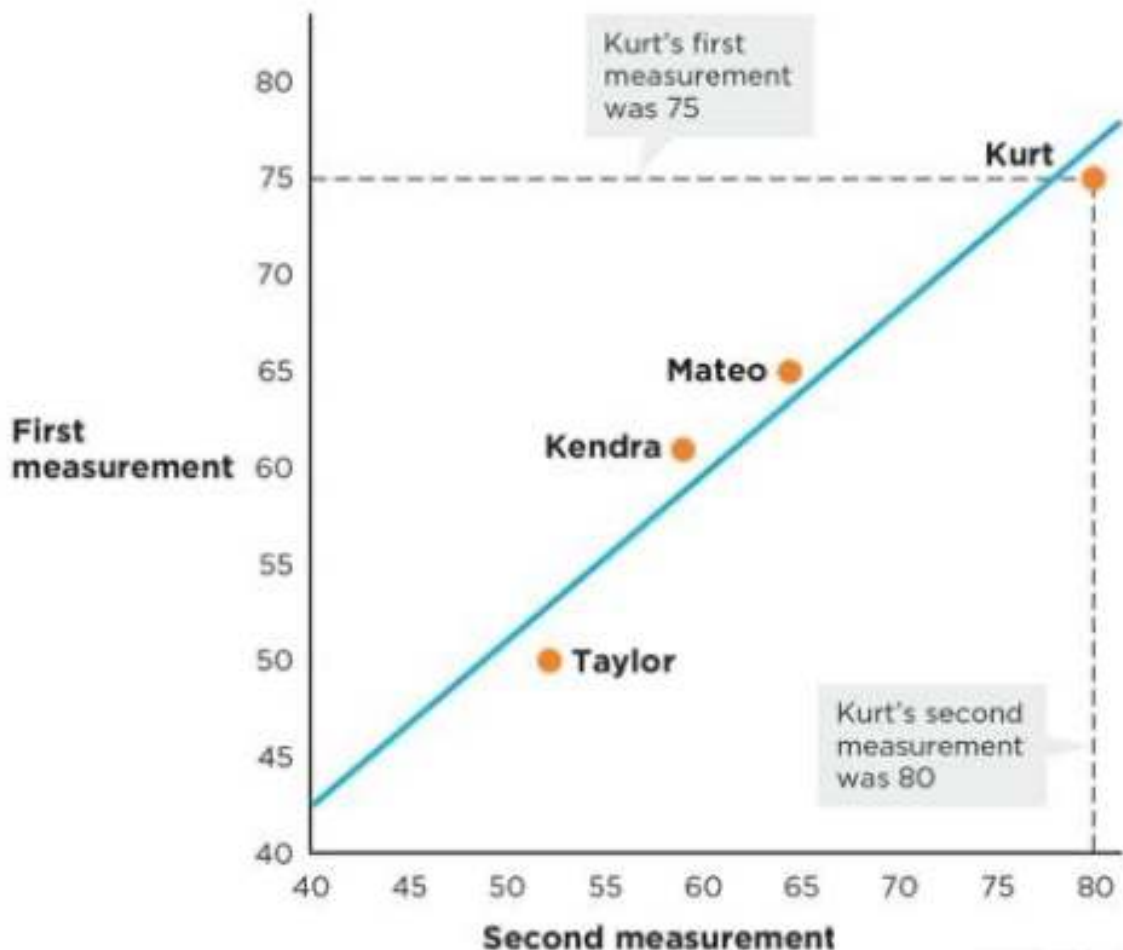
Reliability

- Example of Red dots. In first case almost everyone reports 25 dots, and very little variation in answer. In second case, some say 21, some 29, assume average guess is again 25 in second case.
- Which average value do we trust more? **The first one because there was less variability in answers. So, variability informs us about reliability.**
- **Observed score = true score + error score**
 - **Observed score is the measured value**
 - **True score is the real value**
 - **Error score is the difference between the measured value and the real value which is affected by**
 - **The test situation**
 - **The person**
- **The closer is the measured value is to the true value, the more reliable it is.**
- So, **reliability is the**
 - **Consistency of measurement. Error or variability should be minimal.**
 - Without reliability, we cannot say if a conclusion is valid or not. So, **reliability is the prime condition for validity.**

- **Types of reliability**

- **Test reliability** – a test aims to measure a relatively stable quality for e.g. IQ.
 - **Test reliability indicates the extent to which a stable score can be obtained. Usually obtained on the basis of correlation**
 - **Test-retest reliability** is the **consistency in results we obtain**, every time the test is used. Every time we compare the results of earlier and later tests, we get test-retest reliability. If the first and second scores are very different, we say that the test is not reliable.
 - We can **use scatterplot** to see if there is **variability between first and second reliability.**

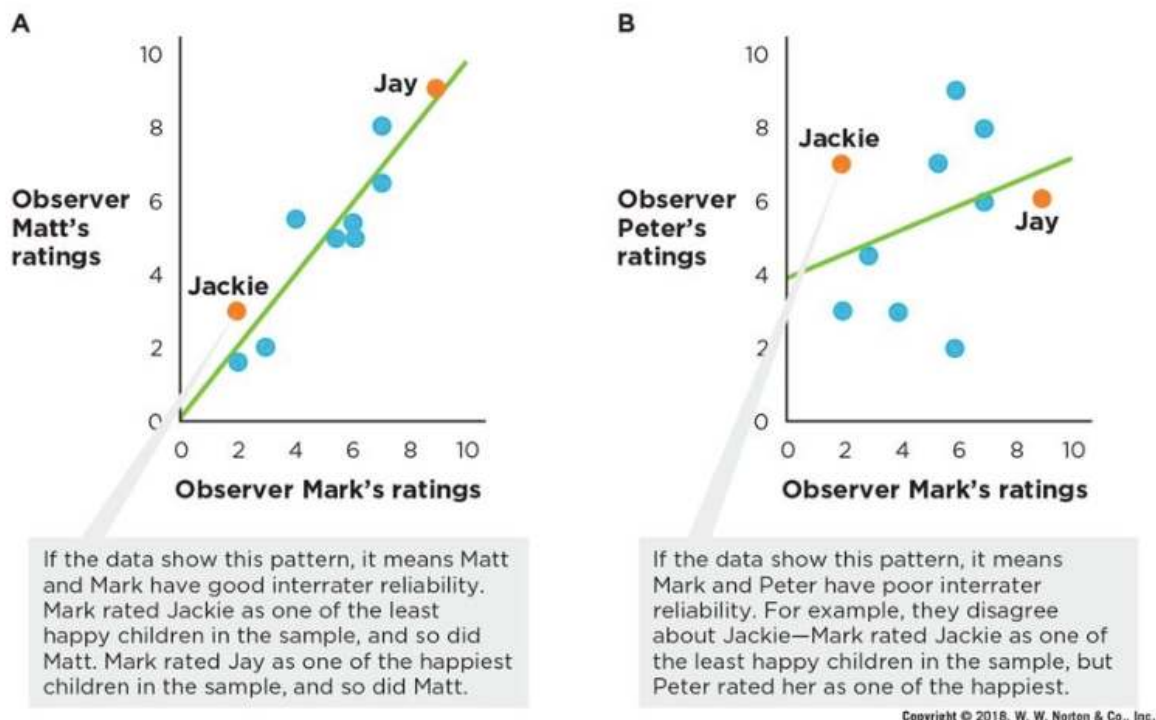
Reliability



- Example – **Self-report answers are notoriously unreliable.** Students self-reporting how much time they spent on a chapter vary widely in test-reliability test. This is due to our memory being imperfect.
- often, **practice effect** alters the test scores as participants have **already had the practice of doing those questions in the test.** During

retest, participants may score more just because they knew what to do this time. So, retests must be created to avoid this, without altering the test too much.

- **Internal reliability –**
 - **Consistency in the pattern of answers in a measure that contains several items.**
 - **Split-half reliability** – divide the test halves arbitrarily. If the scores on one half are almost equal to the other half, the test is reliable.
 - **Cronbach's alpha** – collect data on a test scale from large sample of participants, then we compute all possible correlations among all the items. The formula for Cronbach's alpha returns one number from average of inter-item correlation (inter-item correlation/ number of items). If the number is high, the test is reliable.
- **Inter-rater reliability**
 - **Measures the degree of similarity between ratings of two observers of the same behavior.**
 - E.g. We want to find leadership qualities of a person. Two observers are required to score the person on certain parameters. If the scores vary widely, the inter-rater reliability of such a test is low. So, scoring items should have directly observable specific parameters, this way reliability of test will be higher.



- Example – the 9/11 witness reports are highly variable, and have very low inter-rater reliability.
- **Experimental reliability** – identifies the variability between the results obtained in an experiment. It is calculated by replicating the experiment, to be sure that outcomes will be the same.

Descriptive Research

- **Enumerates what behaviors occur, and in what quantity and frequency**
- **Types**
 - **Observational research** – The process of watching people or animals and systematically recording how they behave or what they are doing.
 - **Survey research** – A method of posing questions to people on the telephone, in personal interviews, on written questionnaires, or via the internet.
- **Observational research**
 - Why we do it – to describe nature, like Darwin's expedition to observe natural life, and found a hypothesis on evolution.
 - **Bystander effect** – people are less likely to help, when they are in a crowd rather than when they are alone. John Darling generated the hypothesis of bystander effect. As per Darling, presence of others can alter a person's experience of the events. If people ignore a person in need, we assume the person must be ok since everyone else is ignoring them. Easier to pass the buck or diffusion of responsibility.
 - Two ideas conflict – the idea that do whatever everyone else is doing, and the urge to help. Another problem can be fear of getting involved.
 - When someone tries to help, the people around tend to help.
 - **Problems**
 - **People are not always good at observation.** So, threat of low inter-rater reliability.
 - **Low internal validity** – the extent to which it is possible to rule out alternative explanations for causal claims. But observational research does not allow causal conclusions but it generates frequency claims. And frequency claims have low internal validity.
 - **Threats to construct validity** – indication of how a conceptual variable is measured or manipulated
 - **Observer bias** – when observer's expectations influence their observation. With code books, practice and training, the observer bias can be reduced. Also, focus on observing clear and specific things.
 - **Observer effect** – when participants change their behavior in presence of the observer in the direction of observer's expectations.
 - **Reactivity** – a change in behavior of participants when they know they are being watched. This may or may not be in the direction of observer's expectations. But a change is surely there. This can be avoided by unobtrusive observation.
 - One-way screen, hidden cameras can be used.
 - Another strategy is participant observation. Wait long enough that people stop changing behavior in observer's presence. Jane Goodall stayed so long with primates that they stopped bothering about her presence.
 - Observe not the behavior but the outcome, like observing the garbage left after a party to find any drug use.

- **Survey research – formats**
 - **Question formats**
 - **Open ended questions** – such as “What do you think about Computational Thinking class ?”
 - **Forced choice format** – like “Did you like this class or not ?” Yes/No
 - **Likert scale** – people are given a statement and given a scale on 1 to 5 to show agreement or disagreement
 - **Semantic differential format** – the participants are asked to rate a target object on a numerical scale, anchored by adjectives such as ‘very easy’ to ‘very difficult’
 - **Problems in survey research**
 - **Low internal validity** – because they are frequency claims
 - **Threats to construct validity** –
 - **Question wording** may alter the interpretation
 - **Leading question** may guide the participant to alter their response in the next question.
 - **Double barreled questions** – two questions at once like “Do you enjoy swimming and wearing sunscreen?”
 - **Double negative worded questions.** “People who don’t drive with suspended license should never be punished?”
 - **Question order** – earlier questions may influence later answers. So, we need to make alternative versions of a survey with different orders.
 - **Use of response sets** – like answering every question with strongly agree or taking safe position by choosing middle option most of the time.
 - People may try **to fake being good or bad**. So, they fake their response. Create options that are equally socially accepted or tell the participants that the survey is anonymous.
 - **People are not always accurate in recalling memory or expressing their emotions or thoughts.**
- **Certain aspects of Descriptive research**
 - It leads to frequency claims.
 - Internal validity of frequency claims is low
 - It is important to question:
 - How well was each variable measured or how good is construct validity.
 - Can the results generalize or is the study externally valid.
 - Do the numbers support the conclusion
- **Threats to external validity of descriptive research.**
 - **Biased or unrepresentative sample** – Even though these claims are made for an entire population, but they always stem from a sample population. The sample must be random, representative and unbiased. Only then we can generalize it. How a sample gets distorted:
 - **Convenience** – the study may use only those group of people who are interested in participating. This leaves out many groups of people.

- **Self-selection** – volunteers may only be people of certain personality type.
- **How to get representative sample and improve validity**
 - **Random sample**
 - **Rewards for people unwilling to participate.** But again, different rewards can alter behavior.
 - **Replicate the study with different set of participants.** This can reduce accidental factors
- **Threats to statistical validity**
 - Taking a **sample too small.** Small samples lead to larger margin of error.

TABLE 7.4

Margins of Error Associated with Different Random Sample Sizes

IF THE PERCENTAGE IS ESTIMATED ON A RANDOM SAMPLE OF SIZE	MARGIN OF ERROR ON THE PERCENTAGE IS
2,000	Plus or minus 2%
1,500	Plus or minus 3%
1,000	Plus or minus 3%
500	Plus or minus 4%
200	Plus or minus 7%
100	Plus or minus 10%
50	Plus or minus 10%

Note: Margin of error varies as a function of sample size and the percentage result of the poll. In this table, estimates were based on a 50% polling result (e.g., if 50% of people supported a candidate).

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Psychology Notes Topic 12

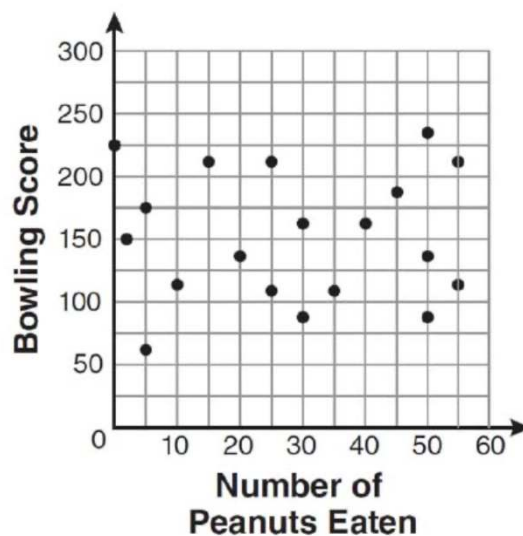
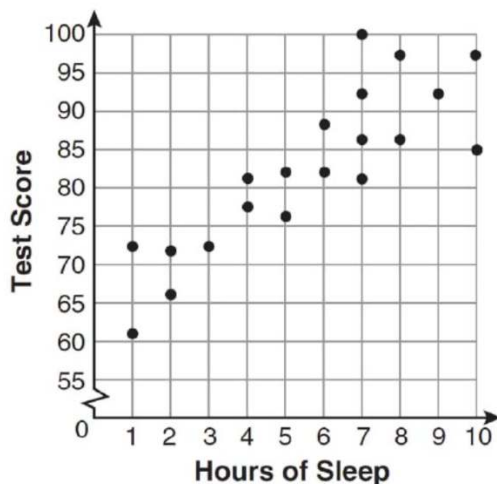
By Somansh – no jokes in exam week

Correlations and Experiment

- Correlational research
- Basics of experimenting
- Experimental research

Correlational Research

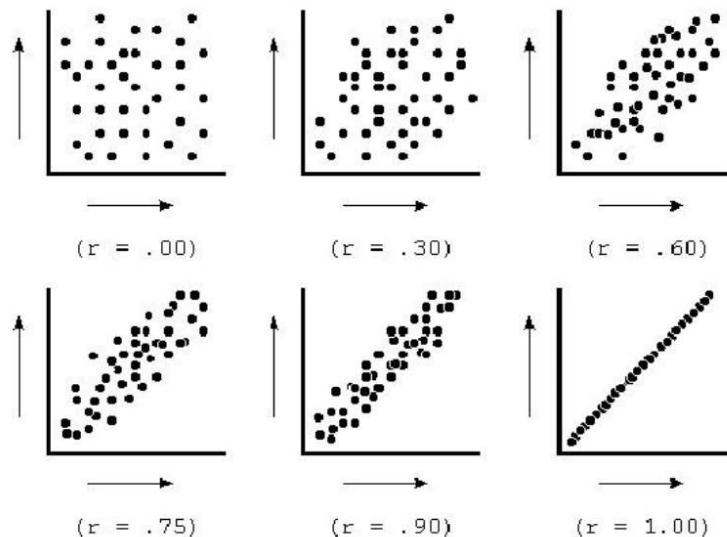
- It **generates association claims**. Association claims are **not supported by particular statistical tests or graphs**, but, by research design where all the variables are measured and **none of the variables are manipulated. It just shows correlation** between two variables, in a sample.
- Examples
 - "People who often watch violence on TV are more violent than others"
 - "Mental disorders are more common among unemployed people than among people with a job"
 - "Couples who meet online have better marriages"
 - "In countries with a high number of divorces, people generally have a high life expectancy"
- It **does not allow us to draw a causal conclusion very easily.**
- The correlation is **measured by a correlation coefficient (r)**
 - For e.g. **Pearson's Product -moment correlation**
 - It is a **measure indicating the degree of linear relation between two variables.**
 - It varies between **-1.00 to +1.00**
 - **Value indicates the strength of correlation**
 - **Sign indicates direction of the relation**
 - **Scatter diagram is used to visualise correlation.** It helps us see, if there is correlation or not.



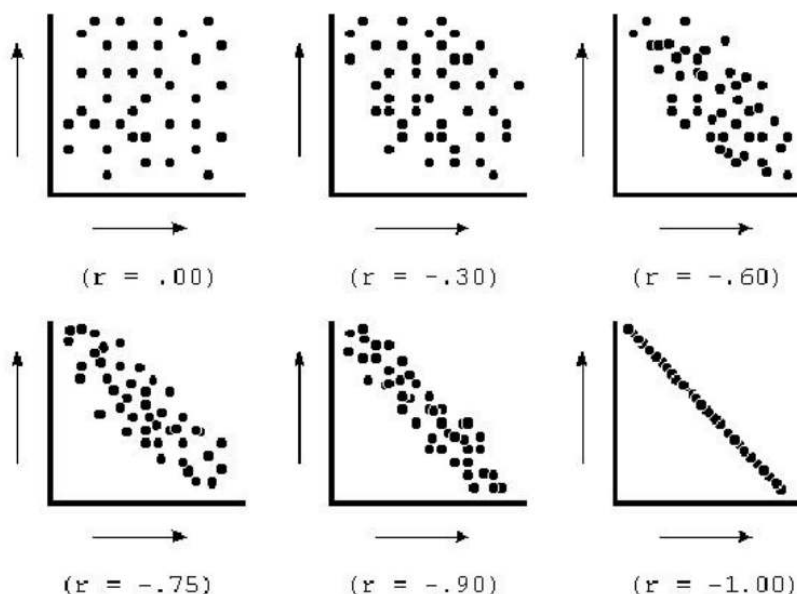
- **Hours of sleep** – Since higher sleep is linked to test score, we can see they are correlated. It allows us to predict one variable with the help

of another variable if they are correlated. It may be that good sleep causes a good score, but we can't say that from correlation study.

- This way, **correlation method is superior to the descriptive method**, as we can
 1. **make as assessment of the degree of relationship** between two variables.
 2. **Predict value of one variable**, based on value of other variable.
- **Bowling** – it has no correlation with peanuts eaten.

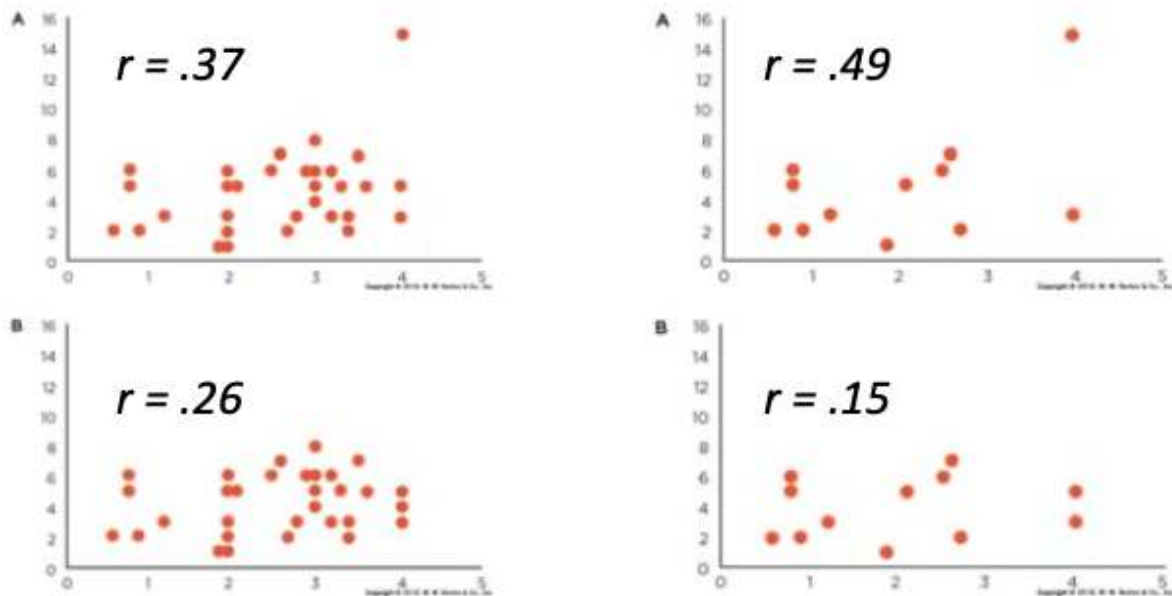


- In above graphs, value of r increases as correlation strengthens, but sign remains positive. In graphs below, correlation is negative, i.e. when one variable increases, other variable decreases.

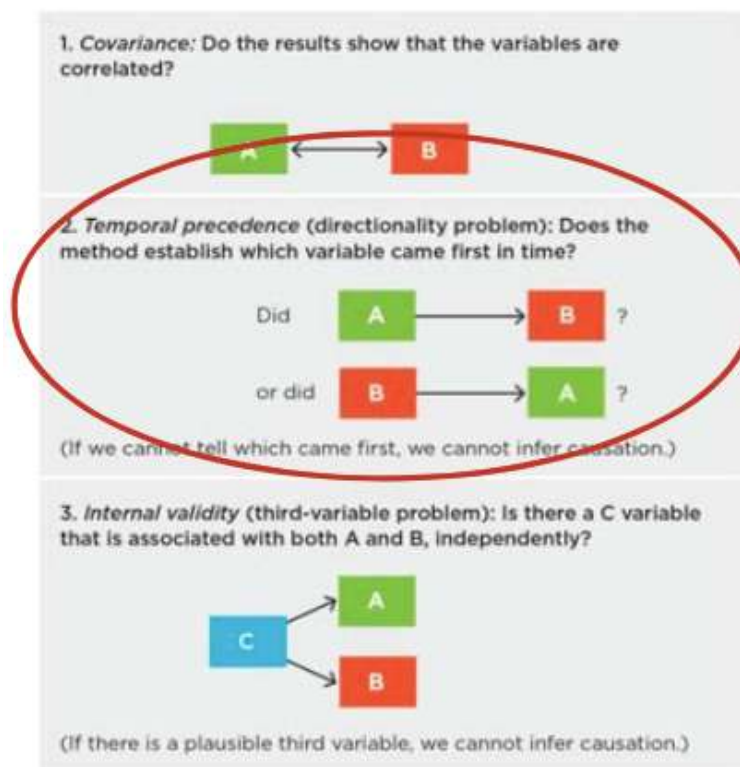


- **Statistical Validity** – does the data support the claim?

- It is the extent to which the statistical conclusions derived from the study are accurate and reasonable.
- **Correlation strength** – It is more statistically valid if the correlation is stronger. So, our prediction will also be more accurate.
- **Outliers** – if there are outliers, they may significantly alter the correlation coefficient, especially in small sample studies. This affects statistical validity.



- Is the correlation statistically significant?
 - It indicates the chance that the results from a sample could've come from a population in which there's no association or very low (0.05 or lower)
 - Probability to obtain statistically significant results depends on strength of correlation, effect size? and the sample size.
- **Correlation PROBLEMS**
 - **directionality problem** – Which Var. affects the other Var.?
 - Health is positively correlated to hours spent on sports per week.
 - Does this imply that sport improve health or health increases hours spent on sports.
 - We can't say for sure, so there is unknown direction of causation. We call this directionality problem.
 - **third variable problem**
 - Number of ice creams sold per month is positively correlated to people drowning. Such correlations are called spurious correlation. Needs a 3rd Var
 - Does ice cream increase drowning rate or drowning increases ice cream demand? Both ideas are absurd.
 - Third variable – temperature – explains it well. In summers, people go to the beach or swim more often, so drown more often. Also, in summers, ice cream sales increase.
 - This is third variable problem

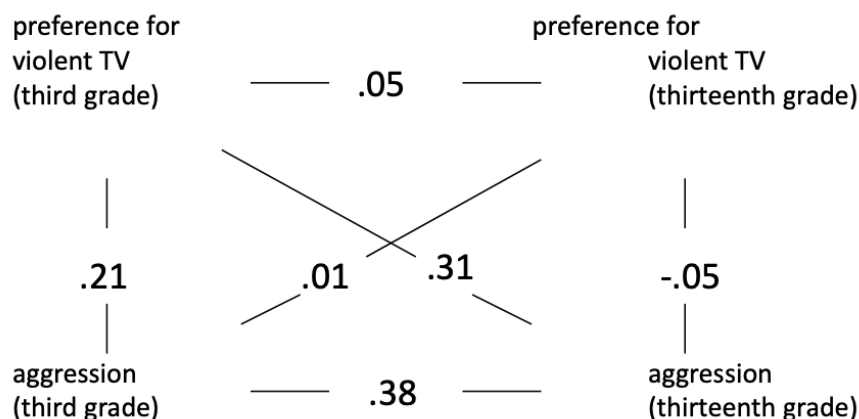


▪ **Criteria to establish causation – three criteria**

- **Covariance** – Needs to be an association between causing variable and affected variable. It's there in Correlational research.
- **Temporal precedence** – Did A happen first, or B happened first. If A we can't tell A came first or B, we can't infer causation. Not very easy to establish in Correlational research.
 - Sometimes researchers try to establish temporal precedence, on basis of correlational research, by using cross-lagged-panel procedure.

by using the cross-lagged-panel procedure

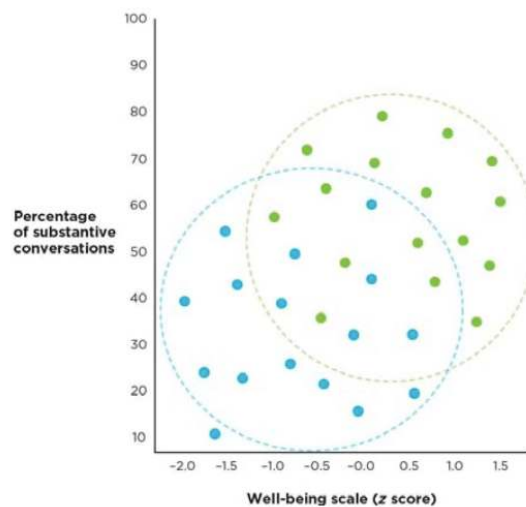
- A → B if $r(A_{t1}-B_{t2}) > r(A_{t1}-B_{t1})$



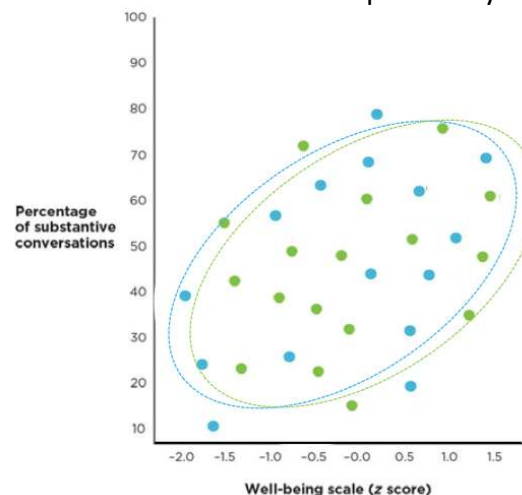
- We look at patterns of correlation over time, so we can establish temporal precedence, and find which one came first. We assume that

when there's a causal relation, correlation between variables V1 (Preference for Violent TV at 3rd grade) at time t_1 , and V2 (Aggression at 13th grade) at time t_2 is larger than correlation between variable V1 and V3 (Aggression at 3rd grade) at time t_1 . This means that child was less correlated with violence earlier, but over time, became more correlated with violence, and this was due to violent TV. But the study takes a lot of time.

- **Internal validity** – Is there a third variable, independently causing A and B or not.
 - This issue is dealt by separating the participants into subgroups on the basis of reasonable 3rd variable.
 - Example



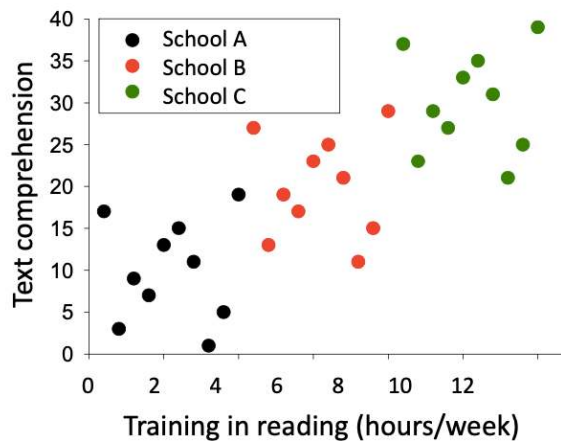
1. Are people, who engage in deep talk are happier. But 3rd variable may be, education. Educated people may be happier and talk more deeply.
2. So, 2 groups were created, green were more educated, and blue were less educated.
3. But within those subgroups, correlation was very weak. So, the overall pattern was likely due to 3rd variable i.e. education.
4. Below graph is how data would look like, if education didn't affect the variables independently.



5. Moreover, there can be another 3rd variable like wealth, location etc. that influenced the variables.

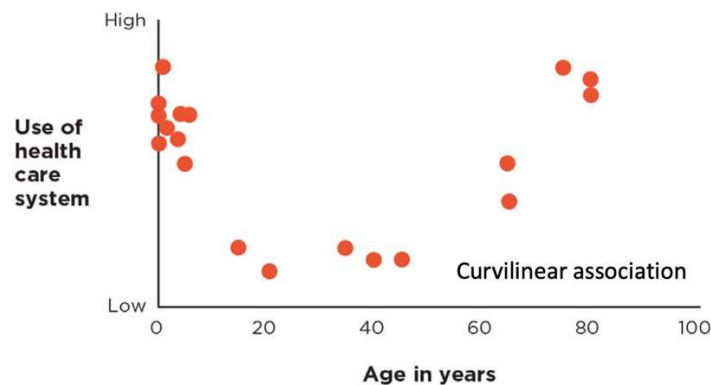
- What if correlation is zero – is it possible to rule out a causal relation if correlation is zero.

- Not necessarily, as it may be due to restricted/truncated range.



Per school: $r = .02$;
Across schools: $r = .78$

- If we had limited the research to school A, the correlation would be zero. But broadening the range to schools where kids got training in reading, correlation would improve.
- Relationship between variables may not be linear. The relation may be curvilinear
 - In the graph below, correlation value is 0.01, yet, there is a strong correlation of age and healthcare use.



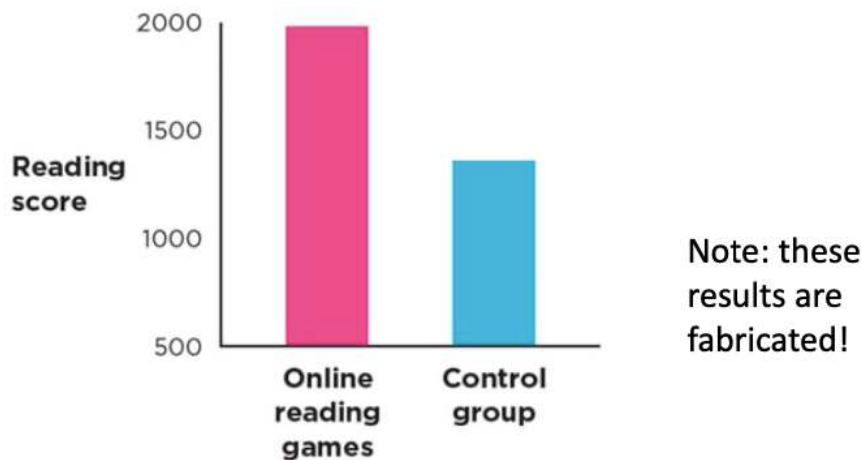
- In conclusion
 - The presence of a correlation indicates an association but does not indicate causality!
 - What if the correlation is zero?
 - Restriction of range (or truncated range)
 - Non-linear relationship
 - No association
- The internal validity of associations claims is low
 - It is important to question how well each variable was measured (construct validity)
 - How well the results generalize (external validity)

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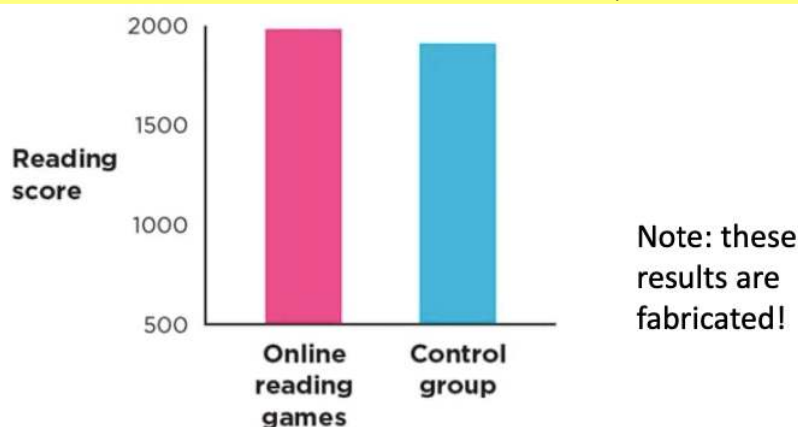
- How well the numbers support the conclusions (statistical validity)

The basics of experimenting

- An experiment occurs when a particular comparison is produced while other aspects of the situation are held constant.
- **A hypothetical experiment**
 - Do online games make kids (7 y/o) better readers
 - Independent variable – Reading Training
 - One group got online reading games. Other group (Control group) got no reading games.
 - Dependent variable – their reading ability. This was operationalized via reading test scores. Better score meant better reading ability.

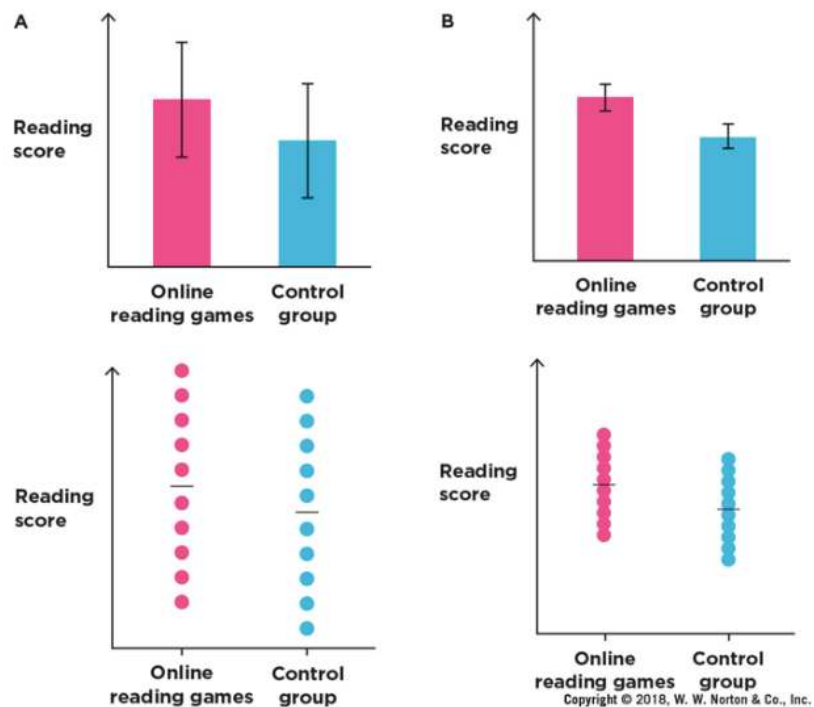


- This proves the claim that online games cause/enhance reading. This passes the causal tests as we have covariance (strong correlation of reading and online games); temporal precedence (games came before improvement); and Internal validity (having a control group takes care of randomness and 3rd variable problem).
- If there wasn't a difference between the results, we call it a null result.



- **Null result indicates –**
 - This may indicate that independent variable [here, online games] did not affect the dependent variable [Reading ability].

- The **manipulation was too weak**. Like, 1 week may not be enough to see an improvement in reading ability. Maybe it takes a month or two.
- The **measures** [the reading test] **may not be sensitive** to capture the change in reading ability
 - **Ceiling effect** – test was too easy. Even without learning much, everyone could score higher.
 - **Floor effect** – test was too hard. Even with learning, no one could score high.
- **Unreliable measurements** – may be due to too much **variability in measurement**.



- **Not enough power** – when we fail to find significant differences, as participants were too few.

Two Experiments

- Experiment by Brady, Porter, Conrad, & Mason (1958)
 - **Sometimes** an experiment may be internally valid in principle, it may need more precautions.
 - **Hypothesis** – people in charge are more in stress, and therefore have more stomach ulcers.
 - **Used** – monkeys to experiment. Monkeys were divided to active and passive groups. Pre-test was done to determine the group. Monkeys that rapidly pressed the button to avoid shock were assigned to active group. Others to passive group.
 - **Experiment** – exposed a pair of active and passive monkeys to electric shock. Only active monkey could stop the shock for both of them. Both monkeys were given same quality and quantity of shock, so there was no difference in the experience. Then measured ulcers in active monkeys. They concluded, management stress caused ulcers.

- **But**, the experiment was flawed, because monkeys were not assigned in groups randomly, but there was a pre-test filter. So, there was a confounding variable – What if the monkeys who pressed button more rapidly, were more prone to ulcers.
- **Bad experiment**
- **Experiment by Jarrard (1963)** – how different doses of LSD affects learning/learned behavior in rats.
 - **Gave** different doses (mg/kg) to trained rats – 0.00, 0.05, 0.10, 0.20, 0.40, and 0.80
 - Suppose he gave the doses on consecutive days and measured the learning behavior each time. But this is **not internally valid** experiment because
 - Susceptibility to LSD might change over every dose. The rats **may get used to LSD dose**, and it may not affect them at all after 3 days.
 - **Sequence, and time of giving LSD is a confounding variable.** What if at that time, rats also had food. Or in those days of experiment, some rats were diseased.
 - **But Jarrard was a smart bloke.** He kept changing the sequence and dosage across rats, in an ordinal precision.
 - This is a **good experiment** design where external effects and uncontrolled variation are minimized.
- How to **assign subjects to different levels of independent variables**
 - **Between-subjects design** (independent-groups design)
 - **Separate groups of subjects receive different levels of the independent variable(s).** Like Monkeys getting charge of button or not.
 - **Within-subjects design** (within-groups /repeated-measures design)
 - **All subjects receive all levels of the independent variable(s).** In Jarrard's experiment, all rats got the dose.
 - Examples

Examples

Independent variable	Dependent variable
Location of lesion in the cortex	% correct
Rotation speed of radar arm	% detected
Learning method	Performance on test
Odor	Score concerning the attractiveness of person



- In testing effect of location of lesions on percentage of visual search, we can't really give a person lesion or take it away for the experiment, so, here we choose Between subject design.
- In rotation speed of radar example – we use within-subject design. We use within-subject design, as the power (number of subjects) is higher here. Also, we can easily raise or reduce speed of rotation of the same radar, and study the effects. So, with 9 radars and 3 speeds, if we used between-subject design, we would have 3 radars for each speed. But in within-subject design, we get 9 radars for each speed.
- Learning method example – we need to choose between subject design. We can't undo a person's learning.
- Odour – we can change a person's odour easily with perfume. So, we use within-subjects design.

Psychology Notes Topic 13

By Somansh

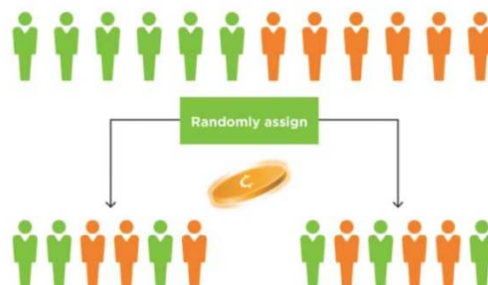
Overview

- Single factor designs
- Factorial designs
- Experimental research

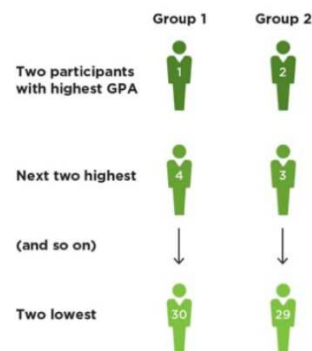
Single Factor design

- **Between-subjects design or independent group design** – If there are 2 levels, then there are 2 groups. One group may be experimental and other may be control group. Sometimes there may be 3 or more groups.
 - **Advantages**
 - There is no chance that one level of the independent variable affects performance under the other level of the independent variable.
 - For example, whatever drugs given to the experiment group won't affect the control group. Thus, any differences can be accounted to the effect of drug.
 - Similarly, while studying the effects of a reading training, the control group is not affected by the training session.
 - **Disadvantage**
 - There is a chance that groups differ from one another.
 - For e.g. a bad method of choosing groups may create control group consisting of older age, while experiment group has younger people.
 - This selection bias can be corrected by random assignment of participants to the group.
 - **Matching** – matching participants, based on certain variables which may be confounding variables like age or IQ. If IQ may confound an experiment, it will be helpful to create groups with people of similar IQ. **However**, people of similar IQ must be randomly assigned to groups.

Random assignment



Matching



- **Use**



- We use it **when independent variable has a permanent effect** – at least in the time frame of experiment – such as training to read, or having lesion.
- When **large risk of ordered effects** i.e. where participants' performance varies over time due to order of things such as practice effect. Or as in carryover effect, where doing one activity affects the upcoming activities.
 - **Practice effect means getting better at an activity over time due to practice, thus affecting the experiment.** So, if we are testing ability to solve a puzzle, the practice of solving the puzzle may improve results rather than just the independent variable.
 - **A carryover effect is an effect that "carries over" from one experimental condition to another.** Whenever subjects perform in more than one condition (as they do in within-subject designs) there is a possibility of carryover effects. For example, consider an experiment on the effect of rate of presentation on memory. Subjects are presented with a list of words and asked to recall as many words as they can. In one condition, the words are presented one word per second; in the other condition, the words are presented two words per second. The question is whether or not having performed in one condition affects performance in the second condition. Perhaps learning the first list of words will interfere with learning the second list because it will be hard to remember which words were in each list. Or maybe the practice involved learning one list will make it easier to learn a second list. In either case, there would be a carryover effect: performance on the second list would be affected by the experience of being given the first list.
- **Within subject design** – Independent variables are applied to everyone in the group. As done in Jarrard's LSD study with rats. So, all participants are exposed to both the conditions.
 - **Advantages** – There is no chance of group differences, as each subject serves as his own control.
 - **Disadvantage** – There is a chance of order effects such as practice effect or it can be carry over effect.
 - When do we use the within-subjects design:
 - When **independent variable does not have a permanent effect** on the participants. For example, where participants have lesion.
 - Use it preferably **when order effects such as practice and carry over effects can be avoided by counter balancing or random order**
 - **Generally, within-subject designs are more efficient than between-subjects design.** They also have more power or statistical significance as the participants are much more.

- So, researchers use following methods to counter order effects
 - Complete counterbalanced design
 - Partial counterbalancing design
 - Latin Square test
 - Balanced Latin Square design
 - Random order
- **Complete counterbalanced design**
 - Where all possible permutations of orders are used.

1 2 3 4	1 2 4 3	1 3 2 4	1 3 4 2
1 4 2 3	1 4 3 2	2 1 3 4	2 1 4 3
2 3 1 4	2 3 4 1	2 4 1 3	2 4 3 1
3 1 2 4	3 1 4 2	3 2 1 4	3 2 4 1
3 4 1 2	3 4 2 1	4 1 2 3	4 1 3 2
4 2 1 3	4 2 3 1	4 3 1 2	4 3 2 1
 - For example, we want to test the effect of 4 font types on reading speed. Using all possible orders with all participants will remove the order effect. Participants may also improve reading speed during the experiment due to practice. So, filter the practice effect by using different text for each font for each participant, such that everyone has the same practice time for each font. Therefore we are using different texts in different fonts in different orders for different participants.
 - Often it is the best option but not always possible. Especially as number of levels rise. For example, for 6 fonts 720 orders are possible.
- **Latin square design**
 - Each condition happens equally as often at each ordinal position.
 - Orders will be – 1234, 2341, 3412, 4123.
 - Need 4 or multiples of 4 participants for this to work. So let's say 16 participants will read the fonts in order 4123, other 16 will read 1234, other 16 will read 2341, and other 16 will read 3412.
 - If we find a difference between different combinations, it cannot be attributed to practice effect because every font in every occurred equally as often for each participant.
- **Balanced Latin square design**
 - Every condition precedes and follows every other condition, equally often.
 - Combinations are
 - 1234, 2413, 3142, 4321
 - So, within these 4 combinations, not only did we get every type of font or independent variable at each ordinal position, but we also got combinations where every font followed and preceded the other ones. For example, 2 occurred after 1, 3 and 4, as well as 2 is before 1, 3 and 4.
 - Use
 - When one specific level of independent variable has large effect on the next variable

- Let's say in a research, level 2 independent variable is to do a task in cold areas without gloves. Level 3 is doing it with thick gloves. Here, 2 would have frozen the participant's hand, and would affect the next level.
- We can't use it when there are many independent variables. Here, we choose the orders randomly.

○ Factorial designs

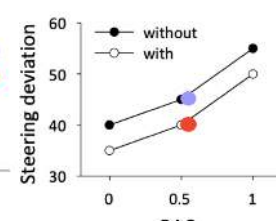
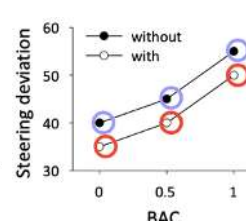
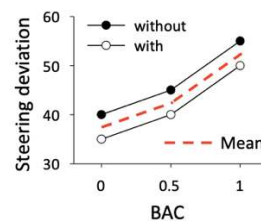
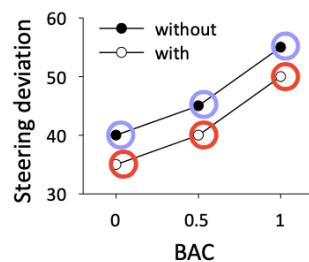
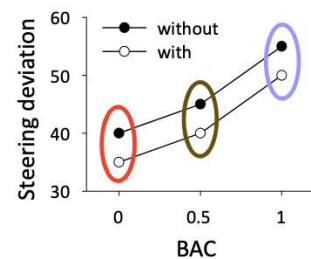
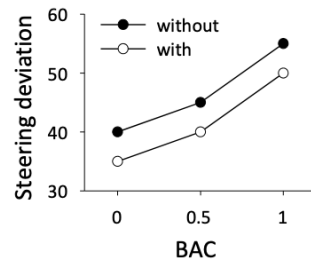
- Design in which each level of every independent variable occurs with all levels of the other independent variable.
- We combine all levels of one variable with other variable
- Example – say we want to find out the effect of blood alcohol levels and power steering on driving. So, blood alcohol and power steering are the independent variables.

Power steering	BAC		
	0	0.5	1.0
without	without, 0	without, 0.5	without, 1.0
with	with, 0	with, 0.5	with, 1.0

	A1	A2	A3
B1	A1B1	A2B1	A3B1
B2	A1B2	A2B2	A3B2

- This is a 3x2 design.
- BAC and Power steering vary independently of each other, and they are not confounded. So, we can attribute effects easily to a certain scenario.
- The letters A and B are the independent variables and called as factors. The figures of the letter (1,2,3) are called levels of a factor.
- In our example, 'A' has 3 levels and 'B' has 2 levels. So, it is a 3x2 factor design. Other designs can have 3 factors and 2 levels each, so that would be 2x2x2 factor design.
- In our example, we can have 3 types of designs
 - Between-group design – 6 groups with one situation for each group.
 - Within-subjects design – 6 ordered combinations for each participant.
 - Mixed design – One of the factors is between -subjects, and another factor is a within-subjects design. In this example, one group gets the power steering and other does not, but both groups get the alcohol in various amounts.
- Advantages of factorial design
 - Efficiency – can do one experiment with two factors instead of two experiments, each with one factor.

- **Generalization** – we can generalize the results. E.g. the effect of alcohol in blood with cars with power steering and cars without it
- **Information on interaction between factors** – does effect of blood alcohol depend on power steering or not. More below:

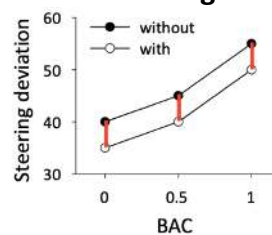


- **BAC** and Power steering are manipulated. We measure the steering deviation with BAC. We can draw a legend or a graph to study the effects and interaction.
- **Main effect** is the overall effect of one independent variable on the dependent variable, averaged over the levels of the other independent variable.
- **Interaction effect** – When the effect of one independent variable depends on the level of the other independent variable(s). So, if interaction of BAC(A) varies as a function of Power steering (B) we say that there is interaction effect AxB.
 - It can be **symmetrical** interaction if: interaction AxB == interaction BxA.
- **Effects** in a two-factorial design
 - **Main effect** of Factor A (BAC). Is there a main effect of BAC? We calculate the average of both

values of BAC at each level, as shown above. We see from the mean line, that dependent variable (Steering deviation) increases with increases in blood alcohol level, clearly there is a main effect.

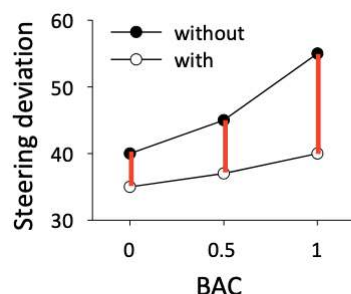
- **Main effect of Factor B (Power steering).** For main effect of Power steering, we average the values of all BAC values for with and without power steering. As we see from the two dots, the steering does deviate more where there is no power steering.

- **Interaction effect: Is there interaction BAC x Power Steering?**



	0	0.5	1.0
without	40	45	55
with	35	40	50
Difference	5	5	5

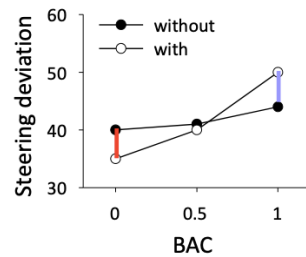
- As the difference in Steering deviation at each BAC remains the same when Power steering is present or absent, we can say that the effect of Power steering does not depend on BAC. But if the lines had been curved, instead of parallel, we would say that there is an interaction between BAC and Power Steering. However, in our example, there is no such interaction.
- If lines (black dotted and white dotted lines) are parallel, then generally there is no interaction.
- Presence or absence of main effect or interaction is generally made in reference to a population. Main effect and interaction are inferred from a statistical test. The graphs in this sample, are not necessarily valid for the entire population. Only after applying statistical tests on our data can we confidently say that a main effect or an interaction exist or not.



	0	0.5	1.0
without	40	45	55
with	35	37	40
Difference	5	8	15

- Main effect of BAC? ✓
- Main effect of Power steering? ✓
- Interaction BAC x Power steering? ✓

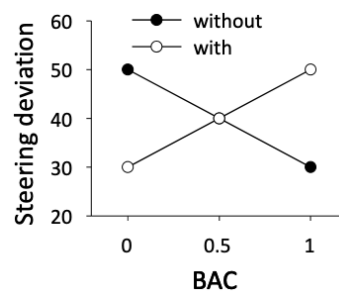
- In the above example, the lines (black dotted and white dotted lines) are not parallel, so there is an interaction.
- **More examples**



	0	0.5	1.0
without	40	41	44
with	35	40	50
Difference	5	1	-6

- Main effect of BAC? ✓
- Main effect of Power steering? ✗
- **Interaction BAC x Power steering?** ✓

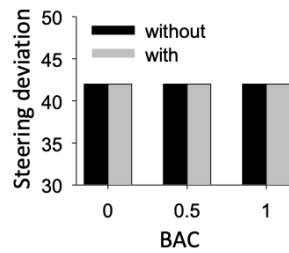
Compare the graphs



- Main effect of BAC? ✗
- Main effect of Power steering? ✗
- **Interaction BAC x Power steering?** ✓

		BAC		
		0	0.5	1.0
Power steering	without	65	60	50
	with	35	40	50

- Main effect of BAC? ✗
- Main effect of Power steering? ✓
- **Interaction BAC x Power steering?** ✓



- Main effect of BAC? ✗
- Main effect of Power steering? ✗
- Interaction BAC x Power steering? ✗
- Which effects can be distinguished in a factorial design with three independent variables A, B, C?
 - Main effect A
 - Main effect B
 - Main effect C
 - Interaction A x B
 - Interaction A x C
 - Interaction B x C
 - Interaction A x B x C (three-way interaction)
- Experimental research
 - On basis of experimental research we can make causal claims
 - Establishes internal validity
 - Requires an appropriate design
 - Removing confounds.
 - Between-subjects variables: Did the researcher control for selection effects using random assignment or matching?
 - Within-subjects variables: Did the researcher control for order effects by (partial) counterbalancing or randomization?
 - How well was each variable manipulated and measured
 - Are the results externally valid
 - Are the results statistically valid

Psychology Notes Topic 14

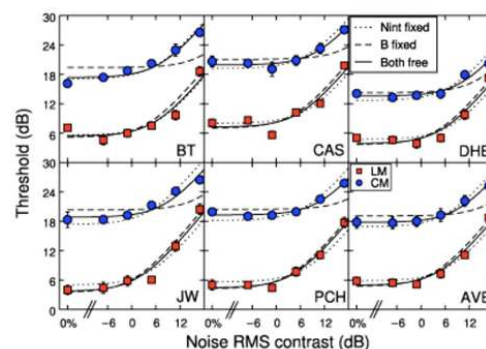
By Somansh – Uilenstede to VU bicycle race champion 2020-21

Overview – Small-N and quasi-experimental designs

- Small-N studies
- Quasi-experimental designs
- Questions

Small-N study

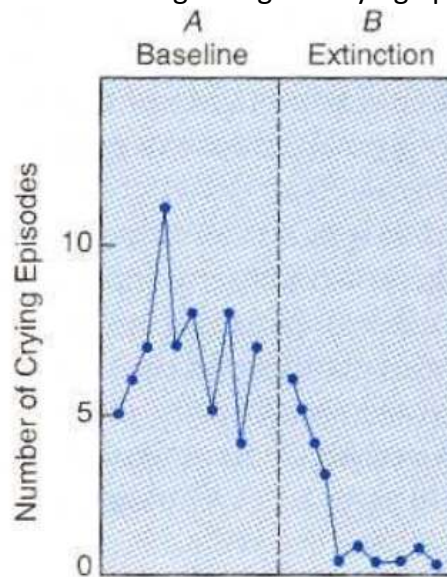
- Study using **only one or few participants**
- Uses
 - Used when **only a few individual differences**.
 - When people want to change behavior of one or few persons.
 - **Used in clinical psychology in form of case studies**
 - **Typically used in psychophysics** i.e. study of relationship between stimuli and perceptual experience.
 - **Psychophysics**



- Psychophysical study
- $N = 5$
- Participants: BT, CAS, DHB, JW, PCH
- AVE = average
- *"if you've seen one, you've seen them all"*

- Since individual differences are small, research is done on few people and can be **generalized on the general population**. So, if there's a pattern in few participants, it practically exists in the general population.
- **Behavior-change studies** – Small-N can also be used to learn **if specific interventions work for specific people**, often patients. Here, individual differences may be large, but we just want to claim if the intervention can be suitable for a single person.
 - One type of design is **Stable-baseline design or AB design**
 - Here, researcher observes the behavior during an extended baseline period **before the treatment (period A)**, and **during or after the treatment (period B)**
 - Example. Does attending the crying baby lead to more tantrums? We suggest parents to selectively reward other types of behavior and ignore when the baby is crying. To see if it works, we use AB design
 - We first measure behavior at normal times – parents reward attention to crying baby. Period A

- Measure behavior after changing attention to other behavior and ignoring the crying episodes.



- Does this sudden change in behavior happen due to parent's behavior or was it due to external factor such as change in school or something else?
- So, it has **very low internal validity** as the **confounding variables** may be present.
- To counter this, we can use **Reversal design**
- **Reversal design or ABAB or ABA design**
 - **Similar to Ab design but we keep taking away the treatment repeatedly in between. This reduces the confounding variable's effects,** and we can more confidently say that any change in behavior was due to the therapy and not due to external factors.

Small-N studies

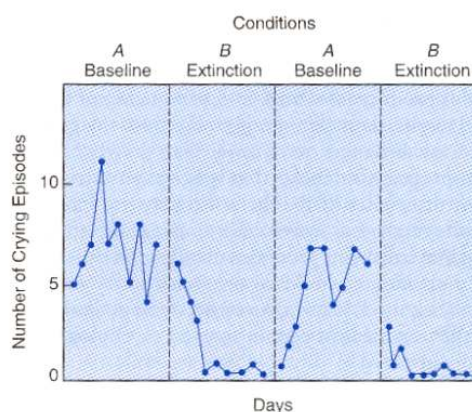
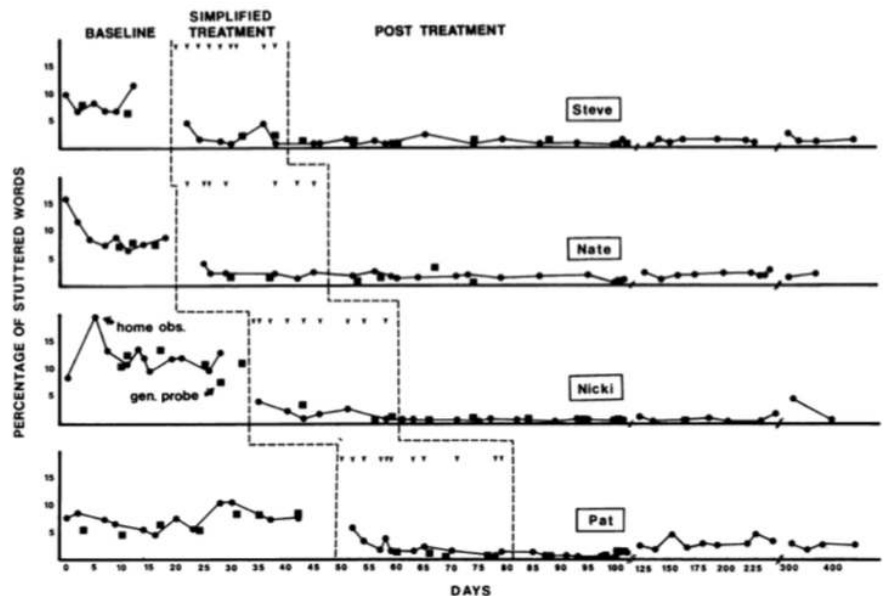


Figure 11-1 The number of crying episodes exhibited by Bill, a nursery-school student, during the four phases of an ABAB design initiated to control his problem crying (From Wolf & Risley, 1969, p. 316).

- The reversal design **can't be used when the treatment or therapy has permanent effect.** For instance, educating the participants cannot be reversed.

- **Multiple-baseline design** – a phased introduction of a treatment
 - The treatment may be a treatment of behavior of different subjects or treatment of several behavior of a single patient.
 - It would work if the treated behaviors were changed and intreated behaviors were unchanged.
 - Wagaman’s 1993 study studied the effects of a therapy on stuttering, used multiple-baseline design. Behavior covaries with the therapy. We are more confident about the effect of therapy. Because there is very little chance that for all 4 kids the stuttering reduced at the same time of therapy, due to some factor, other than therapy.

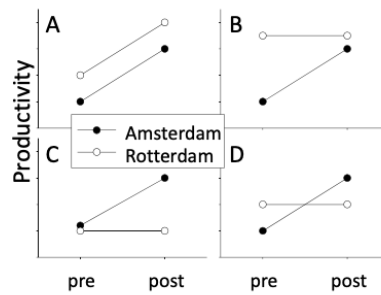


- **Case studies**
 - Often involve a detailed examination of one individual – who may be anxious, depressed or some other psychological condition
 - Often use AB design, where A is historical behavior and B is the present behavior
 - It has low internal validity, and it involves detective work because we never know for sure what is causing the behavior. But they may, at best, suggest future areas for research. Further studies would be needed to confirm or deny the hypothesis generated by a case study. They tell us what ‘can happen’ as opposed to what ‘typically happens’.
 - Achievement’s
 - Broca had a case where patient could only say “10”, but could think normally. He found that the patient’s brain was damaged in left frontal area, thus he concluded that this part was critical to speech production.
- More on small-N studies
 - Their internal validity is not quite high. Except in psychophysics, the external validity is not guaranteed.
 - We must ensure
 - Construct validity – each variable must be properly manipulated and measured.

- External validity – how well results generalize.
- Statistical validity

Quasi-experimental design

- Researcher doesn't have full control over the experiment.
 - The researcher selects rather than manipulate the independent variable. May not also assign participants randomly to different groups.
 - Used
 - When researcher is interested in studying the effects of specific environmental variables. They won't be able to control factors. For e.g. studying the effects of socio-economic status on mental health. Here, researcher can't make anyone poor or rich, but only select rich and poor participants
 - Investigate the effects of Subject variables – for instance age, gender or brain damage. Again, researcher can't assign age and gender to the subjects randomly.
 - So, the internal validity is low. We tackle it with
 - AB design or Pre-test/post-test design
 - O₁T O₂ form – Observation, Treatment, and Observation.
 - Still, it has weak internal validity. Also, there's chances of contamination by other variables:
 - Maturation
 - History
 - Drop-out or death of participants
 - Example. Study on the effect of flower essence on depression.
 - Not much internally valid because people dropped out of study. Participants may've regressed to the mean. Season may've changed, and that may have affected the depression.
 - So, this design is wrong
 - We should use a non-Equivalent control group pre-test/ post-test design
 - The experimental group receives the treatment
 - Non-equivalent control group is not receiving the treatment.
 - Here, the groups should be matched on potential confounding variables such as age, and gender. So, that control group is as equivalent as possible to the experimental group.
 - We should also get a measurement before the treatment, so that we know that both the groups as similar, prior to the treatment.
 - Example. Is there a change in productivity, after introducing flex working schedule, where company controls the schedule, not the researcher. There is flex schedule is in Amsterdam branch.
- more participants/generalized compared to small-n design*

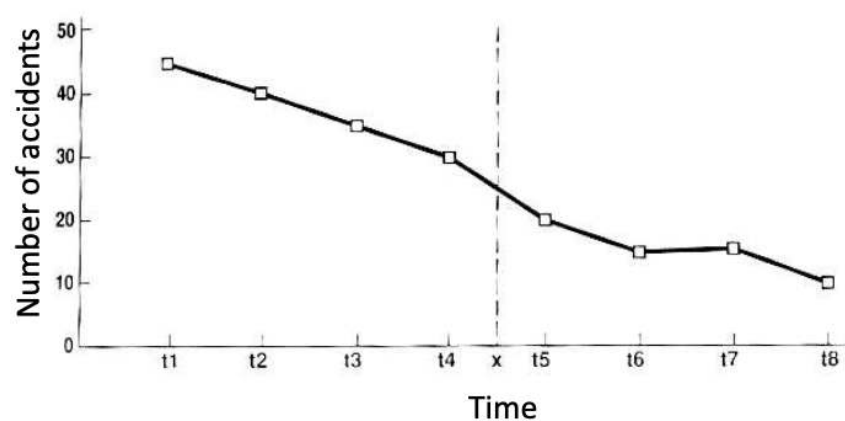


A. Equal change in E and C: no evidence for effect of T
B. possible ceiling effect Rotterdam: no good evidence for effect of T
C & D. Good evidence for effect of T (but be aware of Hawthorne effect!)

- A shows no effect due to flex schedule.
- In B, the productivity was already high, so, there might be a ceiling effect in Rotterdam. So, we don't have good evidence about the effect of treatment.
- Both C and D show good evidence of the effect
- **Hawthorne effect** – Hawthorne company was a company where a famous study was done. They noted that regardless of changes, the worker productivity boosts, because workers knew they were in a study.

○ Interrupted time series design

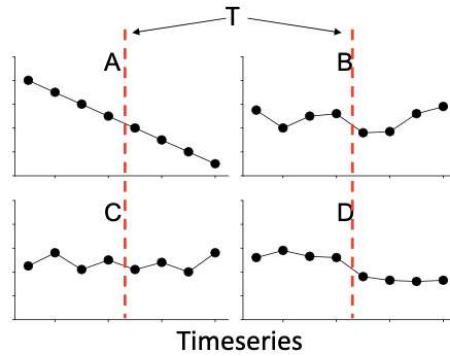
- Multiple observations prior and after the treatment.
- Allows time series analysis
- But **weak internal validity** due to **lack of control group**
- Example. Effect of fluoride in drinking water on school performance. The performance increased with fluoride levels. But, was the effect or indirect: may be students went to the dentist less often due to more fluoride, and got better grades.
- Example. Effect of new traffic reforms on the number of accidents. But there's no way to tell, if traffic measures caused lesser accidents. It is just a correlation.



X = Introduction of new traffic measurement

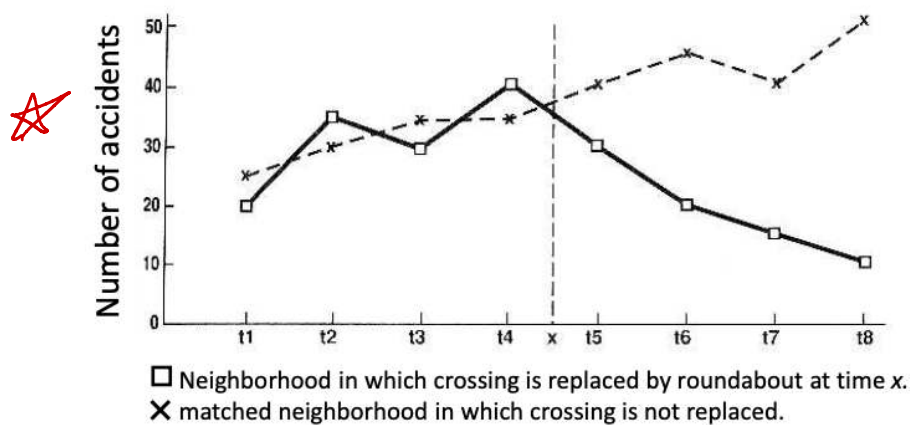
- Other trends example. Effects of anti-smoking campaign on teens smoking.

read examples

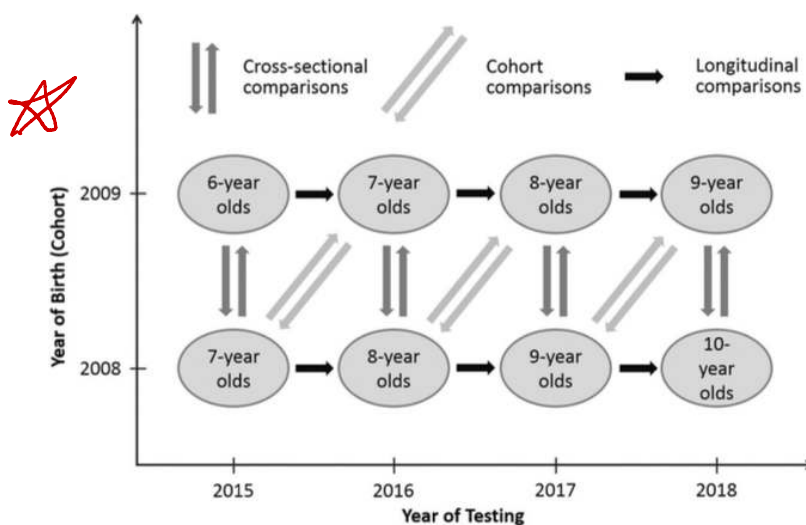


- A.** Continuation of linear trend: No evidence for effect of T.
- B.** Maybe temporary effect of T.
- C.** Continuation of periodic fluctuations: no evidence for effect of T.
- D.** Best possible evidence for effect of T.

- To improve the internal validity of ITS design, we use Non-equivalent control group design (It's called non-equivalent because, groups are matched on certain criteria, and not randomly assigned)
- Non-equivalent control group interrupted time series design
 - It combines the strength of ITS and Non-Equivalent Pre-Test Post-Test design.
 - It is the best quasi-experiment design.
 - Effect replacement of crossing by roundabout



- Design for studying variable of Age



- Cross-sectional – Effect of age can be studied by picking Subjects from each age group

- But here, **confounding variable is generational gap**. Differences in the groups may not be due to age but generational gap
- **Longitudinal – study each subject over their life or years.**
 - Apart from being very time consuming, there is **a confounding variable of time**. Times of testing may affect the variable we are trying to study.
 - **To reduce effect of time, we use cohort comparisons**
- **Cohort comparisons**, the confounding variable of generation arises again.
- **In cross-sequential design, we use a combination of above 3 comparison methods.** We can **statistically separately effects of age, generation and time of testing**. But we'll need to do a lot of calculations and measurements at a time and across time. Thus, it's costly and time consuming.
- **Quasi-experimental design highlights:**
 - Their **internal validity is not quite high**.
 - We must ensure
 - **Construct validity** – each variable must be properly manipulated and measured.
 - **External validity** – how well results generalize.
 - Statistical validity
- Questions at the end of lecture, not included in the notes.

ALL THE BEST

